Supplement to Spatial scale space from hemigroup axioms: rotation-invariant self-decomposable kernels in dimension d

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This supplement belongs to the paper of the same title. The paper prints the statements, the proofs of the classification and of Proposition 3.22, and the proofs of the statements that are proved in print only. Six lemmas that Lean checked are stated in §S3 and not in the paper, which cites them by their numbers here. The supplement has what the formal verification covers: the account of the trust base (§S1), the seven cited facts as the formal development assumes them (§S2), the proofs of the statements that Lean checked (§S3), and the list of the declarations of the formal development with Lean's report (§S4). The statements, sections, equations and tables of the supplement are numbered S.1, S.2, and so on. A number without an S, as in Theorem 3.6, §3.3 or Appendix A, is a number of the paper, and the notation is that of the paper (Table 2).

S1 The trust base in full

§1.1 of the paper says what is machine-checked, lists the statements that are proved in print only, names the seven cited facts, and says in a sentence each how the statements restated from the earlier papers stand, what the formal statement on the time-causal families takes as given, and what is computed. This section adds the detail: the status of the definitions and of the remark, the statements restated from the earlier papers, the form of five formal statements, what is outside the formal verification, and how the formal statements were reviewed.

A definition counts as machine-checked when it is formalised in Lean, that is, when it has a formal counterpart that the formal statements use. Remark 3.26 is commentary and has no formal counterpart.

The statements of §2. The formal development proves what it needs of the statements restated in §2, from the seven cited facts, so the formal proofs of this paper rest on none of them. In their own papers part of them is established in print. Of the line paper, that the admissible exponents are those of the symmetric self-decomposable laws is cited from the literature, and of the cone paper the dimension filtration is proved in prose ([2], §1.1; [3], §1.1). Appendix B has the account statement by statement.

The form of five formal statements. Theorem 3.13, Propositions A.3, A.4 and 3.18 and Lemma S.9 are printed for the law with a given Lévy measure. Their formal statements are about every law with the corresponding transform, and need no such law to exist. That there is one, which the printed "let μ be its law" supposes, is C3(iii). The upper bound and the two forms of the rate are likewise verified for every radially nonincreasing density of the law, and that there is one is Proposition 3.15.

Outside the formal verification. Three things in the paper are proved in print or computed. The first is the statements of the second list of §1.1. Among them is the closed form of a constant. So the constants Q and B of the rate through the profile are verified in the form in which Proposition 3.18 states them, and their closed forms, clause (4) of Lemma 3.17, are proved in print. The second is what the text computes from the statements. That a kernel with a Gaussian part has a bounded smooth density is shown before Theorem 3.13. Example 3.19 applies the statements to the gamma delay and quotes the closed form of the density and its behaviour at the centre, which are cited. The list of c for the familiar kernels, the profile with an unbounded factor, the dependence of the spike on the scale and the stage of the Matérn kernel of index 1 in §3.3, the Matérn exponent in the dictionary and the shadow of a step in §3.4, and the closed form of the constant A_β after Proposition A.4 are computed in the text. The third is the two figures and the numerical values in the text, which are produced by scripts (Figures 1 and 2). Among them are the critical values of the separating family, and the example $w = 4$ of §3.4 is read from them. Of that example Lean has $w_3 < 4$, so that $F^{(4)}$ is admitted in no dimension $D \geq 3$. That it is admitted on the line and in the plane, which is $4 \leq w_2$, is proved in print in the cone paper ([3, Lem. 4.12(2) and (3)]), and the value $w_2 \approx 5.24$ is computed. The statements are independent of the computed values.

The review of the formal statements. The agreement between each formal statement and the statement printed here was reviewed by AI agents, and the review is a reading, not a machine check. One pass compared each formal statement with the printed one. In further passes, agents with no part in the writing restated the main statements from the printed text alone, for comparison with the formal ones, and searched the formal ones for hypotheses that cannot be met. The review found no formal statement weaker than the printed one, none that holds vacuously and none that is true only by a convention of the proof assistant, such as the value it gives to a division by zero, and no cited fact that states more than the page of its source supports. Propositions 3.16, 3.18 and 3.22, Lemmas S.12 and A.5 and Theorem 3.25 were added after that review. Agents with no part in writing them compared each with its formal statement, clause by clause. They found that each printed clause says what its formal counterpart says, or less, with three notes beside the one on the law above. The cases with an unbounded factor K are proved, and the profile with such a factor in §3.3 is computed in the text. Clause (1) of Proposition 3.22 is two formal statements, which are the clause only together. In Lemma A.5(1), that $\bar{k}(r)$ is a mean has no formal counterpart: it is the definition read with the normalisation. Corollary 3.9 and Proposition 3.20 were added later still and compared in the same way. Where the comparison found a printed sentence with no formal counterpart, one was written and proved: the relabelling of the scales, which is clause (4) of the corollary, and in the proposition the density $p_{s,t}$ of clause (2), item (c), and that a stage which is absolutely continuous has no point mass (clause (3), item (b)).

S2 The cited facts

Seven theorems of the literature are used and not proved. Each is stated as it is used, with its page. They are named C1–C7, C for cited, to keep them apart from the axioms (A1)–(A8) of the line. From here on a cited fact is referred to by its name alone, and Table 4 says which of them each statement rests on. C2 and C3 keep the letters of their source, because the proofs that use them quote them. Inside C2 and C3, A is a matrix, ν a Lévy measure and γ a drift vector, and λ and k_η are the measure and the profile of the polar criterion. Elsewhere in the paper these letters have the meanings given where they are introduced (Table 2).

Theorem S.1 (C1: Fourier integrability, [6], Prop. 28.1, p. 190)**.** *If a probability measure μ on $\mathbb{R}^d$ satisfies $\int_{\mathbb{R}^d} |\widehat{\mu}(z)|\,|z|^n\,dz < \infty$ for some integer $n \geq 0$, then μ has a density of class C^n on $\mathbb{R}^d$, and the partial derivatives of the density of orders $0, \ldots, n$ tend to 0 as $|x| \to \infty$.*

The *generating triplet* of an infinitely divisible law is the triple of its Lévy–Khintchine representation, written out in C3 below: a Gaussian covariance matrix A, a Lévy measure ν and a drift γ.

Theorem S.2 (C2: the polar criterion, [6], Thm. 15.10, p. 95)**.** *An infinitely divisible law μ on $\mathbb{R}^d$ with generating triplet (A, ν, γ) is self-decomposable if and only if*

$$\nu(B) = \int_{S^{d-1}} \lambda(d\eta) \int_0^\infty \mathbf{1}_B(r\eta)\, k_\eta(r)\, \frac{dr}{r}$$

for every Borel set B, for a finite measure λ on S^{d-1} and a function $k_\eta(r) \geq 0$, measurable in η and nonincreasing in $r > 0$.

The criterion puts no restriction on A or γ. For a symmetric law on the line it is the folded-profile form of [2, Prop. 2.13]. Sato attributes it to Lévy on the line and to Wolfe and to Sato in $\mathbb{R}^d$ ([6], p. 117).

Theorem S.3 (C3: the Lévy–Khintchine representation, [6], Thm. 8.1, pp. 37–38)**.** *On $\mathbb{R}^d$:*

(i) An infinitely divisible law μ on $\mathbb{R}^d$ has

$$\widehat{\mu}(z) = \exp\Big[-\tfrac{1}{2}\langle z, Az\rangle + i\langle \gamma, z\rangle + \int_{\mathbb{R}^d} \big(e^{i\langle z,x\rangle} - 1 - i\langle z, x\rangle \mathbf{1}_{\{|x|\leq 1\}}\big)\, \nu(dx)\Big]$$

for a symmetric nonnegative definite A, a $\gamma \in \mathbb{R}^d$ and a measure ν with $\nu(\{0\}) = 0$ and $\int (|x|^2 \wedge 1)\, \nu(dx) < \infty$.

(ii) The triplet (A, ν, γ) is unique.

(iii) Conversely, every such triplet is the generating triplet of an infinitely divisible law.

Theorem S.4 (C4, [6], Prop. 15.5, p. 93)**.** *If a probability measure μ on $\mathbb{R}^d$ is self-decomposable, that is if for every $b > 1$ there is a probability measure ρ_b on $\mathbb{R}^d$ with $\widehat{\mu}(z) = \widehat{\mu}(b^{-1}z)\, \widehat{\rho}_b(z)$ for every z, then μ is infinitely divisible.*

C5 needs one definition. For a nonempty compact convex set $K \subset \mathbb{R}^d$ with $K = -K$, let V be its linear span, of some dimension $k \leq d$, and let u_K be the *uniform law* on K for the k-dimensional Lebesgue measure of V. A segment, a disc in a plane and the origin itself are such sets, and $u_{\{0\}} = \delta_0$. A law is *symmetric unimodal in Kanter's sense* if it is a mixture of these uniform laws ([5], Def. 1.2, p. 66, and Def. 3.1, p. 70). Wolfe states his theorem for the symmetric laws of class L, which are the symmetric self-decomposable laws ([6], Def. 15.1, p. 90, and Thm. 15.3, p. 91).

Theorem S.5 (C5: Wolfe's unimodality theorem, [9], Thm. 2, p. 144)**.** *Let μ be a symmetric self-decomposable law on $\mathbb{R}^d$. Then there is a probability measure Λ on the uniform laws u_K of the nonempty compact convex symmetric sets $K \subset \mathbb{R}^d$ with*

$$\mu(A) = \int u_K(A)\, \Lambda(du_K) \qquad \text{for every Borel set } A \subseteq \mathbb{R}^d.$$

C5 says nothing about densities: a uniform law on a lower-dimensional set has none. Its formal statement has the mixing measure on all the probability measures, almost every one a uniform law. The source has it on the set of the uniform laws, which is measurable (the end of the proof of Proposition 3.15, §S3.3). Wolfe's printed proof covers the laws with no Gaussian part: it ends with a theorem of Kanter that is stated for them ([5], Thm. 4.1, p. 78, with the convention of p. 77). The others are convolutions of such a law with a centred Gaussian law (C2 to C4), and two results of the same paper cover them: a symmetric law with a log-concave

density on a subspace is symmetric unimodal (Lemma 3.1, p. 70), and so is the convolution of two symmetric unimodal laws (Thm. 3.2, p. 74).

C6 is Schoenberg's theorem on the radial functions that are negative definite in every dimension, read with the representation of a Bernstein function. The two are Theorems 13.14 and 3.2 of one book, and C6 composes them: the first makes such a function a Bernstein function, the second writes a Bernstein function in the form displayed below. Inside C6 the letters f, a, b, c_j, m and t are the source's.

Theorem S.6 (C6: Schoenberg's radial theorem, with the representation of a Bernstein function, [7], Thm. 13.14, p. 212, and Thm. 3.2, p. 21)**.** *Let f be a real function on $[0,\infty)$ with $f \geq 0$ on $(0,\infty)$. Suppose that for every integer $d \geq 0$ the function $\xi \mapsto f(|\xi|^2)$ on $\mathbb{R}^d$ is continuous and negative definite:*

$$\sum_{j,k=1}^{n} \big(f(|\xi_j|^2) + f(|\xi_k|^2) - f(|\xi_j - \xi_k|^2)\big)\, c_j\, \overline{c_k} \geq 0$$

for every n, all $\xi_1, \ldots, \xi_n \in \mathbb{R}^d$ and all $c_1, \ldots, c_n \in \mathbb{C}$. Then there are $a, b \geq 0$ and a measure m on $(0,\infty)$ with $\int_{(0,\infty)} (1 \wedge t)\, m(dt) < \infty$ such that

$$f(\sigma) = a + b\sigma + \int_{(0,\infty)} \big(1 - e^{-\sigma t}\big)\, m(dt) \qquad \text{for every } \sigma > 0.$$

The book states the hypothesis for every $d \geq 1$. C6 writes it for $d \geq 0$, which adds nothing: at $d = 0$ it says only $f(0) \geq 0$, which is the hypothesis at $d = 1$ with $n = 1$, $\xi_1 = 0$ and $c_1 = 1$. C6 leaves out the uniqueness of (a, b, m), which the book states with the representation ([7], Thm. 3.2, p. 21). The value $f(0)$ is the right limit $f(0+)$, which the continuity of the hypothesis forces and which is the book's convention wherever it fixes the value. The definition of a negative definite function is the book's Definition 4.3, p. 36, in its form (4.8) on $\mathbb{R}^d$, pp. 45–46, written out in the display.

The last cited fact is Karamata's Tauberian theorem, in its form at the origin. A positive measurable function ℓ on $(0,\infty)$ is *slowly varying* if $\ell(\lambda x)/\ell(x) \to 1$ as $x \to \infty$ for every $\lambda > 0$ ([1], p. 6). The logarithm is slowly varying on $(1,\infty)$, and so is every positive function with a positive limit at infinity. The theorem concerns a measure on the half-line. It passes from the behaviour of its Laplace transform at infinity to the mass that the measure gives to small intervals $[0, x]$. A passage in this direction, from a transform back to the measure, needs a condition and is called Tauberian. Here it turns the transform of a kernel into the mass of the small balls round its centre (Lemma S.9). Inside C7 the letters ν, ℓ, c, ρ, s and x are local: c and ρ are two positive constants, and c is not the profile at the origin.

Theorem S.7 (C7: Karamata's Tauberian theorem at the origin, [1], Thm. 1.7.1′, p. 38)**.** *Let ν be a finite measure on $[0,\infty)$, let ℓ be a positive measurable function on $(0,\infty)$ with $\ell(\lambda x)/\ell(x) \to 1$ as $x \to \infty$ for every $\lambda > 0$, and let $c > 0$ and $\rho > 0$. If*

$$\frac{\int_{[0,\infty)} e^{-sx}\, \nu(dx)}{c\, s^{-\rho}\, \ell(s)} \to 1 \quad (s \to \infty), \qquad \text{then} \qquad \frac{\nu([0,x])}{c\, x^{\rho}\, \ell(1/x)/\Gamma(1+\rho)} \to 1 \quad (x \downarrow 0).$$

The book states the theorem as an equivalence, for every nondecreasing function on the line that vanishes on the negative half-line and whose Laplace–Stieltjes transform is finite for large s, and for $c \geq 0$ and $\rho \geq 0$. It attributes it to Feller. C7 is one direction, from the transform to the measure, for the distribution function $x \mapsto \nu([0,x])$ of a finite measure, for positive c and ρ, and for a slowly varying function defined on all of $(0,\infty)$, where the book requires it near infinity only. So C7 assumes more than its source and concludes less.

S3 Proofs of the machine-checked statements

This section has the proofs of the statements of the paper that Lean checked, in the order of §3. The proofs of the classification are in the paper (§A.2), and so is the proof of Proposition 3.22 (§A.4). Six lemmas are stated here and not in the paper: Lemmas S.8, S.9, S.10, S.11, S.12 and S.13. The paper cites them by these numbers, and where a statement or a figure of its main text needs a formula of one of them, the paper gives the formula.

S3.1 The axioms

The proof is the line paper's, carried from the line to $\mathbb{R}^d$: clause (1) is proved as [2, Lem. 4.1] is, with balls in place of intervals, and the passage between the operators and the kernels follows the proofs of [2, Lem. 4.2 and Lem. 6.1], with $\mathrm{GL}(d)$ in place of the dilations. It is written out in full because those statements are on the line.

Proof of Lemma 3.3. For $A \in \mathrm{GL}(d)$ write $D_A f := |\det A|^{-1} f(A^{-1}\,\cdot)$ on $L^1(\mathbb{R}^d)$. Then $R_Q = D_Q$ for $Q \in \mathrm{O}(d)$, since $|\det Q| = 1$, and $D_\lambda = D_{\lambda I}$. Let $\varphi(x) := (2\pi)^{-d/2} e^{-|x|^2/2}$ be the standard Gaussian density, and for $\sigma > 0$ write $\varphi_\sigma(x) := \sigma^{-d}\varphi(x/\sigma)$, the centred Gaussian density with covariance $\sigma^2 I$.

Two facts are used throughout. For a probability measure μ and $f \in L^1(\mathbb{R}^d)$, $\mu * f$ is the $L^1(\mathbb{R}^d)$-valued Bochner integral $\int T_y f \, \mu(dy)$: the map $y \mapsto T_y f$ is continuous because translation acts continuously on $L^1(\mathbb{R}^d)$, and it is bounded by $\|f\|_1$. And with $\widehat{h}(\xi) := \int e^{i\langle \xi, x\rangle} h(x)\, dx$ for $h \in L^1(\mathbb{R}^d)$, Fubini gives the pairing

$$\widehat{\mu * f}(\xi) = \widehat{\mu}(\xi)\, \widehat{f}(\xi), \qquad |\widehat{h}(\xi)| \le \|h\|_1,$$

and $\widehat{\varphi}(\xi) = e^{-|\xi|^2/2} > 0$.

(1), the interchange identity. For $f, g \in L^1(\mathbb{R}^d)$, $f * g = \int f(y)\, T_y g \, dy$ as a Bochner integral, dominated by $|f(y)|\,\|g\|_1$. A bounded operator commutes with an $L^1(\mathbb{R}^d)$-valued integral, so (A1) and (A2) give $\Phi(f * g) = \int f(y)\, T_y \Phi g \, dy = f * \Phi g$.

The approximants. Let $B_\varepsilon := \{|x| < \varepsilon\}$, $|B_\varepsilon|$ its volume, $\rho_\varepsilon := |B_\varepsilon|^{-1} \mathbf{1}_{B_\varepsilon}$ and $h_\varepsilon := \Phi \rho_\varepsilon$. By (A4) and (A5), $h_\varepsilon \ge 0$ with $\int h_\varepsilon = 1$, so $\nu_\varepsilon := h_\varepsilon\, dx$ is a probability measure on $\mathbb{R}^d$. For every f, $\|f * \rho_\varepsilon - f\|_1 \le \sup_{|y|<\varepsilon} \|T_y f - f\|_1 \to 0$ as $\varepsilon \downarrow 0$.

Tightness. By the interchange identity $\rho_1 * h_\varepsilon = \Phi(\rho_1 * \rho_\varepsilon)$, which tends to $\Phi\rho_1$ in $L^1(\mathbb{R}^d)$ by (A1). Along $\varepsilon_n := 1/(n+1)$ the functions $\rho_1 * h_{\varepsilon_n}$ therefore have uniformly small tails: given $\eta > 0$ there is R with $\int_{|x|>R} (\rho_1 * h_{\varepsilon_n}) < \eta$ for every n (choose N with $\|\rho_1 * h_{\varepsilon_n} - \Phi\rho_1\|_1 < \eta/2$ for $n \ge N$, then R for $\Phi\rho_1$ and the first N terms). Convolution with a nonnegative unit-mass density carried by the unit ball moves mass by at most 1: for $|y| > R+1$ the translate $\rho_1(\cdot - y)$ is carried by $\{|x| > R\}$, so by Tonelli

$$\int_{|x|>R} (\rho_1 * h_\varepsilon)(x)\, dx = \int h_\varepsilon(y) \int_{|x|>R} \rho_1(x-y)\, dx\, dy \ge \nu_\varepsilon\big(\{|y| > R+1\}\big).$$

So the sequence (ν_{ε_n}) is tight, with the closed balls of radius $R+1$ as compacts.

The limit. By Prokhorov's theorem a subsequence $\nu_{\varepsilon_{n_k}}$ converges weakly to a probability measure μ on $\mathbb{R}^d$.

Identification. Let $\Psi \in L^\infty(\mathbb{R}^d)$, acting on $L^1(\mathbb{R}^d)$ by $\Psi(h) := \int \Psi h$, and $f \in L^1(\mathbb{R}^d)$. The map $y \mapsto \Psi(T_y f)$ is continuous and bounded by $\|\Psi\|_\infty \|f\|_1$, hence a test function for weak convergence. Since $f * h_\varepsilon = \int h_\varepsilon(y)\, T_y f \, dy$ and Ψ commutes with the integral,

$$\Psi(f * h_{\varepsilon_{n_k}}) = \int \Psi(T_y f)\, \nu_{\varepsilon_{n_k}}(dy) \xrightarrow{k\to\infty} \int \Psi(T_y f)\, \mu(dy) = \Psi(\mu * f).$$

On the other hand $f * h_\varepsilon = \Phi(f * \rho_\varepsilon) \to \Phi f$ in $L^1(\mathbb{R}^d)$, by the interchange identity and (A1). Hence $\Psi(\Phi f - \mu * f) = 0$ for every Ψ; taking Ψ the sign of $\Phi f - \mu * f$ gives $\Phi f = \mu * f$.

Uniqueness. If μ and ν are probability measures with $\mu * \varphi = \nu * \varphi$, the pairing gives $\widehat{\mu}(\xi)\,\widehat{\varphi}(\xi) = \widehat{\nu}(\xi)\,\widehat{\varphi}(\xi)$, so $\widehat{\mu} = \widehat{\nu}$ because $\widehat{\varphi} > 0$, and $\mu = \nu$ because a finite measure on $\mathbb{R}^d$ is determined by its Fourier transform. In particular two measures representing the same operator agree.

Converse. For a probability measure μ, $\|\mu * f\|_1 \le \|f\|_1$ by Minkowski's integral inequality (A1); $T_a(\mu * f) = \mu * T_a f$, both being $x \mapsto \int f(x - a - y)\,\mu(dy)$ (A2); $\mu * f \ge 0$ when $f \ge 0$ (A4); and $\int \mu * f = \int f$ for $f \ge 0$ by Tonelli (A5).

(2), from L^1 to the transforms. Suppose $\mu_n * f \to \mu * f$ in $L^1(\mathbb{R}^d)$ for every f, and take $f = \varphi$. By the pairing, $|\widehat{\mu}_n(\xi) - \widehat{\mu}(\xi)|\,\widehat{\varphi}(\xi) \le \|\mu_n * \varphi - \mu * \varphi\|_1 \to 0$; divide by $\widehat{\varphi}(\xi) > 0$. The test density must have a zero-free transform, which is why the indicator of a ball would not do.

(2), from the transforms to L^1: the smoothed measures. For a probability measure ν and $f, g \in L^1(\mathbb{R}^d)$ three facts are used, each by Fubini's theorem: $\|\nu * f\|_1 \le \|f\|_1$; $\|g * f\|_1 \le \|g\|_1 \|f\|_1$; and $\nu * (g * f) = (\nu * g) * f$, where $\nu * g$ is the integrable function $x \mapsto \int g(x - y)\,\nu(dy)$. The Gaussian integral gives, for every $z \in \mathbb{R}^d$,

$$\varphi_\sigma(z) = (2\pi)^{-d} \int e^{-i\langle \xi, z\rangle} e^{-\sigma^2 |\xi|^2/2}\, d\xi.$$

Put $z = x - y$ and integrate in $\nu(dy)$. The integrand is bounded by $e^{-\sigma^2|\xi|^2/2}$, which is integrable for $d\xi \otimes \nu$, so Fubini's theorem gives, for every probability measure ν and every $x \in \mathbb{R}^d$,

$$(\nu * \varphi_\sigma)(x) = (2\pi)^{-d} \int e^{-i\langle \xi, x\rangle}\, \widehat{\nu}(\xi)\, e^{-\sigma^2|\xi|^2/2}\, d\xi.$$

Put $h_n := \mu_n * \varphi_\sigma$ and $h := \mu * \varphi_\sigma$. Since $|\widehat{\mu}_n| \le 1$ and $\widehat{\mu}_n \to \widehat{\mu}$ pointwise, dominated convergence gives $h_n(x) \to h(x)$ for every x. The h_n and h are nonnegative with integral 1 (Tonelli), so $\int (h_n - h) = 0$ and

$$\|h_n - h\|_1 = 2 \int (h - h_n)^+ \xrightarrow{n \to \infty} 0,$$

by dominated convergence again, since $0 \le (h - h_n)^+ \le h$ and $(h - h_n)^+ \to 0$ pointwise.

(2), the approximation. By Minkowski's integral inequality and the substitution $y = \sigma z$,

$$\|\varphi_\sigma * f - f\|_1 \le \int \varphi_\sigma(y)\, \|T_y f - f\|_1\, dy = \int \varphi(z)\, \|T_{\sigma z} f - f\|_1\, dz,$$

which tends to 0 as $\sigma \downarrow 0$ by dominated convergence: the integrand is bounded by $2\|f\|_1 \varphi(z)$ and tends to 0 pointwise, translation being continuous on $L^1(\mathbb{R}^d)$. Now split $f = (f - \varphi_\sigma * f) + \varphi_\sigma * f$. By the three facts, $\mu_n * (\varphi_\sigma * f) = h_n * f$ and $\mu * (\varphi_\sigma * f) = h * f$, so

$$\|\mu_n * f - \mu * f\|_1 \le 2\,\|f - \varphi_\sigma * f\|_1 + \|(h_n - h) * f\|_1 \le 2\,\|f - \varphi_\sigma * f\|_1 + \|h_n - h\|_1\, \|f\|_1.$$

Given $\varepsilon > 0$, choose σ so that the first term is below ε, and then n_0 so that the second is below ε for $n \ge n_0$.

Transport. For a probability measure μ, $A \in \mathrm{GL}(d)$, $f \in L^1(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$, substituting $z = Ay$,

$$\begin{aligned} \big((A_*\mu) * D_A f\big)(x) &= \int |\det A|^{-1} f\big(A^{-1}(x - z)\big)\, (A_*\mu)(dz) \\ &= |\det A|^{-1} \int f(A^{-1}x - y)\, \mu(dy) = D_A(\mu * f)(x). \end{aligned}$$

D_A is invertible on $L^1(\mathbb{R}^d)$ with inverse $D_{A^{-1}}$, so applying the identity to $D_A^{-1} g$ gives $D_A\,(\mu * D_A^{-1} g) = (A_*\mu) * g$ for every g: the operator $D_A\,(\mu * \cdot\,)\,D_A^{-1}$ is convolution by $A_*\mu$. It is convolution by a probability measure ν if and only if $A_*\mu = \nu$, by the uniqueness in (1).

(3). By (1), each $\Phi_{s,t}$ is convolution by a unique probability measure $\mu_{s,t}$. *(I1).* $\Phi_{t,t} = \mathrm{Id}$ is convolution by δ_0, so $\mu_{t,t} = \delta_0$ by uniqueness. By Fubini $\mu_{s,t} * (\mu_{r,s} * f) = (\mu_{s,t} * \mu_{r,s}) * f$, so (A6) says that $\mu_{r,t}$ and $\mu_{s,t} * \mu_{r,s}$ represent the same operator, and they are equal by uniqueness. *(I2).* $\{0 \le s \le t\}$ is a subset of $\mathbb{R}^2$, so continuity on it is sequential. If $(s_n, t_n) \to (s, t)$ in it, (A7) gives $\mu_{s_n,t_n} * f \to \mu_{s,t} * f$ in $L^1(\mathbb{R}^d)$ for every f, and (2) gives $\widehat{\mu}_{s_n,t_n}(\xi) \to \widehat{\mu}_{s,t}(\xi)$ for every ξ. *(I3).* $R_Q = D_Q$, and (A3) reads $D_Q \Phi_{s,t} D_Q^{-1} = \Phi_{s,t}$; the transport with $\nu = \mu = \mu_{s,t}$ gives $Q_* \mu_{s,t} = \mu_{s,t}$. *(I4).* $D_\lambda = D_{\lambda I}$, and (A8) reads $D_{\lambda I} \Phi_{s,t} D_{\lambda I}^{-1} = \Phi_{S_\lambda s,\, S_\lambda t}$; the transport gives $(\lambda I)_* \mu_{s,t} = \mu_{S_\lambda s,\, S_\lambda t}$. *(ND).* If $\mu_{s,t} = \delta_0$ with $s < t$ then $\Phi_{s,t} = \mathrm{Id}$, against (ND).

(4). Each $\Phi_{s,t}$ is linear and satisfies (A1), (A2), (A4) and (A5) by the converse in (1). *Cascade.* $\Phi_{t,t} = \delta_0 * \cdot = \mathrm{Id}$ by (I1), and for $r \le s \le t$, by the associativity of convolution (Fubini) and (I1), $\Phi_{s,t}\Phi_{r,s} f = \mu_{s,t} * (\mu_{r,s} * f) = (\mu_{s,t} * \mu_{r,s}) * f = \Phi_{r,t} f$, which is (A6). *Continuity.* $\{0 \le s \le t\}$ is a subset of $\mathbb{R}^2$, so continuity on it is sequential. If $(s_n, t_n) \to (s, t)$ in it, (I2) gives $\widehat{\mu}_{s_n,t_n}(\xi) \to \widehat{\mu}_{s,t}(\xi)$ for every ξ, and (2) gives $\Phi_{s_n,t_n} f \to \Phi_{s,t} f$ in $L^1(\mathbb{R}^d)$ for every f, which is (A7). *(ND).* If $\Phi_{s,t} = \mathrm{Id}$ with $s < t$, then $\mu_{s,t}$ and δ_0 represent the same operator, so $\mu_{s,t} = \delta_0$ by uniqueness, against (ND) of the family. *Covariance.* For $Q \in \mathrm{O}(d)$, $D_Q = R_Q$, and (I3) with the transport gives $R_Q \Phi_{s,t} R_Q^{-1} = \Phi_{s,t}$, that is $\Phi_{s,t} R_Q = R_Q \Phi_{s,t}$, which is (A3). For $\lambda > 0$, $D_\lambda = D_{\lambda I}$, and (I4) with the transport gives $D_\lambda \Phi_{s,t} D_\lambda^{-1} = \Phi_{S_\lambda s,\, S_\lambda t}$, which is (A8). *The measures.* $\Phi_{s,t}$ is convolution by $\mu_{s,t}$, so by the uniqueness in (1) the measure that (1) gives for it is $\mu_{s,t}$. □

S3.2 Reduction to the line

In the next proof the argument for the continuity (A7) is the one by which the line paper verifies that axiom in the sufficiency half of its classification (the proof of [2, Thm. 7.3]).

Proof of Lemma 3.5. The kernels. By (I3) each $\mu_{s,t}$ is rotation invariant, so by Lemma 3.4(1) each $\kappa_{s,t}$ is a symmetric probability measure with $\widehat{\mu}_{s,t}(\xi) = \widehat{\kappa}_{s,t}(|\xi|)$.

(A1), (A2), (A4), (A5). For a probability measure κ on $\mathbb{R}$ and $f \in L^1(\mathbb{R})$, $(\kappa * f)(x) = \int f(x - y)\, \kappa(dy)$. Tonelli gives $\|\kappa * f\|_1 \le \|f\|_1$, so the operator is bounded, and Fubini gives $\int \kappa * f = \int f$. A nonnegative f gives a nonnegative $\kappa * f$, and $\tau_a(\kappa * f) = \kappa * \tau_a f$ is a change of variables.

(A3). As $\kappa_{s,t}$ is symmetric, $R(\kappa_{s,t} * f)(x) = \int f(-x - y)\, \kappa_{s,t}(dy) = \int f(-x + y)\, \kappa_{s,t}(dy) = (\kappa_{s,t} * Rf)(x)$.

(A6). By (I1) and Lemma 3.4(3), $\kappa_{t,t} = \pi_* \delta_0 = \delta_0$ and $\kappa_{r,t} = \pi_*(\mu_{s,t} * \mu_{r,s}) = \kappa_{s,t} * \kappa_{r,s}$. So $\Phi_{t,t} = \mathrm{Id}$, and $\Phi_{s,t}\Phi_{r,s} f = \kappa_{s,t} * (\kappa_{r,s} * f) = (\kappa_{s,t} * \kappa_{r,s}) * f = \Phi_{r,t} f$, by Fubini in the middle step.

(A8). By (I4) and Lemma 3.4(3), $(\lambda I)_* \kappa_{s,t} = \pi_*\big((\lambda I)_* \mu_{s,t}\big) = \kappa_{S_\lambda s, S_\lambda t}$. For a probability measure κ on $\mathbb{R}$, substituting $z = \lambda y$,

$$D_\lambda(\kappa * f)(x) = \int \lambda^{-1} f\Big(\frac{x - \lambda y}{\lambda}\Big) \kappa(dy) = \int (D_\lambda f)(x - z)\, \big((\lambda I)_* \kappa\big)(dz),$$

so $D_\lambda \Phi_{s,t} f = \kappa_{S_\lambda s, S_\lambda t} * D_\lambda f = \Phi_{S_\lambda s, S_\lambda t} D_\lambda f$.

(ND). Let $s < t$. By (ND) of the family $\mu_{s,t} \ne \delta_0$, so $\kappa_{s,t} \ne \delta_0$ by Lemma 3.4(2). If $\kappa_{s,t} * f = f$ for every $f \in L^1(\mathbb{R})$, take f the standard Gaussian density: its transform is positive, and the transform of the identity reads $\widehat{\kappa}_{s,t} \widehat{f} = \widehat{f}$, so $\widehat{\kappa}_{s,t} \equiv 1$ and $\kappa_{s,t} = \delta_0$, a contradiction. So $\Phi_{s,t} \ne \mathrm{Id}$.

(A7), the collapsing increment. For $f \in L^1(\mathbb{R})$ put $\Theta_f(y) := \|\tau_y f - f\|_1$, a function that is bounded by $2\|f\|_1$, continuous, and zero at $y = 0$, because translation acts continuously on $L^1(\mathbb{R})$. For a probability measure κ, $\kappa * f - f = \int (\tau_y f - f)\, \kappa(dy)$, so Minkowski's integral inequality gives

$$\|\kappa * f - f\|_1 \le \int \Theta_f \, d\kappa. \tag{S.1}$$

Moreover $\Theta_{\kappa'*g} \le \Theta_g$ for every probability measure κ' and every $g \in L^1(\mathbb{R})$, because $\tau_y(\kappa' * g) - \kappa' * g = \kappa' * (\tau_y g - g)$ and convolution by κ' does not increase the norm. Now let pairs (s_n, t_n) with $s_n \le t_n$ converge to a diagonal point (t_0, t_0). By (I2) at $\xi = \omega e$, $\widehat{\kappa}_{s_n,t_n}(\omega) = \widehat{\mu}_{s_n,t_n}(\omega e) \to \widehat{\mu}_{t_0,t_0}(\omega e) = 1$ for every $\omega \in \mathbb{R}$. Lévy's continuity theorem on the line (§2.5) says that if the transforms of probability measures on $\mathbb{R}$ converge at every point to the transform of a probability measure, the measures converge weakly to it. So $\kappa_{s_n,t_n} \to \delta_0$ weakly. Since Θ_g is bounded and continuous, $\int \Theta_g \, d\kappa_{s_n,t_n} \to \Theta_g(0) = 0$ for every g.

(A7), the general pair. Let $(s_n, t_n) \to (s, t)$ with $s_n \le t_n$. If $s = t$ then $\Phi_{s,t} = \mathrm{Id}$ and (S.1) with the previous paragraph gives $\Phi_{s_n,t_n} f \to f$. If $s < t$ put $u_n := \max\{s_n, s\}$ and $v_n := \min\{t_n, t\}$, so that $u_n \to s$, $v_n \to t$ and, for all large n, $s_n \le u_n \le v_n \le t_n$ and $s \le u_n \le v_n \le t$. By (A6),

$$\Phi_{s_n,t_n} f = \Phi_{v_n,t_n} \, \Phi_{u_n,v_n} \, \Phi_{s_n,u_n} f, \qquad \Phi_{s,t} f = \Phi_{v_n,t} \, \Phi_{u_n,v_n} \, \Phi_{s,u_n} f.$$

The left endpoints are moved first, where contraction absorbs them: with $g_n := \Phi_{u_n,v_n} \Phi_{s_n,u_n} f$ and $h_n := \Phi_{u_n,v_n} \Phi_{s,u_n} f$,

$$\|g_n - h_n\|_1 \le \|\Phi_{s_n,u_n} f - f\|_1 + \|f - \Phi_{s,u_n} f\|_1 \to 0,$$

both increments collapsing onto (s, s). The right endpoints are moved last, where (S.1) is read at h_n and $\Theta_{h_n} \le \Theta_f$:

$$\|\Phi_{v_n,t_n} h_n - \Phi_{v_n,t} h_n\|_1 \le \int \Theta_f \, d\kappa_{v_n,t_n} + \int \Theta_f \, d\kappa_{v_n,t} \to 0,$$

both increments collapsing onto (t, t). Hence $\|\Phi_{s_n,t_n} f - \Phi_{s,t} f\|_1 \le \|g_n - h_n\|_1 + \|\Phi_{v_n,t_n} h_n - \Phi_{v_n,t} h_n\|_1 \to 0$. □

Proof of Proposition 3.12. (1). Each $\mu_{s,t}$ is a mixture of probability measures, and $u \mapsto g_u^{(d)}(B)$ is measurable for every Borel B, so it is a probability measure. By Fubini and $\widehat{g_u^{(d)}}(\xi) = e^{-u|\xi|^2/2}$,

$$\widehat{\mu}_{s,t}(\xi) = \int e^{-u|\xi|^2/2} \, \mu^{\mathrm{I}}_{s^2,t^2}(du) = \exp\bigl[-\bigl(F_{\mathrm{I}}(t^2|\xi|^2/2) - F_{\mathrm{I}}(s^2|\xi|^2/2)\bigr)\bigr],$$

which is $\exp[-(F(t|\xi|) - F(s|\xi|))]$. The axioms are read off this transform exactly as in the sufficiency half of Theorem 3.6, with $\chi = \mathrm{id}$ and $S_\lambda t = \lambda t$. That half used three properties of F, and they hold here. F is continuous on $[0, \infty)$, because F_{I} is continuous by dominated convergence: its integrand is at most $\max\{1, \Sigma\}(1 \wedge u) \, k_{\mathrm{I}}(u)/u$ on $\sigma \le \Sigma$, integrable by the two conditions. $F(0) = F_{\mathrm{I}}(0) = 0$. And $F \not\equiv 0$, because $F_{\mathrm{I}} \not\equiv 0$ and $r^2/2$ runs over $[0, \infty)$. So $(\mu_{s,t})$ is an isotropic cascade family, and Theorem 3.6 applies. Its conclusion is met by (id, F) once $F \in \mathcal{A}_d$, and the argument of its necessity half gives that. F is admissible by Proposition 2.5. And $\mu_{0,1}$ has transform $e^{-F(|\cdot|)}$ and, by (I1), residuals $\mu_{\lambda,1}$ in (2.2), as $\widehat{\mu}_{0,\lambda}(\xi) = e^{-F(\lambda|\xi|)} = \widehat{\mu}_{0,1}(\lambda\xi)$. So $F \in \mathcal{A}_d$, and d was arbitrary.

(2) and (3) are the exponent identities $\gamma \log(1 + r^2/2)$ and $(r^2/2)^\alpha = 2^{-\alpha} r^{2\alpha}$, read at $r = t|\xi|$ through (1). At $\alpha = 1$, $\widehat{\mu}_{s,t}(\xi) = e^{-(t^2 - s^2)|\xi|^2/2}$, the transform of $g^{(d)}_{t^2 - s^2}$.

(4). The mixture is a probability measure as in (1), and by Fubini its transform is $E\, e^{-t^2 T_1 |\xi|^2/2}$. On the line, [3, Prop. 5.10(2)] says that the law of the mixture of $g_u^{(1)}$ against T_1 has transform $\frac{2^{1-\vartheta}}{\Gamma(\vartheta)} |\omega|^\vartheta K_\vartheta(|\omega|)$. By the same Fubini that transform is $E\, e^{-T_1 \omega^2/2}$. Reading it at $\omega = t|\xi|$ gives the display. When $\mu^{\mathrm{I}}_{0,1}$ is the law of T_1, by (1) and its Laplace transform at $t^2|\xi|^2/2$, $\widehat{\mu}_{0,t}(\xi) = e^{-F_{\mathrm{I}}(t^2|\xi|^2/2)} = E\, e^{-t^2 T_1 |\xi|^2/2}$, which is the transform of the mixture. Two probability measures on $\mathbb{R}^d$ with the same transform are equal, so $\mu_{0,t}$ is the mixture, and its exponent is in $\mathcal{A}_d$ by (1). □

S3.3 The kernel at its centre

The proof of Theorem 3.13 rests on two lemmas. The first goes back to an observation of Feller, on the line: a nonnegative characteristic function is integrable exactly when the law has a bounded density ([4], XV.3, Corollary, p. 510). The lemma is one half of it in $\mathbb{R}^d$. It gives the value at the centre of the law blurred by a Gaussian as an integral of the transform, and from it the direction that the unbounded side of the theorem uses: a bound on the law near the origin makes the transform integrable.

Lemma S.8 (a nonnegative transform: bounded near the origin only if integrable)**.** *Let $d \geq 1$ and let μ be a probability measure on $\mathbb{R}^d$ whose characteristic function $\widehat{\mu}(\xi) = \int e^{i\langle \xi, x\rangle}\mu(dx)$ is real and nonnegative at every $\xi \in \mathbb{R}^d$. For $t > 0$ let $g_t(x) = (2\pi t)^{-d/2}e^{-|x|^2/2t}$ be the heat kernel.*

1. *For every $t > 0$,*
$$\int_{\mathbb{R}^d} g_t \, d\mu = (2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{\mu}(\xi)\, e^{-t|\xi|^2/2}\, d\xi,$$
and as $t \downarrow 0$ this increases to $(2\pi)^{-d}\int_{\mathbb{R}^d}\widehat{\mu}(\xi)\,d\xi \in (0,\infty]$.

2. *If there are $\delta > 0$ and $M < \infty$ with $\mu(B) \leq M|B|$ for every Borel set $B \subseteq \{|x| < \delta\}$, then $\int_{\mathbb{R}^d}\widehat{\mu}(\xi)\,d\xi \leq (2\pi)^d M$. The hypothesis holds in particular when μ has a density with a representative bounded by M on $\{0 < |x| < \delta\}$.*

With C1 (Theorem S.1), which gives the other direction, the lemma is the form in $\mathbb{R}^d$ of Feller's equivalence: a law with a nonnegative transform has a density bounded near the origin exactly when its transform is integrable. The law of every isotropic origin datum has the positive transform e^{-F_d}. The second lemma, Lemma A.2, bounds that transform from below.

Proof of Lemma S.8. Clause (1). The one-dimensional Gaussian integral $\int_{\mathbb{R}} e^{-isy}e^{-ts^2/2}\,ds = (2\pi/t)^{1/2}e^{-y^2/2t}$, multiplied over the d coordinates, gives $g_t(x) = (2\pi)^{-d}\int_{\mathbb{R}^d} e^{-i\langle\xi,x\rangle}e^{-t|\xi|^2/2}\,d\xi$. The integrand has modulus $e^{-t|\xi|^2/2}$, which is integrable against $d\xi \otimes \mu(dx)$ because μ is finite, so Fubini's theorem gives

$$\int g_t\,d\mu = (2\pi)^{-d}\int e^{-t|\xi|^2/2}\int e^{-i\langle\xi,x\rangle}\mu(dx)\,d\xi = (2\pi)^{-d}\int e^{-t|\xi|^2/2}\,\widehat{\mu}(-\xi)\,d\xi,$$

and the substitution $\xi \mapsto -\xi$ gives the display. Since $\widehat{\mu} \geq 0$, the integrand increases as t decreases, and monotone convergence gives the limit. It is positive because $\widehat{\mu}$ is continuous with $\widehat{\mu}(0) = 1$.

Clause (2). Put $U = \{|x| < \delta\}$. The hypothesis gives $\int_U h\,d\mu \leq M\int_U h\,dx$ for every nonnegative Borel h, for indicator functions of Borel subsets of U directly and in general by simple functions and monotone convergence. With $h = g_t$, $\int_U g_t\,d\mu \leq M\int_{\mathbb{R}^d} g_t\,dx = M$. Off U, $g_t \leq (2\pi t)^{-d/2}e^{-\delta^2/2t}$, so $\int_{\mathbb{R}^d\setminus U} g_t\,d\mu \leq (2\pi t)^{-d/2}e^{-\delta^2/2t}$, which tends to 0 as $t \downarrow 0$. By clause (1), $(2\pi)^{-d}\int\widehat{\mu} = \lim_{t\downarrow 0}\int g_t\,d\mu \leq M$. For the last sentence: if q is a density of μ with $q \leq M$ on $U\setminus\{0\}$, then $\mu(B) = \int_{B\setminus\{0\}} q \leq M|B|$ for every Borel $B \subseteq U$, since the origin is a Lebesgue-null set. □

Proof of Lemma A.2. Since the Gaussian coefficient is 0, (3.3) and (3.2) give, with σ_{d-1} the surface measure of S^{d-1},

$$F_d(\xi) = \frac{1}{|S^{d-1}|}\int_{S^{d-1}} \varphi\big(\langle\xi,\eta\rangle\big)\,\sigma_{d-1}(d\eta), \qquad \varphi(s) := \int_0^\infty \big(1-\cos(rs)\big)\,k_d(r)\,\frac{dr}{r} \quad (s \in \mathbb{R}).$$

φ is even and finite, because $1-\cos(rs) \leq \min\{2, r^2s^2/2\}$ and the two integrability conditions of Definition 3.10 hold. As k_d is nonincreasing with right limit c at the origin, $k_d \leq c$ on $(0,\infty)$. And $k_d(r) \to 0$ as $r \to \infty$, since a nonincreasing $k_d \geq \ell > 0$ would make $\int_1^\infty k_d(r)\,dr/r$ diverge.

The logarithmic bounds on Cin. Write $\mathrm{Cin}(X) = \int_0^X (1-\cos s)\, s^{-1} ds$ for $X \geq 0$, a non-decreasing function. For $X \geq 1$, $\mathrm{Cin}(X) = \mathrm{Cin}(1) + \log X - \int_1^X s^{-1}\cos s\, ds$. Here $0 \leq \mathrm{Cin}(1) \leq \int_0^1 \frac{s}{2}\, ds = \frac{1}{4}$, because $1-\cos s \leq s^2/2$. Integrating by parts, $\left|\int_1^X s^{-1}\cos s\, ds\right| = \left|[s^{-1}\sin s]_1^X + \int_1^X s^{-2}\sin s\, ds\right| \leq 1 + \sin 1 + 1 \leq 3$. For $0 < X < 1$ the lower bound below holds because its right side is negative. Hence

$$\log X - 3 \leq \mathrm{Cin}(X) \quad (X > 0), \qquad \mathrm{Cin}(X) \leq \log X + \tfrac{13}{4} \quad (X \geq 1). \tag{S.2}$$

Clause (1). Write $\rho = |\xi| \geq 1$. For $|s| \leq \rho$, split $\varphi(s)$ at $r = 1$. On $(0,1)$ use $k_d \leq c$, and on $[1,\infty)$ use $1 - \cos \leq 2$:

$$\varphi(s) \leq c\ \mathrm{Cin}(|s|) + 2C_k \leq c\ \mathrm{Cin}(\rho) + 2C_k.$$

On the sphere $|\langle \xi, \eta\rangle| \leq \rho$, and the normalized surface measure has mass 1. So $F_d(\xi) \leq c\ \mathrm{Cin}(\rho) + 2C_k$, and the upper bound of (S.2) gives the bound on F_d. The bound on e^{-F_d} follows.

Clause (2), one variable. Let $a \geq 1$ and $s > 0$. Substituting $u = ar$ in $\varphi(as)$,

$$\varphi(as) - \varphi(s) = \int_0^\infty \big(1 - \cos(us)\big)\, h_a(u)\, du, \qquad h_a(u) := \frac{k_d(u/a) - k_d(u)}{u} \geq 0,$$

where both terms on the left are finite. For $0 < \varepsilon < N$,

$$\int_\varepsilon^N h_a(u)\, du = \int_{\varepsilon/a}^{N/a} k_d(v)\, \frac{dv}{v} - \int_\varepsilon^N k_d(v)\, \frac{dv}{v} = \int_{\varepsilon/a}^{\varepsilon} k_d(v)\, \frac{dv}{v} - \int_{N/a}^{N} k_d(v)\, \frac{dv}{v},$$

and every integral here is finite because $k_d \leq c$. The first term lies between $k_d(\varepsilon)\log a$ and $c\log a$, and tends to $c\log a$ as $\varepsilon \downarrow 0$. The second lies between 0 and $k_d(N/a)\log a$, and tends to 0 as $N \to \infty$. By monotone convergence $\int_0^\infty h_a = c\log a$, so h_a is integrable on $(0,\infty)$, and the Riemann–Lebesgue lemma gives $\int_0^\infty \cos(us)\, h_a(u)\, du \to 0$ as $s \to \infty$. Hence

$$\varphi(as) - \varphi(s) \to c\log a \quad (s \to \infty), \qquad 0 \leq \varphi(as) - \varphi(s) \leq 2c\log a \quad (s > 0).$$

For $0 < a < 1$ apply this with $1/a$ at the point as, which tends to infinity with s: $\varphi(as) - \varphi(s) = -\big(\varphi(a^{-1}\cdot as) - \varphi(as)\big) \to c\log a$, and $|\varphi(as) - \varphi(s)| \leq 2c|\log a|$.

Clause (2), the sphere average. Write $\xi = \rho\widehat{\xi}$ with $|\widehat{\xi}| = 1$. Since φ is even,

$$F_d(a\xi) - F_d(\xi) = \frac{1}{|S^{d-1}|}\int_{S^{d-1}} \Big[\varphi\big(a\rho|\langle\widehat{\xi},\eta\rangle|\big) - \varphi\big(\rho|\langle\widehat{\xi},\eta\rangle|\big)\Big]\sigma_{d-1}(d\eta).$$

For every η with $\langle\widehat{\xi},\eta\rangle \neq 0$ the bracket tends to $c\log a$ as $\rho \to \infty$. The remaining η form a σ_{d-1}-null set: a great subsphere when $d \geq 2$, and no point at all when $d = 1$. The bracket is bounded by $2c|\log a|$. By dominated convergence $F_d(a\rho\widehat{\xi}) - F_d(\rho\widehat{\xi}) \to c\log a$ as $\rho \to \infty$. The left side does not depend on $\widehat{\xi}$, because F_d is rotation invariant, so the limit holds as $|\xi| \to \infty$. For the equivalent form, $g(au) - g(u) = F_d(\sqrt{a}\,\sqrt{u}\,e) - F_d(\sqrt{u}\,e) \to c\log\sqrt{a} = \frac{c}{2}\log a$. □

Proof of Theorem 3.13. Clause (1). Write $\rho = |\xi|$, and let Cin be as in the proof of Lemma A.2. Since $a = 0$, (3.3) and (3.2) give

$$-\log\big|\widehat{\mu}(\xi)\big| = F_d(\xi) = \frac{1}{|S^{d-1}|}\int_{S^{d-1}} \varphi\big(\langle\xi,\eta\rangle\big)\, \sigma_{d-1}(d\eta), \qquad \varphi(s) := \int_0^\infty \big(1-\cos(rs)\big)\, k_d(r)\, \frac{dr}{r}.$$

F_d is radial: if $|\xi| = |\xi'|$, the reflection Q in $(\xi - \xi')^\perp$ carries ξ to ξ', preserves Lebesgue measure and the radial density of ν_d, so substituting $x = Qy$ in (3.3) gives $F_d(\xi') = F_d(\xi)$. It is therefore enough to bound $F_d(\rho e_1)$. Fix a finite $\beta < c$. If $\beta \leq 0$ the bound holds with $C_\beta = 1$, since

$|\widehat{\mu}| \le 1 \le (1+|\xi|)^{-\beta}$. Let $\beta > 0$ and choose β_1 with $\beta < \beta_1 < c$. As k_d is nonincreasing with right limit c at the origin, there is $r_\beta \in (0,1]$ with $k_d \ge \beta_1$ on $(0, r_\beta)$. The integrand of φ is nonnegative, so discarding (r_β, ∞) and substituting gives $\varphi(s) \ge \beta_1 \operatorname{Cin}(r_\beta |s|)$.

For $\varepsilon > 0$ let $K_\varepsilon := \{\eta \in S^{d-1} : |\langle e_1, \eta\rangle| \ge \varepsilon\}$. As $\varepsilon \downarrow 0$ these caps increase to the complement of the great subsphere $\{\langle e_1, \eta\rangle = 0\}$, which is σ_{d-1}-null: $\sigma_{d-1}(E)$ is d times the Lebesgue measure of the cone $(0,1)\cdot E$, and the cone over the great subsphere lies in the hyperplane $e_1^\perp$. So there is $\varepsilon \in (0,1]$ with $p := \sigma_{d-1}(K_\varepsilon)/|S^{d-1}| \ge \beta/\beta_1$. Put $L := 3 - \log(\varepsilon r_\beta) \ge 0$. For $\eta \in K_\varepsilon$, $r_\beta |\langle \rho e_1, \eta\rangle| \ge \rho\varepsilon r_\beta$, so the lower bound of (S.2) gives $\varphi(\langle \rho e_1, \eta\rangle) \ge \beta_1(\log\rho - L)$. Discarding the rest of the sphere, $F_d(\rho e_1) \ge p\,\beta_1(\log\rho - L) \ge \beta(\log\rho - L)$ when $\log\rho \ge L$, and the outer inequality holds as well when $\log\rho < L$, because $F_d \ge 0$. Hence $|\widehat{\mu}(\xi)| \le e^{\beta L}\rho^{-\beta}$ for every $\xi \ne 0$. For $\rho \ge 1$, $1 + \rho \le 2\rho$ gives $\rho^{-\beta} \le 2^\beta(1+\rho)^{-\beta}$; for $\rho < 1$, $|\widehat{\mu}(\xi)| \le 1 \le 2^\beta(1+\rho)^{-\beta}$. This is the stated bound with $C_\beta = 2^\beta(1 + e^{\beta L})$.

Clause (2). Let $c > d + N$ and choose a finite β with $d + N < \beta < c$. By clause (1),

$$\int_{\mathbb{R}^d} \bigl|\widehat{\mu}(\xi)\bigr|\,|\xi|^N\,d\xi \le C_\beta \int_{\mathbb{R}^d} (1+|\xi|)^{-\beta}|\xi|^N\,d\xi < \infty.$$

The integral converges at infinity because $\beta - N > d$, and at the origin because $N \ge 0$. That is the hypothesis of C1, whose conclusion is the clause: a density $g \ge 0$ of class C^N whose partial derivatives of orders $0, \dots, N$ tend to 0 at infinity. At $N = 0$ the density is continuous and tends to 0 at infinity, so it is bounded: it is at most 1 outside a compact set, and continuous on that set. For $c = \infty$ every N is admissible, and the densities obtained for different N coincide: two continuous densities of μ agree almost everywhere, hence everywhere, since a nonempty open set has positive Lebesgue measure. So the density at $N = 0$ is of class C^N for every N, that is, of class C^∞.

Clause (3). First, $c = 2\,k_{\mathrm{I}}(0+)$. As $r \downarrow 0$ the integrand of (3.4) increases to $v^{d/2-1}e^{-v}k_{\mathrm{I}}(0+)$, because k_{I} is nonincreasing. So monotone convergence gives $k_d(0+) = 2\,\Gamma(d/2)^{-1}\Gamma(d/2)\,k_{\mathrm{I}}(0+)$. Next, [3, Prop. 4.9(2)], restated, which is Lemma A.6 for the delay datum $(0, k_{\mathrm{I}})$: the datum's exponent is $F_d(\xi) = F_{\mathrm{I}}(|\xi|^2/2)$. The law of W has Laplace transform $e^{-F_{\mathrm{I}}}$ by hypothesis. So $\widehat{\mu}(\xi) = e^{-F_{\mathrm{I}}(|\xi|^2/2)} = \int_0^\infty e^{-u|\xi|^2/2}\Pr(W \in du)$, which is the transform of the law of $\sqrt{W}Z$, and μ is that law. Since $k_{\mathrm{I}}(0+) = c/2 > 0$, clause (1) read along a ray, at $|\xi| = \sqrt{2\sigma}$, gives $e^{-F_{\mathrm{I}}(\sigma)} \to 0$, so $\Pr(W = 0) = \lim_{\sigma\to\infty} e^{-F_{\mathrm{I}}(\sigma)} = 0$. So $W > 0$ almost surely, and conditioning on W gives (3.6). For $x \ne 0$ the integrand is bounded in u, so $p(x) < \infty$. Each integrand is radial and nonincreasing in $|x|$, and so is p. Monotone convergence as $x \to 0$ gives

$$p_0 := \lim_{x\to 0} p(x) = \int_0^\infty (2\pi u)^{-d/2}\Pr(W \in du) \in (0, \infty].$$

The two other forms of p_0 in the clause follow from Tonelli. Since $\widehat{\mu} \ge 0$, its integral is defined in $[0,\infty]$, and $\int_{\mathbb{R}^d} e^{-u|\xi|^2/2}d\xi = (2\pi/u)^{d/2}$ gives

$$(2\pi)^{-d}\int_{\mathbb{R}^d}\widehat{\mu}(\xi)\,d\xi = (2\pi)^{-d}\int_0^\infty \Bigl(\frac{2\pi}{u}\Bigr)^{d/2}\Pr(W \in du) = p_0.$$

And $\int_0^\infty \sigma^{d/2-1}e^{-\sigma u}d\sigma = \Gamma(d/2)u^{-d/2}$ gives

$$\int_0^\infty \sigma^{d/2-1}e^{-F_{\mathrm{I}}(\sigma)}\,d\sigma = \Gamma(d/2)\int_0^\infty u^{-d/2}\Pr(W \in du) = \Gamma(d/2)\,(2\pi)^{d/2}\,p_0.$$

It remains to decide when p_0 is finite, and the argument reads the transform $\widehat{\mu}$ itself. If $c > d$, pick a finite $\beta \in (d, c)$. By clause (1), $\widehat{\mu}(\xi) \le C_\beta(1+|\xi|)^{-\beta}$, which is integrable over $\mathbb{R}^d$ because $\beta > d$; so $p_0 < \infty$. If $c \le d$, then c is finite, and Lemma A.2(1) gives $\widehat{\mu}(\xi) \ge C_0|\xi|^{-c}$ for $|\xi| \ge 1$, with $C_0 = e^{-13c/4-2C_k}$. In polar coordinates $\int_{\mathbb{R}^d}\widehat{\mu} \ge C_0|S^{d-1}|\int_1^\infty \rho^{d-1-c}\,d\rho = \infty$, because $d - 1 - c \ge -1$. So $p_0 = \infty$, the threshold $c = d$ included.

That makes *every* representative unbounded near the origin, by comparing measures rather than functions, as in Corollary 2.8. If a representative q of μ's density had $q \le M$ on $\{0 < |x| < \delta\}$, then $\mu(B) = \int_B q \le M|B|$ for every Borel $B \subseteq \{|x| < \delta\}$, the origin being a Lebesgue-null set, and Lemma S.8(2) would give $\int_{\mathbb{R}^d} \widehat{\mu} \le (2\pi)^d M < \infty$, a contradiction.

Clause (4). By Definition 3.10, $\widehat{\mu} = e^{-F_d}$ with F_d real and finite, so $\widehat{\mu} > 0$ everywhere and Lemma S.8 applies. By Lemma A.2(1), $\widehat{\mu}(\xi) \ge C_0 |\xi|^{-c}$ for $|\xi| \ge 1$, with $C_0 = e^{-13c/4 - 2C_k} > 0$ and C_k as in that lemma. In polar coordinates,

$$\int_{\mathbb{R}^d} \widehat{\mu}(\xi)\, d\xi \ge C_0\, |S^{d-1}| \int_1^\infty \rho^{d-1-c}\, d\rho = \infty,$$

because $d - 1 - c \ge -1$. If there were δ and M with $\mu(B) \le M|B|$ for every Borel $B \subseteq \{|x| < \delta\}$, Lemma S.8(2) would make this integral at most $(2\pi)^d M$. So there are none. A representative q of a density with $q \le M$ on $\{0 < |x| < \delta\}$ would give such δ and M, by the last sentence of Lemma S.8(2). So there is none.

For the rate, Lemma S.8(1) and the same lower bound give, for $0 < t \le 1$,

$$\int g_t\, d\mu \ge (2\pi)^{-d} C_0\, |S^{d-1}| \int_1^\infty \rho^{d-1-c} e^{-t\rho^2/2}\, d\rho.$$

If $c < d$, the substitution $\rho = \sigma/\sqrt{t}$ turns the last integral into $t^{-(d-c)/2} \int_{\sqrt{t}}^\infty \sigma^{d-1-c} e^{-\sigma^2/2}\, d\sigma \ge t^{-(d-c)/2} \int_1^\infty \sigma^{d-1-c} e^{-\sigma^2/2}\, d\sigma$, a positive multiple of $t^{-(d-c)/2}$. If $c = d$, it is at least $e^{-1/2} \int_1^{t^{-1/2}} \rho^{-1}\, d\rho = \frac{1}{2} e^{-1/2} \log(1/t)$, since $t\rho^2 \le 1$ on that range. For the upper bound let $0 < \beta < c$. Clause (1) and $(1 + |\xi|)^{-\beta} \le |\xi|^{-\beta}$ give, with $\xi = y/\sqrt{t}$,

$$\int g_t\, d\mu \le (2\pi)^{-d} C_\beta \int_{\mathbb{R}^d} |\xi|^{-\beta} e^{-t|\xi|^2/2}\, d\xi = (2\pi)^{-d} C_\beta\, t^{-(d-\beta)/2} \int_{\mathbb{R}^d} |y|^{-\beta} e^{-|y|^2/2}\, dy,$$

and the last integral is finite because $\beta < c \le d$. Now take logarithms and divide by $\log(1/t)$. By the lower bound, the quotient $\log \int g_t\, d\mu / \log(1/t)$ is at least $(d - c)/2 - \varepsilon$ for every $\varepsilon > 0$ once t is small enough. At $c = d$ this uses $\log\log(1/t)/\log(1/t) \to 0$. By the upper bound, the quotient is at most $(d - \beta)/2 + \varepsilon$ for every $\beta < c$ and $\varepsilon > 0$ once t is small enough. So it tends to $(d - c)/2$.

Clause (5). If $c > d$, clause (2) at $N = 0$ gives μ a continuous density that tends to 0 at infinity, hence a bounded one. If $c \le d$, clause (4) excludes every bound $\mu(B) \le M|B|$ on the Borel subsets B of a ball $\{|x| < \delta\}$, and with it every representative of a density bounded near the origin. □

Proof of Corollary 3.14. Let ν_d be the datum's Lévy measure and put $G(\xi) := \int_{\mathbb{R}^d} (1 - \cos\langle \xi, x\rangle)\, \nu_d(dx)$, so that $F_d(\xi) = \frac{1}{2} a|\xi|^2 + G(\xi)$ by (3.3). Then $G \ge 0$, and G is the exponent of the datum $(0, k_d)$, which has the same polar profile and the same profile at the origin c. As in the proof of Lemma A.2, (3.2) gives

$$G(\xi) = \frac{1}{|S^{d-1}|} \int_{S^{d-1}} \varphi\big(\langle \xi, \eta\rangle\big)\, \sigma_{d-1}(d\eta), \qquad \varphi(s) := \int_0^\infty \big(1 - \cos(rs)\big)\, k_d(r)\, \frac{dr}{r}.$$

Clause (1). Fix $\tau \in (0, 1]$ and put $I(\tau) := \int_0^\tau k_d(r)\, r\, dr$ and $T(\tau) := \int_\tau^\infty k_d(r)\, dr/r$. Both are finite: $I(\tau)$ is at most the first integral of Definition 3.10, and $T(\tau) \le k_d(\tau) \log(1/\tau) + \int_1^\infty k_d(r)\, dr/r$ because k_d is nonincreasing. Since $1 - \cos(rs) \le r^2 s^2/2$ on $(0, \tau)$ and $1 - \cos \le 2$ on $[\tau, \infty)$, $\varphi(s) \le \frac{1}{2} s^2\, I(\tau) + 2\, T(\tau)$. On the sphere $|\langle \xi, \eta\rangle| \le |\xi|$, and the normalized surface measure has mass 1. So

$$0 \le G(\xi) \le \tfrac{1}{2} |\xi|^2\, I(\tau) + 2\, T(\tau), \qquad \xi \in \mathbb{R}^d.$$

Let $\varepsilon > 0$. The integrand of $I(1)$ is integrable on $(0,1)$, so $I(\tau) \to 0$ as $\tau \downarrow 0$, and there is a τ with $I(\tau) \le \varepsilon$. For that τ, $G(\xi)/|\xi|^2 \le \frac{1}{2}\varepsilon + 2T(\tau)/|\xi|^2 \le \varepsilon$ once $|\xi|^2 \ge 4T(\tau)/\varepsilon$. Hence $G(\xi)/|\xi|^2 \to 0$, and $F_d(\xi)/|\xi|^2 = \frac{1}{2}a + G(\xi)/|\xi|^2 \to \frac{1}{2}a$.

Clause (2). The quotient is $G(\xi)/\log|\xi|$, and it is nonnegative for $|\xi| > 1$. *From below.* Let β be finite with $0 < \beta < c$. By Theorem 3.13(1) for the datum $(0, k_d)$ there is C_β with $e^{-G(\xi)} \le C_\beta(1+|\xi|)^{-\beta}$ for every ξ, and $C_\beta \ge 1$ by the bound at $\xi = 0$. That clause states the bound for $|\widehat{\mu}|$, μ being a law of the datum $(0, k_d)$. Its proof bounds e^{-G}, which is $|\widehat{\mu}|$ by the definition of the law, so the bound holds whether or not a law is given. Taking logarithms, $G(\xi) \ge \beta \log|\xi| - \log C_\beta$ for $|\xi| \ge 1$. *From above.* Let $c < \infty$. By Lemma A.2(1) for the datum $(0, k_d)$, $G(\xi) \le c\log|\xi| + \frac{13}{4}c + 2C_k$ for $|\xi| \ge 1$, with $C_k = \int_1^\infty k_d(r)\,dr/r$. *The limit.* Divide by $\log|\xi|$, which tends to infinity. For every $\varepsilon > 0$ the quotient is at least $\beta - \varepsilon$ once $|\xi|$ is large enough, for every finite β with $0 < \beta < c$, and at most $c + \varepsilon$ once $|\xi|$ is large enough when $c < \infty$. If $c = \infty$ the first bound makes the quotient tend to $+\infty$. If $0 < c < \infty$ the two bounds give the limit c. If $c = 0$ the quotient lies between 0 and ε, and tends to 0. For the last sentence of the clause, $a = 0$ gives $F_d = G$, and $|\widehat{\mu}| = e^{-F_d}$ by the definition of the law.

Clause (3). Since $d, d' \ge 1$ there are unit vectors $e \in \mathbb{R}^d$ and $e' \in \mathbb{R}^{d'}$, and $F_d(re) = F(r) = F'_{d'}(re')$ for every $r > 0$. Clause (1) for the two data, read along $\xi = re$ and $\xi' = re'$, gives $F(r)/r^2 \to \frac{1}{2}a$ and $F(r)/r^2 \to \frac{1}{2}a'$ as $r \to \infty$. So $a' = a$. Clause (2), read along the same rays, then gives $\big(F(r) - \frac{1}{2}ar^2\big)/\log r \to k_d(0+)$ and $\big(F(r) - \frac{1}{2}ar^2\big)/\log r \to k'_{d'}(0+)$ as $r \to \infty$. A limit in $[0,\infty]$ is unique, so $k'_{d'}(0+) = k_d(0+)$. □

Proof of Proposition A.3. Write $|E|$ for the Lebesgue measure of E, $B(z,r)$ and $\bar{B}(z,r)$ for the open and the closed ball, and ν_d for the datum's Lévy measure. As k_d is nonincreasing with right limit c at the origin, $k_d \le c$ on $(0,\infty)$ and $k_d(r) \uparrow c$ as $r \downarrow 0$, so ε_0 exists. Every integral of ν_d below is computed by (3.2), applied to a nonnegative Borel function vanishing at the origin.

The decomposition at scale ε. Fix $\varepsilon \in (0, \varepsilon_0]$ and let ν_1 and $\nu_\ge$ be the restrictions of ν_d to $\{0 < |x| < \tau\varepsilon\}$ and to $\{|x| \ge \tau\varepsilon\}$. Each has a radial density, so each is symmetric, and $\psi_1(\xi) := \int(1-\cos\langle\xi,x\rangle)\,\nu_1(dx)$ and $\psi_\ge(\xi) := \int(1-\cos\langle\xi,x\rangle)\,\nu_\ge(dx)$ are finite with $F_d = \psi_1 + \psi_\ge$, F_d being (3.3) at $a = 0$. The mass of $\nu_\ge$ is finite: as $\tau\varepsilon \le 1$ and $k_d \le c$,

$$m := \nu_\ge(\mathbb{R}^d) = \int_{\tau\varepsilon}^\infty k_d(r)\,\frac{dr}{r} \le c\log\frac{1}{\tau\varepsilon} + C_k, \qquad \text{so} \qquad e^{-m} \ge \tau^c\varepsilon^c e^{-C_k}.$$

For a finite symmetric measure λ of mass m_λ put $\pi_\lambda := e^{-m_\lambda}\sum_{n\ge0}\lambda^{*n}/n!$, with $\lambda^{*0} = \delta_0$; it is a probability measure, and, summing the series of transforms by dominated convergence ($|\widehat{\lambda}^n| \le m_\lambda^n$), its transform is $\exp(\widehat{\lambda}(\xi) - m_\lambda) = \exp\big(-\int(1-\cos\langle\xi,x\rangle)\,\lambda(dx)\big)$, the sine part of $\widehat{\lambda}$ vanishing by symmetry. Put $\pi := \pi_{\nu_\ge}$, of transform $e^{-\psi_\ge}$. The profile $k_d\mathbf{1}_{(0,\tau\varepsilon)}$ is nonincreasing and satisfies the two integrability conditions of Definition 3.10, so $(0, k_d\mathbf{1}_{(0,\tau\varepsilon)})$ is an isotropic origin datum, with Lévy measure ν_1 and exponent ψ_1; let μ_1 be its law (C3(iii)), so $\widehat{\mu}_1 = e^{-\psi_1}$. Then $\mu_1 * \pi$ has transform $e^{-F_d} = \widehat{\mu}$, and by the uniqueness of the Fourier transform

$$\mu = \mu_1 * \pi.$$

Keeping the term $n = 2$ of the series, $\pi \ge \frac{1}{2}e^{-m}\,\nu_\ge * \nu_\ge$, as measures.

The twofold convolution fills a ball. Let $S := \{\varepsilon/2 \le |x| < \varepsilon\}$, which lies in $\{|x| \ge \tau\varepsilon\}$ because $\tau \le 1/2$. On S the density $k_d(|x|)\,(|S^{d-1}||x|^d)^{-1}$ of $\nu_\ge$ is at least $a := c\,(2|S^{d-1}|\varepsilon^d)^{-1}$, because $k_d(|x|) \ge k_d(\varepsilon_0) \ge c/2$ and $|x|^d < \varepsilon^d$; so $\nu_\ge(A) \ge a\,|A \cap S|$ for every Borel A. Let $|z| \le 3\varepsilon/2$. As $d \ge 2$ there is a unit vector w orthogonal to z; put $y_0 := z/2 + sw$ with $s := (9\varepsilon^2/16 - |z|^2/4)^{1/2}$, so that $|y_0| = |z - y_0| = 3\varepsilon/4$. Every $y \in B(y_0, \varepsilon/8)$ has $|y|$ and $|z-y|$ in $(5\varepsilon/8, 7\varepsilon/8)$, inside S, so $|\{y \in S : z - y \in S\}| \ge |B(0,\varepsilon/8)|$. For a Borel $G \subseteq \bar{B}(0, 3\varepsilon/2)$,

Tonelli's theorem and the substitution $x = z - y$ give

$$(\nu_{\geq} * \nu_{\geq})(G) \geq a^2 \iint \mathbf{1}_S(x)\, \mathbf{1}_S(y)\, \mathbf{1}_G(x+y)\, dx\, dy$$
$$= a^2 \int_G \big|\{y \in S : z - y \in S\}\big|\, dz \geq a^2 \big|B(0, \varepsilon/8)\big|\, |G| = \kappa_0\, \varepsilon^{-d} |G|.$$

Hence $\pi(G) \geq \frac{1}{2} e^{-m} \kappa_0 \varepsilon^{-d} |G|$ for every Borel $G \subseteq \bar{B}(0, 3\varepsilon/2)$.

The small jumps move little. By $1 - \cos u \leq u^2/2$ and (3.2),

$$\psi_1(\xi) \leq \frac{|\xi|^2}{2} \int |x|^2\, \nu_1(dx) = \frac{|\xi|^2}{2} \int_0^{\tau\varepsilon} k_d(r)\, r\, dr \leq \frac{c\, \tau^2 \varepsilon^2}{4}\, |\xi|^2.$$

Let γ be the centred Gaussian law on $\mathbb{R}^d$ with covariance $\varepsilon^{-2} I$, whose transform at w is $e^{-|w|^2/2\varepsilon^2}$ and whose second moment is $\int |\xi|^2\, \gamma(d\xi) = d\, \varepsilon^{-2}$. By Fubini's theorem, $\widehat{\mu}_1$ being real, and by $1 - e^{-\psi} \leq \psi$,

$$\int \big(1 - e^{-|w|^2/2\varepsilon^2}\big)\, \mu_1(dw) = \int \big(1 - \widehat{\mu}_1(\xi)\big)\, \gamma(d\xi) \leq \int \psi_1\, d\gamma \leq \frac{c\, \tau^2 \varepsilon^2}{4} \cdot \frac{d}{\varepsilon^2} = \frac{c\, \tau^2 d}{4}.$$

The integrand on the left is at least $1 - e^{-1/2} \geq 1/3$ where $|w| \geq \varepsilon$, and $c\tau^2 d/4 \leq 3/(16d) \leq 3/32$ by the choice of τ and $d \geq 2$, so

$$\mu_1\big(\{|w| \geq \varepsilon\}\big) \leq \tfrac{9}{32} \leq \tfrac{1}{2}.$$

One scale. Let $E \subseteq \bar{B}(0, \varepsilon/2)$ be Borel. For $|w| < \varepsilon$ the translate $E - w$ lies in $\bar{B}(0, 3\varepsilon/2)$ and has measure $|E|$. So, the integrand being nonnegative,

$$\mu(E) = \int \pi(E - w)\, \mu_1(dw) \geq \int_{\{|w| < \varepsilon\}} \pi(E - w)\, \mu_1(dw) \geq \tfrac{1}{2} e^{-m} \kappa_0 \varepsilon^{-d} |E| \cdot \tfrac{1}{2}.$$

With $e^{-m} \geq \tau^c \varepsilon^c e^{-C_k}$ this is

$$\mu(E) \geq \kappa_1\, \varepsilon^{c-d}\, |E|, \qquad \kappa_1 := \tfrac{1}{4} \kappa_0 \tau^c e^{-C_k}, \qquad E \subseteq \bar{B}(0, \varepsilon/2),\ 0 < \varepsilon \leq \varepsilon_0.$$

All scales. For $n \geq 0$ put $\varepsilon_n := 2^{-n} \varepsilon_0$ and $A_n := \{\varepsilon_n/4 < |x| \leq \varepsilon_n/2\}$; these sets are disjoint and cover $B(0, \delta) \setminus \{0\}$. For $x \in A_n$, $2|x| \leq \varepsilon_n < 4|x|$, so $\varepsilon_n^{c-d} \geq \theta |x|^{c-d}$ with $\theta := \min\{2^{c-d}, 4^{c-d}\}$ (the first bound when $c \geq d$, the second when $c < d$). For a Borel $E \subseteq B(0, \delta)$, the origin being a Lebesgue-null set, the previous step at $\varepsilon = \varepsilon_n$ gives

$$\mu(E) \geq \sum_{n \geq 0} \mu(E \cap A_n) \geq \kappa_1 \sum_{n \geq 0} \varepsilon_n^{c-d} |E \cap A_n| \geq \kappa_1 \theta \sum_{n \geq 0} \int_{E \cap A_n} |x|^{c-d}\, dx = c_1 \int_E |x|^{c-d}\, dx,$$

which is (A.1).

The consequences. Write $\mu = \mu_{\mathrm{ac}} + \mu_{\mathrm{s}}$ for the Lebesgue decomposition and let N be a Borel set with $\mu_{\mathrm{s}}(N) = 0$ and $|\mathbb{R}^d \setminus N| = 0$. For a Borel $E \subseteq B(0, \delta)$, $\int_E q\, dx \geq \int_{E \cap N} q\, dx = \mu_{\mathrm{ac}}(E \cap N) = \mu(E \cap N) \geq c_1 \int_{E \cap N} |x|^{c-d}\, dx = c_1 \int_E |x|^{c-d}\, dx$. A function whose integral over every Borel subset of $B(0, \delta)$ is at least that of the measurable $c_1 |x|^{c-d}$ is at least $c_1 |x|^{c-d}$ almost everywhere on $B(0, \delta)$, both functions being nonnegative and measurable and Lebesgue measure being σ-finite: on a set $E \subseteq B(0, \delta)$ of finite measure where $q + 1/n \leq c_1 |x|^{c-d} \leq n$, the inequality gives $\int_E q\, dx + |E|/n \leq \int_E q\, dx$, the integral being finite as $q \leq n$ on E, so $|E| = 0$, and such sets exhaust the set where $q < c_1 |x|^{c-d}$. That is clause (1); for a density q of μ the same argument runs with $\int_E q\, dx = \mu(E)$, and $c_1 |x|^{c-d} > 0$ off the origin, which is null. As $x \to 0$, $|x| \to 0$ through positive values, and $|x|^{c-d} \to \infty$ when $c < d$. Clause (2) is (A.1) at $E = \{|x| < r\}$, in polar coordinates: $\int_{\{|x| < r\}} |x|^{c-d} dx = |S^{d-1}| \int_0^r \rho^{c-1} d\rho = |S^{d-1}|\, r^c / c$. $\square$

The last paragraph of the next proof shows that the uniform laws of C5 form a Borel set. Its argument is a standard one in the space of the compact convex sets with the Hausdorff metric, by Blaschke's selection theorem and the support function, and the facts are cited there from [8].

Proof of Proposition 3.15. The argument reads four properties of a law μ on $\mathbb{R}^d$: it is rotation invariant, it is self-decomposable, its transform is real, and $\widehat{\mu}(\xi) \to 0$ as $|\xi| \to \infty$. The first two paragraphs check them for the two clauses. The rest derives the density from them.

Clause (1): the four properties. Put $\mu := \mu_{0,t}$. By Theorem 3.6, $\widehat{\mu}(\xi) = e^{-F(\chi(t)|\xi|)}$ for a gauge χ and an $F \in \mathcal{A}_d$ that does not vanish identically on $[0,\infty)$, and $s := \chi(t) > 0$. By the definition of $\mathcal{A}_d$ there is a self-decomposable law μ' on $\mathbb{R}^d$ with $\widehat{\mu'}(\xi) = e^{-F(|\xi|)}$, so $\widehat{\mu}(\xi) = \widehat{\mu'}(s\xi)$. So $\widehat{\mu}$ is real and radial: for $Q \in \mathrm{O}(d)$, $\widehat{Q_*\mu}(\xi) = \widehat{\mu}(Q^{\mathsf{T}}\xi) = \widehat{\mu}(\xi)$, and a law is determined by its transform, so μ is rotation invariant. For $\lambda \in (0,1)$ let ρ_λ be a residual of μ', $\widehat{\mu'}(\xi) = \widehat{\mu'}(\lambda\xi)\,\widehat{\rho}_\lambda(\xi)$. Then

$$\widehat{\mu}(\xi) = \widehat{\mu'}(s\xi) = \widehat{\mu'}(\lambda s\xi)\,\widehat{\rho}_\lambda(s\xi) = \widehat{\mu}(\lambda\xi)\,\widehat{(sI)_*\rho_\lambda}(\xi),$$

which is (2.2) with residual $(sI)_*\rho_\lambda$. It remains to see that $e^{-F(r)} \to 0$ as $r \to \infty$. F is admissible, of the form (2.1) with Gaussian coefficient a_{V} and folded profile k. The pair $(2a_{\mathrm{V}}, k)$ is an isotropic origin datum in dimension 1: its two integrability conditions are those of (2.1), and by (3.2) at $d = 1$, where $|S^0| = 2$, its exponent at ω is $a_{\mathrm{V}}\omega^2 + \int_{\mathbb{R}} (1 - \cos\omega x)\, k(|x|)\, (2|x|)^{-1} dx = F(|\omega|)$. Its law is the image of μ' under $x \mapsto \langle e_1, x\rangle$, whose transform is $\widehat{\mu'}(\omega e_1) = e^{-F(|\omega|)}$, as in Lemma 3.4(1). The datum is not trivial: a trivial datum has exponent 0, and F does not vanish identically on $[0,\infty)$. So the last two sentences of the next paragraph, at $d = 1$, give $e^{-F(r)} \to 0$, and $\widehat{\mu}(\xi) \to 0$ as $|\xi| \to \infty$.

Clause (2): the four properties. Let μ be the datum's law, so that $\widehat{\mu} = e^{-F_d}$ is real, positive and radial by (3.3). For $Q \in \mathrm{O}(d)$, $\widehat{Q_*\mu}(\xi) = \widehat{\mu}(Q^{\mathsf{T}}\xi) = \widehat{\mu}(\xi)$, and a law is determined by its transform, so μ is rotation invariant. Put $F(r) := F_d(re_1)$. Then $\xi \mapsto F(|\xi|)$ is the exponent F_d of the datum, so by Proposition 3.11(1), in its converse direction, it is the exponent of a self-decomposable law, which is μ. Finally $F_d(\xi) \geq \frac{1}{2}a|\xi|^2$, which tends to infinity if $a > 0$. If $a = 0$, then k_d is not almost everywhere 0, so $c = k_d(0+) > 0$, k_d being at most c on $(0,\infty)$, and Theorem 3.13(1) with a finite $\beta \in (0,c)$ gives $\widehat{\mu}(\xi) \leq C_\beta(1 + |\xi|)^{-\beta}$.

No atom at the origin. Let μ have the four properties. For $\varepsilon > 0$, multiply the identity of Lemma S.8(1) at $t = \varepsilon$ by $(2\pi\varepsilon)^{d/2}$ and substitute $\xi = y/\sqrt{\varepsilon}$:

$$\int_{\mathbb{R}^d} e^{-|x|^2/2\varepsilon}\, \mu(dx) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} \widehat{\mu}\big(y/\sqrt{\varepsilon}\big)\, e^{-|y|^2/2}\, dy.$$

Let $\varepsilon \downarrow 0$. On the left the integrand tends to the indicator of the origin, and on the right $\widehat{\mu}(y/\sqrt{\varepsilon}) \to 0$ for every $y \neq 0$. The integrand on the left is bounded by 1, which is integrable against μ, and the one on the right by $e^{-|y|^2/2}$, which is integrable against dy, as $|\widehat{\mu}| \leq 1$; so by dominated convergence $\mu(\{0\}) = 0$.

The mixture. The law μ is symmetric, by rotation invariance at $Q = -I$, and self-decomposable. By C5 it is symmetric unimodal: there is a probability measure Λ on the uniform laws u_K of the compact convex symmetric sets $K \subset \mathbb{R}^d$ with

$$\mu(A) = \int u_K(A)\, \Lambda(du_K) \qquad \text{for every Borel } A \subseteq \mathbb{R}^d.$$

Here u_K is uniform for the Lebesgue measure Vol_V of the linear span V of K, whose dimension k may be anything from 0 to d, and K is nonempty. Λ is a probability measure on the measurable space of the probability measures on $\mathbb{R}^d$, whose σ-algebra is generated by the evaluations $\nu \mapsto \nu(A)$, and almost every measure under it is a u_K; nothing below asks the set W_d of the u_K to

be measurable. The source puts its mixing measure on W_d itself, with the Borel sets of weak convergence, which on the probability measures of $\mathbb{R}^d$ are the sets of that σ-algebra. Reading it as almost every measure being a u_K uses that W_d is such a set. It is one, being a countable union of weakly compact sets, as the paragraph on W_d at the end of the proof shows. The formal statement puts Λ on all measures on $\mathbb{R}^d$, which loses nothing: the probability measures form a measurable set, and almost every measure under Λ, being a u_K, lies in it. A set with $k < d$ carries no density in $\mathbb{R}^d$, and the next two paragraphs deal with that.

The radial part of one uniform law. Let ω_d be the volume of the unit ball and $\varphi(x) := \omega_d|x|^d$ the volume of the ball of radius $|x|$. For $v \geq 0$ put $r_v := (v/\omega_d)^{1/d}$ and $B(v) := \varphi^{-1}((0,v]) = \{0 < |x| \leq r_v\}$, the punctured closed ball of volume v, and put $G_K(v) := u_K(B(v))$. We show that G_K is concave on $[0,\infty)$. If $k = 0$, then $K = \{0\}$, $u_K = \delta_0$ and $G_K \equiv 0$, as $B(v)$ misses the origin. Let $k \geq 1$, let S_V be the unit sphere of V and σ_V its surface measure, so that $\mathrm{Vol}_V(A) = \int_{S_V} \sigma_V(d\theta) \int_0^\infty \mathbf{1}_A(s\theta)\, s^{k-1}\, ds$ for every Borel $A \subseteq V \setminus \{0\}$ (polar coordinates in V). Fix $\theta \in S_V$. The set $J_\theta := \{s > 0 : s\theta \in K\}$ contains $(0,s]$ with each of its points s: K contains $0 = \frac{1}{2}y + \frac{1}{2}(-y)$ for any $y \in K$, so $s\theta \in K$ and $0 < s' < s$ give $s'\theta = \frac{s'}{s}\, s\theta + \big(1 - \frac{s'}{s}\big)0 \in K$ by convexity. It is bounded, K being compact. With $\rho_\theta := \sup J_\theta$ ($\rho_\theta := 0$ if J_θ is empty), $(0,\rho_\theta) \subseteq J_\theta \subseteq (0,\rho_\theta]$, and for $s > 0$, $s\theta \in B(v)$ exactly when $s \leq r_v$. So the ray in the direction θ contributes

$$\int_0^\infty \mathbf{1}_{K\cap B(v)}(s\theta)\, s^{k-1}\, ds = \int_0^{\min\{\rho_\theta,\, r_v\}} s^{k-1}\, ds = \frac{1}{k}\min\Big\{\rho_\theta^k,\ \Big(\frac{v}{\omega_d}\Big)^{k/d}\Big\}.$$

As $k \leq d$, $v \mapsto (v/\omega_d)^{k/d}$ is concave on $[0,\infty)$, and so is its minimum with a constant. Integrating over θ against σ_V and dividing by $\mathrm{Vol}_V(K) \in (0,\infty)$, $G_K(v) = \mathrm{Vol}_V(K \cap B(v))/\mathrm{Vol}_V(K)$ is concave in v. The volume is positive because K spans V, so it contains a basis $v_1, \ldots, v_k$ of V and, being convex and symmetric, the convex hull of the $\pm v_i$, which has positive volume in V; it is finite because K is compact. (Concavity survives multiplication by any nonnegative constant, so the formal proof does not need this; that u_K is a probability measure is checked separately.) In words: each ray of K contributes the minimum of a constant and a concave power of the volume, the power being $k/d \leq 1$ because K has dimension at most d.

The density. Put $G(v) := \mu(B(v)) = \int G_K(v)\, \Lambda(du_K)$. A mixture of concave functions is concave, the inequality $aG_K(v) + bG_K(w) \leq G_K(av + bw)$ integrating against Λ, so G is concave on $[0,\infty)$. Let m be the image of μ under φ, a finite measure on $\mathbb{R}$ with $m((0,v]) = G(v)$ and $m((-\infty,0]) = \mu(\{0\}) = 0$. G is nondecreasing; let $D(v)$ be its right derivative at $v > 0$. It is finite, nonnegative and nonincreasing, and G is continuous on $(0,\infty)$, so by the fundamental theorem of calculus for right derivatives $m((a,b]) = G(b) - G(a) = \int_a^b D(w)\, dw$ for $0 < a \leq b$. Letting $a \downarrow 0$ by continuity from below, and as m does not charge $(-\infty,0]$, the measures m and $D(w)\, dw$ on $(0,\infty)$ agree on every interval $(a,b]$, so $m = D(w)\, dw$ on $(0,\infty)$. Put $q(r) := D(\omega_d r^d)$, a nonincreasing function on $(0,\infty)$ with values in $[0,\infty)$, so that $q(|x|) = D(\varphi(x))$. The image of Lebesgue measure on $\mathbb{R}^d$ under φ is Lebesgue measure on $(0,\infty)$, because $\varphi^{-1}((-\infty,v])$ is the closed ball of volume v. So the image of $q(|x|)\, dx$ under φ is $D(w)\, dw = m$, the image of μ, and as $|x| = (\varphi(x)/\omega_d)^{1/d}$ the two measures have the same image under $x \mapsto |x|$. Both are finite and rotation invariant ($q(|x|)\, dx$ because Lebesgue measure is and the density is radial), and both vanish at the origin. A measure with these properties is determined by its image under $x \mapsto |x|$. For let m be one of them and $\varepsilon > 0$, and let m_ε be its restriction to $U_\varepsilon := \{|x| > \varepsilon\}$, a finite measure invariant under every $Q \in \mathrm{O}(d)$, as U_ε is. Then $\widehat{m_\varepsilon}(\xi) = \widehat{m_\varepsilon}(|\xi|\theta)$ for every unit vector θ, a reflection carrying ξ to $|\xi|\theta$. Averaging over θ against $\sigma_{d-1}/|S^{d-1}|$ and exchanging the integrals gives

$$\widehat{m_\varepsilon}(\xi) = \frac{1}{|S^{d-1}|}\int_{U_\varepsilon} \Phi(|\xi|,|x|)\, m(dx), \qquad \Phi(s,r) := \int_{S^{d-1}} e^{isr\langle e_1,\theta\rangle}\, \sigma_{d-1}(d\theta),$$

since $\int e^{is\langle x,\theta\rangle}\sigma_{d-1}(d\theta)$ depends on x only through $|x|$, by the invariance of σ_{d-1}. The right side depends on m only through its image under $x \mapsto |x|$. So the two measures have restrictions to

U_ε with the same transform, hence equal; letting $\varepsilon \downarrow 0$ along $1/(n+1)$, they agree off the origin, and as neither charges it, they are equal. Hence $\mu = q(|x|)\,dx$, and $p(x) := q(|x|)$ is a density of μ.

The set W_d is a Borel set. This paragraph proves the reading of the source made in the paragraph on the mixture, and no other step uses it. Let $\bar{B}$ be the closed unit ball of $\mathbb{R}^d$, and give the nonempty compact convex sets $K = -K$ of $\mathbb{R}^d$ the Hausdorff metric. For integers $k \geq 1$ and $n \geq 1$ let $\mathcal{K}_{k,n}$ be the set of those K whose span V has dimension k and which satisfy

$$\tfrac{1}{n}\bar{B} \cap V \subseteq K \subseteq n\bar{B}.$$

Every K of dimension $k \geq 1$ is in $\mathcal{K}_{k,n}$ for some n. It is bounded, and it contains the convex hull of the points $\pm v_i$ of the paragraph on one uniform law, which contains a ball of V round the origin.

Each $\mathcal{K}_{k,n}$ is compact. Let $K_j \in \mathcal{K}_{k,n}$, with spans V_j. By Blaschke's selection theorem ([8], Thm. 1.8.7, p. 63, where a convex body is any nonempty compact convex set, p. 8), the nonempty compact convex subsets of a ball form a compact space in the Hausdorff metric. So a subsequence of (K_j) converges to a nonempty compact convex set K, and $K = -K$. Let P_j be the orthogonal projection onto V_j. Along a further subsequence the matrices P_j converge, and their limit P is the orthogonal projection onto a subspace V' of dimension k, since $P_j^2 = P_j = P_j^{\mathsf{T}}$ and $\operatorname{tr} P_j = k$ pass to the limit. A point $x \in K$ is a limit of points $x_j \in K_j$, and $x_j = P_j x_j \to Px$, so $x \in V'$. Hence $K \subseteq V' \cap n\bar{B}$. For $y \in V'$ with $|y| \leq 1/n$ the points $P_j y$ lie in $\frac{1}{n}\bar{B} \cap V_j \subseteq K_j$ and converge to y, so $y \in K$. Hence the span of K is V', and $K \in \mathcal{K}_{k,n}$.

The map $K \mapsto u_K$ is continuous on $\mathcal{K}_{k,n}$, for the weak convergence of measures. Let $K_j \to K$ in $\mathcal{K}_{k,n}$, with spans V_j and V. Choose a basis $y_1, \ldots, y_k$ of V in K and points $y_{i,j} \in K_j$ with $y_{i,j} \to y_i$ as $j \to \infty$. For large j the points $y_{1,j}, \ldots, y_{k,j}$ are linearly independent, their Gram determinant converging to that of the y_i, which is positive. So they are a basis of V_j. Let T_j be the linear map from V onto V_j with $T_j y_i = y_{i,j}$, and let ε_j be the supremum of $|T_j z - z|$ over the $z \in V$ with $|z| \leq 1$. Then $\varepsilon_j \to 0$. A linear bijection carries the uniform law of a set to the uniform law of its image, so u_{K_j} is the image of $u_{K_j'}$ under T_j, where $K_j' := T_j^{-1} K_j \subseteq V$. For $w \in K_j'$, $|T_j w - w| \leq \varepsilon_j |w|$ and $|T_j w| \leq n$, so $|w| \leq n/(1-\varepsilon_j)$, and the Hausdorff distance of K_j' and K_j is at most $\varepsilon_j n/(1-\varepsilon_j)$. So $K_j' \to K$. Let η_j be the Hausdorff distance of K_j' and K, and put $r := 1/n$ and $B_V := \bar{B} \cap V$, so that $rB_V \subseteq K$. Then

$$K_j' \subseteq K + \eta_j B_V \subseteq \big(1 + \eta_j/r\big)K,$$

K being convex. In the other direction, write $h_A(e) := \sup_{x \in A} \langle e, x \rangle$ for the support function of a compact convex set $A \subseteq V$ at a unit vector e of V. Then $h_K \leq h_{K_j'} + \eta_j$, the Hausdorff distance of two compact convex sets being the supremum over the unit vectors of the absolute difference of their support functions ([8], Lemma 1.8.14, p. 66), and $h_K \geq r$, so $h_{K_j'} \geq (1 - \eta_j/r)\, h_K$. A compact convex set contains every compact convex set with a smaller support function, and $h_{\lambda A} = \lambda h_A$ for $\lambda \geq 0$ ([8], p. 45), so $(1 - \eta_j/r)K \subseteq K_j'$ once $\eta_j < r$. The two inclusions hold for K in place of K_j' as well. So K_j' and K differ by a set of volume at most $\big((1+\eta_j/r)^k - (1-\eta_j/r)^k\big)\mathrm{Vol}_V(K)$, which tends to 0, and $u_{K_j'}(A) \to u_K(A)$ uniformly in the Borel set A. For a bounded continuous function φ on $\mathbb{R}^d$,

$$\int \varphi \, du_{K_j} = \int \varphi(T_j w)\, u_{K_j'}(dw),$$

and $\varphi(T_j w) \to \varphi(w)$ uniformly for w in the bounded set $2K$, which contains K_j' for large j. So $\int \varphi \, du_{K_j} \to \int \varphi \, du_K$.

So each set $\{u_K : K \in \mathcal{K}_{k,n}\}$ is the continuous image of a compact set. It is weakly compact, hence weakly closed. With $\{\delta_0\}$, the case $k = 0$, these countably many sets make up W_d. So W_d is a Borel set for weak convergence. □

Proof of Proposition A.4. Write $B_r := \{|x| < r\}$ for the open ball and $|E|$ for the Lebesgue measure of E, so that $|B_r| = V_d\, r^d$.

Clause (1). Fix β with $0 < \beta < c$, and let C'_β be a constant with $\int g_t\, d\mu \le C'_\beta\, t^{-(d-\beta)/2}$ for all $t > 0$, g_t being the heat kernel of Lemma S.8. There is one by Theorem 3.13(4), since $c \le d$. Let $r > 0$. For $|x| < r$,

$$g_{r^2}(x) = (2\pi r^2)^{-d/2} e^{-|x|^2/2r^2} \ge (2\pi)^{-d/2} e^{-1/2}\, r^{-d}.$$

As $g_{r^2} \ge 0$,

$$\int_{\mathbb{R}^d} g_{r^2}\, d\mu \ge \int_{B_r} g_{r^2}\, d\mu \ge (2\pi)^{-d/2} e^{-1/2}\, r^{-d}\, \mu(B_r).$$

The left side is at most $C'_\beta\, (r^2)^{-(d-\beta)/2} = C'_\beta\, r^{\beta-d}$. Hence $\mu(B_r) \le (2\pi)^{d/2} e^{1/2}\, C'_\beta\, r^\beta$, which is the bound with the stated A_β.

Clause (2). As $c = \sup_{r>0} k_d(r) > 0$, there is $r_0 > 0$ with $k_d(r_0) > 0$, and $k_d \ge k_d(r_0)$ on $(0, r_0]$ because k_d is nonincreasing. So k_d is not almost everywhere 0, the datum is not trivial, and Proposition 3.15(2) gives a density $p(x) = q(|x|)$ with q nonincreasing. Now let $q : (0,\infty) \to [0,\infty)$ be any nonincreasing function such that $x \mapsto q(|x|)$ is a density of μ, and let $r > 0$. For $0 < |x| < r$, $q(|x|) \ge q(r)$. The origin is a Lebesgue-null set, so

$$\mu(B_r) = \int_{B_r} q(|x|)\, dx \ge q(r)\, |B_r| = q(r)\, V_d\, r^d.$$

With clause (1), $q(r) \le A_\beta V_d^{-1}\, r^{\beta-d}$.

Clause (3), the lower bound at every radius. Let $d \ge 2$. Since $c \le d$, c is finite, and Proposition A.3 applies. By its clause (1) the density $x \mapsto q(|x|)$ of μ is at least $c_1|x|^{c-d}$ at almost every $x \in B_\delta$. Suppose that $q(r_0) < c_1 r_0^{c-d}$ for some r_0 with $0 < r_0 < \delta$. The function $r \mapsto c_1 r^{c-d}$ is continuous at r_0, so there is r_1 with $r_0 < r_1 \le \delta$ and $q(r_0) < c_1 r^{c-d}$ for $r_0 < r < r_1$. For such r, $q(r) \le q(r_0) < c_1 r^{c-d}$, because q is nonincreasing. So $q(|x|) < c_1|x|^{c-d}$ on the shell $\{r_0 < |x| < r_1\}$. The shell is a nonempty open subset of B_δ, hence of positive Lebesgue measure, and this contradicts the bound at almost every point. So $q(r) \ge c_1 r^{c-d}$ for every $0 < r < \delta$.

Clause (3), the limits. Both follow from one remark. Let L be real and let f be a function on $(0,\infty)$ such that, for small $r > 0$, $f(r) \ge C r^{-L}$ with a $C > 0$ and, for every $\varepsilon > 0$, $f(r) \le C_\varepsilon r^{-(L+\varepsilon)}$ with a $C_\varepsilon > 0$. Then $\log f(r)/\log(1/r) \to L$ as $r \downarrow 0$. Indeed, for $0 < r < 1$ small, $f(r) > 0$ and $\log(1/r) > 0$, and taking logarithms gives

$$L + \frac{\log C}{\log(1/r)} \le \frac{\log f(r)}{\log(1/r)} \le L + \varepsilon + \frac{\log C_\varepsilon}{\log(1/r)}.$$

As $\log(1/r) \to \infty$, the quotient is at least $L - \varepsilon$ and at most $L + 2\varepsilon$ once r is small enough, for every $\varepsilon > 0$.

For the density take $f = q$ and $L = d - c$. The lower bound is the one just proved, $q(r) \ge c_1 r^{-(d-c)}$ for $0 < r < \delta$. For the upper bound let $\varepsilon > 0$ and choose β with $0 < \beta < c$ and $\beta \ge c - \varepsilon$. By clause (2), for $0 < r < 1$, $q(r) \le A_\beta V_d^{-1}\, r^{\beta-d} \le A_\beta V_d^{-1}\, r^{-(d-c+\varepsilon)}$, as $\beta - d \ge -(d - c + \varepsilon)$ and $r < 1$. So $\log q(r)/\log(1/r) \to d - c$, and with $r = |x|$ this is the first limit, since $|x| \downarrow 0$ through positive values as $x \to 0$.

For the mass of a ball take $f(r) = \mu(B_r)$ and $L = -c$. By Proposition A.3(2), $\mu(B_r) \ge c_1 |S^{d-1}|\, c^{-1}\, r^c$ for $0 < r \le \delta$. By clause (1), with β as before, $\mu(B_r) \le A_\beta\, r^\beta \le A_\beta\, r^{c-\varepsilon}$ for $0 < r < 1$. So $\log \mu(B_r)/\log(1/r) \to -c$, and $\log(1/r) = -\log r$ gives the second limit. □

The constant of Proposition A.4 in closed form. This paragraph computes a constant from the two proofs above. It is not part of the formal development (§1.1). Let $0 < \beta < c \le d$, put $\beta_1 := (\beta + c)/2$, and let $r_\beta \in (0, 1]$ satisfy $k_d(r_\beta) \ge \beta_1$. There is such an r_β, because

$k_d(0+) = c > \beta_1$, and $k_d \geq \beta_1$ on $(0, r_\beta)$, because k_d is nonincreasing. So β_1 and r_β can be used in the proof of clause (1) of Theorem 3.13. That proof also asks for an $\varepsilon \in (0,1]$ for which the cap $K_\varepsilon = \{\eta \in S^{d-1} : |\langle e_1, \eta\rangle| \geq \varepsilon\}$ has at least the share β/β_1 of the sphere. Take

$$\varepsilon := \frac{c-\beta}{(c+\beta)\sqrt{d}}.$$

Then $1 - \beta/\beta_1 = \varepsilon\sqrt{d}$, so it is enough that the band $\{|\langle e_1, \eta\rangle| < \varepsilon\}$ has at most the share $\varepsilon\sqrt{d}$. For $d = 1$ the band is empty. For $d = 2$ its share is $(2/\pi)\arcsin\varepsilon \leq \varepsilon$. For $d \geq 3$ the first coordinate of a uniform point of the sphere has the density $\Gamma(\frac{d}{2})\left(\sqrt{\pi}\,\Gamma(\frac{d-1}{2})\right)^{-1}(1-u^2)^{(d-3)/2}$ on $(-1,1)$. It is at most $\sqrt{(d-1)/(2\pi)}$, since $\Gamma(x+\frac{1}{2}) \leq \Gamma(x)\sqrt{x}$ by the logarithmic convexity of Γ. So the share of the band is at most $2\varepsilon\sqrt{(d-1)/(2\pi)} \leq \varepsilon\sqrt{d}$. With these choices the number L of that proof satisfies $e^L = e^3/(\varepsilon r_\beta) = e^3\sqrt{d}\,(c+\beta)/((c-\beta)\,r_\beta)$, and clause (1) holds with $C_\beta = 2^\beta(1+e^{\beta L})$. The proof of clause (4) gives

$$C'_\beta = (2\pi)^{-d}\,C_\beta \int_{\mathbb{R}^d} |y|^{-\beta} e^{-|y|^2/2}\,dy, \qquad \int_{\mathbb{R}^d} |y|^{-\beta} e^{-|y|^2/2}\,dy = |S^{d-1}|\,2^{(d-\beta)/2-1}\,\Gamma\big(\tfrac{d-\beta}{2}\big),$$

the second in polar coordinates. With $|S^{d-1}| = 2\pi^{d/2}/\Gamma(\frac{d}{2})$, the constant $A_\beta = (2\pi)^{d/2}e^{1/2}\,C'_\beta$ of the proposition is

$$A_\beta = \sqrt{e}\;2^{-\beta/2}\,C_\beta\,\frac{\Gamma\big(\frac{d-\beta}{2}\big)}{\Gamma\big(\frac{d}{2}\big)} = \sqrt{e}\;2^{\beta/2}\,\frac{\Gamma\big(\frac{d-\beta}{2}\big)}{\Gamma\big(\frac{d}{2}\big)}\Big(1+\Big(\frac{e^3\sqrt{d}\,(c+\beta)}{(c-\beta)\,r_\beta}\Big)^\beta\Big).$$

The rate through the polar profile (Proposition 3.18) is proved from the rate through the transform, which is the next lemma. For a law whose transform is a power times a slowly varying factor its clause (1) is a known Tauberian theorem, and §A.1 says where it is in print. For a finite c the transform of a kernel has that form, $|\xi|^{-c}L(|\xi|)$ with L slowly varying (Lemma A.2(2)). In the statement s is a distance from the centre, not a scale, e is a unit vector, and $\widehat{\mu}(e/s)$ is the transform at a frequency of length $1/s$.

Lemma S.9 (the rate at the centre through the transform)**.** *Let $d \geq 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0, let μ be its law, and suppose that $c := k_d(0+)$ satisfies $0 < c \leq d$. Let e be a unit vector of $\mathbb{R}^d$, let $|S^{d-1}|$ be the surface measure of the unit sphere and V_d the volume of the unit ball. The law μ has a density p with $p(x) = q(|x|)$ for $x \neq 0$, where $q : (0,\infty) \to [0,\infty)$ is nonincreasing. Let q be any such function. All limits are as $s \downarrow 0$.*

1. (Below the threshold.) *Let $c < d$ and put*

$$R(d,c) := \frac{\Gamma\big(\frac{d-c}{2}\big)}{2^c\,\pi^{d/2}\,\Gamma\big(\frac{c}{2}\big)}.$$

Then

$$\frac{\mu(\{|x| \leq s\})}{R(d,c)\,|S^{d-1}|\,c^{-1}\,\widehat{\mu}(e/s)} \to 1 \qquad \text{and} \qquad \frac{q(s)}{R(d,c)\,s^{-d}\,\widehat{\mu}(e/s)} \to 1.$$

2. (At the threshold.) *Let $c = d$ and put $I(s) := \int_{\{|\xi| \leq 1/s\}} \widehat{\mu}(\xi)\,d\xi$. Then*

$$\frac{\mu(\{|x| \leq s\})}{(2\pi)^{-d}\,V_d\,s^d\,I(s)} \to 1 \qquad \text{and} \qquad \frac{q(s)}{(2\pi)^{-d}\,I(s)} \to 1.$$

Moreover $I(s) \geq |S^{d-1}|\,e^{-13d/4-2C_k}\log(1/s)$ for $0 < s \leq 1$, with $C_k = \int_1^\infty k_d(r)\,dr/r$. So for every $C < (2\pi)^{-d}\,|S^{d-1}|\,e^{-13d/4-2C_k}$ there is $s_0 > 0$ with $q(s) \geq C\log(1/s)$ for $0 < s < s_0$.

So for $c < d$ the density $p(x)$ is asymptotic, as $x \to 0$, to $R(d,c)\,|x|^{c-d}L(1/|x|)$. Clause (2) is the boundary case. The density at distance s is asymptotic to $(2\pi)^{-d}$ times the integral of the transform cut at the frequency $1/s$, which depends on the law and grows at least like $\log(1/s)$.

Proof of Lemma S.9. As $c = \sup_{r>0} k_d(r) > 0$ and k_d is nonincreasing, k_d is positive on an interval $(0, r_0]$, so the datum is not trivial and Proposition 3.15(2) gives a density $p(x) = q(|x|)$ of μ with q nonincreasing. From here on q is any nonincreasing function on $(0,\infty)$ such that $x \mapsto q(|x|)$ is a density of μ.

Write $\chi(\rho) := \widehat{\mu}(\rho e) = e^{-F_d(\rho e)}$ for $\rho > 0$. As F_d is rotation invariant, $\widehat{\mu}(\xi) = \chi(|\xi|)$ for $\xi \neq 0$. Three properties of χ are used.

(a) $0 < \chi \leq 1$, and χ is continuous, because $F_d \geq 0$ and $\widehat{\mu}$ is a characteristic function.

(b) χ is nonincreasing. In the notation of the proof of Lemma A.2, $F_d(\xi)$ is the average over the sphere of $\varphi(\langle \xi, \eta\rangle)$, and the substitution $u = ar$ made there gives $\varphi(as) = \int_0^\infty (1 - \cos(us))\, k_d(u/a)\, du/u \geq \varphi(s)$ for $a \geq 1$ and every real s, because $k_d(u/a) \geq k_d(u)$. So $F_d(a\xi) \geq F_d(\xi)$ for $a \geq 1$.

(c) $\chi(a\rho)/\chi(\rho) \to a^{-c}$ as $\rho \to \infty$, for every $a > 0$. This is Lemma A.2(2), which applies because $c \leq d$ is finite.

Step 1: a bound on the ratios of χ. Let $\gamma > c$. By (c) at $a = 2$, $\chi(2y)/\chi(y) \to 2^{-c} > 2^{-\gamma}$, so there is $X \geq 1$ with $\chi(y) \leq 2^\gamma \chi(2y)$ for every $y \geq X$. Let $X \leq y \leq x$, and let $n \geq 0$ be the least integer with $2^n y \geq x$, so that $n < \log_2(x/y) + 1$. Iterating the inequality and using (b),

$$\chi(y) \leq 2^{\gamma n}\, \chi(2^n y) \leq 2^{\gamma n}\, \chi(x) \leq 2^\gamma\, (x/y)^\gamma\, \chi(x).$$

It follows that, with $C := 2^\gamma/\chi(X)$,

$$\frac{\chi(Ru)}{\chi(R)} \leq 1 + C\, u^{-\gamma} \qquad (R \geq X,\ u > 0). \tag{S.3}$$

Indeed, for $u \geq 1$ the quotient is at most 1 by (b). For $u < 1$ and $Ru \geq X$ the display with $y = Ru$ and $x = R$ gives the bound $2^\gamma u^{-\gamma}$, and $2^\gamma \leq C$ because $\chi(X) \leq 1$. For $u < 1$ and $Ru < X$ the numerator is at most 1, and the display with $y = X$ and $x = R$ gives $1/\chi(R) \leq 2^\gamma (R/X)^\gamma/\chi(X) \leq C\, u^{-\gamma}$.

Step 2: the Gaussian average of the law. For $\lambda > 0$ put $M(\lambda) := \int_{\mathbb{R}^d} e^{-\lambda|x|^2}\mu(dx)$. At $t = 1/(2\lambda)$ the heat kernel is $g_t(x) = (\lambda/\pi)^{d/2} e^{-\lambda|x|^2}$, and Lemma S.8(1) gives $M(\lambda) = (\pi/\lambda)^{d/2}(2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{\mu}(\xi)\, e^{-|\xi|^2/4\lambda}\, d\xi$. In polar coordinates, at $\lambda = R^2$,

$$M(R^2) = (4\pi)^{-d/2}\, |S^{d-1}|\, R^{-d}\, T(R), \qquad T(R) := \int_0^\infty \chi(\rho)\, \rho^{d-1}\, e^{-\rho^2/4R^2}\, d\rho, \tag{S.4}$$

and the substitution $\rho = Ru$ gives $T(R) = R^d \int_0^\infty \chi(Ru)\, u^{d-1}\, e^{-u^2/4}\, du$.

Step 3: the Gaussian average for $c < d$. Let $c < d$ and take $\gamma := (c+d)/2$ in Step 1, so that $c < \gamma < d$. By the second form of $T(R)$ in Step 2,

$$\frac{T(R)}{R^d \chi(R)} = \int_0^\infty \frac{\chi(Ru)}{\chi(R)}\, u^{d-1}\, e^{-u^2/4}\, du.$$

For each $u > 0$ the quotient under the integral tends to u^{-c} as $R \to \infty$, by (c). For $R \geq X$ the integrand is at most $(1 + C\, u^{-\gamma})\, u^{d-1} e^{-u^2/4}$ by (S.3), which is integrable on $(0,\infty)$ because $\gamma < d$. By dominated convergence

$$\frac{T(R)}{R^d \chi(R)} \to \int_0^\infty u^{d-1-c}\, e^{-u^2/4}\, du = 2^{d-c-1}\, \Gamma\big(\tfrac{d-c}{2}\big) \qquad (R \to \infty),$$

and with $|S^{d-1}| = 2\pi^{d/2}/\Gamma(d/2)$,

$$\frac{M(R^2)}{\chi(R)} \to I_c := (4\pi)^{-d/2}\, |S^{d-1}|\, 2^{d-c-1}\, \Gamma\big(\tfrac{d-c}{2}\big) = \frac{2^{-c}\, \Gamma\big(\frac{d-c}{2}\big)}{\Gamma\big(\frac{d}{2}\big)}. \tag{S.5}$$

Step 3′: the Gaussian average for $c = d$. Let $c = d$, and put $P(y) := \int_0^y \chi(\rho)\,\rho^{d-1}\,d\rho$ for $y > 0$ and $L(\rho) := \rho^d \chi(\rho)$. By (b), for $0 < x \le y$,

$$\chi(y)\,\frac{y^d - x^d}{d} \le P(y) - P(x) \le \chi(x)\,\frac{y^d - x^d}{d}. \tag{S.6}$$

(i) $L(R)/P(R) \to 0$ as $R \to \infty$. The left inequality of (S.6) at $x = y/2$ reads $P(y) - P(y/2) \ge \kappa\, L(y)$ with $\kappa := (1 - 2^{-d})/d$. For each integer $j \ge 0$, $L(R/2^j)/L(R) = 2^{-jd}\chi(2^{-j}R)/\chi(R) \to 1$ as $R \to \infty$, by (c) with $c = d$. Let $n \ge 1$. For all large R, $L(R/2^j) \ge L(R)/2$ for every $j < n$, and then

$$P(R) \ge \sum_{j=0}^{n-1} \big(P(R/2^j) - P(R/2^{j+1})\big) \ge \kappa \sum_{j=0}^{n-1} L(R/2^j) \ge \tfrac{n\kappa}{2}\, L(R).$$

As n is arbitrary, $L(R)/P(R) \to 0$. *(ii) $P(aR)/P(R) \to 1$ as $R \to \infty$, for every $a > 0$.* For $a \ge 1$ the right inequality of (S.6) gives $0 \le P(aR) - P(R) \le (a^d - 1)\, L(R)/d$, and (i) gives the limit. For $a < 1$ the quotient is the reciprocal of $P(a^{-1}R')/P(R')$ at $R' = aR$, which tends to infinity with R. *(iii) $T(R)/P(R) \to 1$ as $R \to \infty$.* Let $\varepsilon > 0$. On $(0, \varepsilon R]$ the Gaussian factor of $T(R)$ is at least $e^{-\varepsilon^2/4}$, so $T(R) \ge e^{-\varepsilon^2/4} P(\varepsilon R)$. On $(0, R]$ it is at most 1, and on (R, ∞), $\chi(\rho) \le \chi(R)$ by (b), so

$$T(R) \le P(R) + \chi(R) \int_0^\infty \rho^{d-1} e^{-\rho^2/4R^2}\,d\rho = P(R) + 2^{d-1}\Gamma\big(\tfrac{d}{2}\big)\, L(R).$$

Hence $e^{-\varepsilon^2/4}\, P(\varepsilon R)/P(R) \le T(R)/P(R) \le 1 + 2^{d-1}\Gamma(d/2)\, L(R)/P(R)$. By (ii) the left side tends to $e^{-\varepsilon^2/4}$ and by (i) the right side tends to 1. As ε is arbitrary, $T(R)/P(R) \to 1$. With (S.4),

$$\frac{M(R^2)}{(4\pi)^{-d/2}\,|S^{d-1}|\,R^{-d}\,P(R)} \to 1 \qquad (R \to \infty). \tag{S.7}$$

Step 4: the mass of a ball, by Karamata's theorem. Let ν be the image of μ under $x \mapsto |x|^2$. It is a probability measure on $[0, \infty)$ with $\int_{[0,\infty)} e^{-\lambda r}\,\nu(dr) = M(\lambda)$ and $\nu([0, r]) = \mu(\{|x| \le \sqrt{r}\})$.

Let $c < d$ and put $\ell(\lambda) := \lambda^{c/2}\chi(\sqrt{\lambda})$ for $\lambda > 0$. It is positive and continuous, and $\ell(a\lambda)/\ell(\lambda) = a^{c/2}\chi(\sqrt{a}\sqrt{\lambda})/\chi(\sqrt{\lambda}) \to 1$ as $\lambda \to \infty$ for every $a > 0$, by (c). By (S.5) at $R = \sqrt{\lambda}$, $M(\lambda)/(I_c\,\lambda^{-c/2}\,\ell(\lambda)) \to 1$. Karamata's Tauberian theorem (Theorem S.7, with the index $c/2 > 0$ and the constant $I_c > 0$) gives $\nu([0, r]) \sim I_c\, r^{c/2}\,\ell(1/r)/\Gamma(1 + c/2)$ as $r \downarrow 0$, and $r^{c/2}\ell(1/r) = \chi(r^{-1/2})$. At $r = s^2$,

$$\frac{\mu(\{|x| \le s\})}{B_c\,\chi(1/s)} \to 1 \quad (s \downarrow 0), \qquad B_c := \frac{I_c}{\Gamma(1 + c/2)} = \frac{R(d, c)\,|S^{d-1}|}{c}, \tag{S.8}$$

the last equality by $\Gamma(1 + c/2) = (c/2)\Gamma(c/2)$ and $|S^{d-1}| = 2\pi^{d/2}/\Gamma(d/2)$. This is the first limit of clause (1).

Let $c = d$ and put $\ell(\lambda) := P(\sqrt{\lambda})$. It is positive and continuous, and $\ell(a\lambda)/\ell(\lambda) \to 1$ by Step 3′(ii). By (S.7), $M(\lambda)/\big((4\pi)^{-d/2}|S^{d-1}|\,\lambda^{-d/2}\,\ell(\lambda)\big) \to 1$. Karamata's theorem with the index $d/2$ gives, at $r = s^2$, $\mu(\{|x| \le s\}) \sim (4\pi)^{-d/2}\,|S^{d-1}|\,s^d\,P(1/s)/\Gamma(1 + d/2)$. In polar coordinates $|S^{d-1}|\,P(1/s) = I(s)$, and $(4\pi)^{-d/2}/\Gamma(1 + d/2) = (2\pi)^{-d}\,\pi^{d/2}/\Gamma(1 + d/2) = (2\pi)^{-d}\,V_d$. This is the first limit of clause (2).

Step 5: from the mass of balls to the density. Write $m(s) := \mu(\{|x| \le s\})$. Let W be a positive function on $(0, \infty)$, $\varkappa > 0$ and $B > 0$, and suppose that $W(as)/W(s) \to a^\varkappa$ for every $a > 0$ and $m(s)/W(s) \to B$, as $s \downarrow 0$. Then

$$\frac{q(s)\,s^d}{W(s)} \to \frac{B\,\varkappa}{|S^{d-1}|} \qquad (s \downarrow 0). \tag{S.9}$$

Indeed, let $0 < a < b$. On the shell $\{as < |x| \le bs\}$, whose volume is $V_d\, s^d(b^d - a^d)$, the density $q(|x|)$ lies between $q(bs)$ and $q(as)$, because q is nonincreasing. So

$$q(bs)\, V_d\, s^d\, (b^d - a^d) \le m(bs) - m(as) \le q(as)\, V_d\, s^d\, (b^d - a^d),$$

and $(m(bs) - m(as))/W(s) \to B\,(b^\varkappa - a^\varkappa)$. With $b = 1$ and $a < 1$ the left inequality shows that, for every $\varepsilon > 0$, $q(s)s^d/W(s) \le B(1 - a^\varkappa)/(V_d(1 - a^d)) + \varepsilon$ for all small s. With $a = 1$ and $b > 1$ the right one shows $q(s)s^d/W(s) \ge B(b^\varkappa - 1)/(V_d(b^d - 1)) - \varepsilon$ for all small s. Now $(t^\varkappa - 1)/(t^d - 1) \to \varkappa/d$ as $t \to 1$, the quotient of the derivatives of $t^\varkappa$ and t^d at 1. So the first bound tends to $B\varkappa/(V_d\, d) + \varepsilon$ as $a \uparrow 1$ and the second to $B\varkappa/(V_d\, d) - \varepsilon$ as $b \downarrow 1$. As ε is arbitrary and $V_d\, d = |S^{d-1}|$, this proves (S.9). It is the argument of the monotone density theorem ([1], Thm. 1.7.2b, pp. 39–40), on spherical shells.

For $c < d$ take $W(s) := \chi(1/s)$, $\varkappa := c$ and $B := B_c$: the first hypothesis is (c) at the ratio $1/a$, the second is (S.8). So $q(s)\, s^d/\chi(1/s) \to B_c\, c/|S^{d-1}| = R(d, c)$, the second limit of clause (1). For $c = d$ take $W(s) := s^d I(s)$, $\varkappa := d$ and $B := (2\pi)^{-d} V_d$: the first hypothesis is Step 3′(ii), since $I(s) = |S^{d-1}|P(1/s)$, and the second is the first limit of clause (2). So $q(s)/I(s) \to (2\pi)^{-d}\, V_d\, d/|S^{d-1}| = (2\pi)^{-d}$, the second limit of clause (2).

The logarithmic bound of clause (2). Let $c = d$ and $0 < s \le 1$. By Lemma A.2(1), $\chi(\rho) \ge e^{-13d/4 - 2C_k}\rho^{-d}$ for $\rho \ge 1$. In polar coordinates

$$I(s) \ge |S^{d-1}| \int_1^{1/s} \chi(\rho)\, \rho^{d-1}\, d\rho \ge |S^{d-1}|\, e^{-13d/4 - 2C_k} \int_1^{1/s} \frac{d\rho}{\rho} = |S^{d-1}|\, e^{-13d/4 - 2C_k} \log(1/s).$$

Let $C < C_* := (2\pi)^{-d}|S^{d-1}|e^{-13d/4 - 2C_k}$. If $C \le 0$ there is nothing to prove, as $q(s) > 0$ for small s by the second limit. Otherwise $q(s) \ge (C/C_*)(2\pi)^{-d} I(s)$ for all small s, by the same limit, and the bound on $I(s)$ gives $q(s) \ge C \log(1/s)$. □

The proofs of Propositions 3.16 and 3.18 follow.

Proof of Proposition 3.16. Write $k := k_d$ and $\bar\sigma := \sigma_{d-1}/|S^{d-1}|$ for the normalized surface measure. Three things are taken from the proof of Lemma A.2. First,

$$F_d(\xi) = \int_{S^{d-1}} \varphi\big(\langle \xi, \eta\rangle\big)\, \bar\sigma(d\eta), \qquad \varphi(s) := \int_0^\infty \big(1 - \cos(rs)\big)\, k(r)\, \frac{dr}{r} \quad (s \in \mathbb{R}),$$

with φ even and finite. Second, $k \le c$ on $(0, \infty)$ and $k(r) \to 0$ as $r \to \infty$. Third, the bounds (S.2), and $0 \le \mathrm{Cin}(X) \le \frac14$ for $0 \le X \le 1$, because Cin is nondecreasing and $\mathrm{Cin}(1) \le \frac14$.

Clause (1). For $0 < u \le 1$ the integrand $(c - k(u))/u$ lies in $[0, c/u]$. So $\int_x^1 (c - k(u))\, du/u$ lies in $[0, c\log(1/x)]$, it is continuous and nonincreasing in x, and it vanishes at $x = 1$. Hence K is continuous and nonincreasing with $1 = K(1) \le K(x) \le x^{-c}$. Let $a > 0$, and let $0 < x \le \min\{1, 1/a\}$, so that x and ax lie in $(0, 1]$. Between x and ax the integrand is at most $(c - k(\max\{a, 1\}x))/u$, because k is nonincreasing. So

$$\big|\log K(ax) - \log K(x)\big| = \Big|\int_{ax}^{x} \big(c - k(u)\big)\frac{du}{u}\Big| \le \big(c - k(\max\{a, 1\}\, x)\big)\, |\log a|,$$

which tends to 0 as $x \downarrow 0$ because $k(0+) = c$. By monotone convergence $\int_x^1 (c - k(u))\, du/u$ increases to $\int_0^1 (c - k(u))\, du/u$ as $x \downarrow 0$. If this integral is finite, $K(x)$ increases to its exponential, which is $K(0+)$. If it is infinite, $K(x) \to \infty$.

Clause (2), the limit g_∞. Put $\log^+ X := \max\{\log X, 0\}$ for $X > 0$, $\log^+ 0 := 0$, and

$$G(X) := \mathrm{Cin}(X) - \log^+ X \qquad (X \ge 0).$$

Then $-3 \le G \le \frac{13}{4}$ on $[0,\infty)$: for $X \ge 1$ this is (S.2), and for $0 \le X < 1$, $G(X) = \mathrm{Cin}(X) \in [0, \frac{1}{4}]$. For $X \ge 1$, $\mathrm{Cin}(X) = \mathrm{Cin}(1) + \log X - \int_1^X u^{-1} \cos u \, du$, so

$$G(X) = \mathrm{Cin}(1) - \int_1^X \frac{\cos u}{u}\, du = \mathrm{Cin}(1) - \Big[\frac{\sin u}{u}\Big]_1^X - \int_1^X \frac{\sin u}{u^2}\, du.$$

The bracket tends to $-\sin 1$ as $X \to \infty$, and $u^{-2}\sin u$ is absolutely integrable on $[1,\infty)$. So $G(X)$ has a limit as $X \to \infty$. As $G(X) = \mathrm{Cin}(X) - \log X$ for $X \ge 1$, this limit is g_∞, and $-3 \le g_\infty \le \frac{13}{4}$.

Clause (2), the mean of the logarithm over the sphere. Let B be the closed unit ball of $\mathbb{R}^d$ and e_1 the first basis vector. Lebesgue measure is the image of $\sigma_{d-1} \otimes r^{d-1} dr$ under $(\eta, r) \mapsto r\eta$, so for every Borel $h : S^{d-1} \to [0,\infty]$,

$$\int_B h\big(x/|x|\big)\, dx = \int_0^1 r^{d-1}\, dr \int_{S^{d-1}} h\, d\sigma_{d-1} = \frac{1}{d}\int_{S^{d-1}} h\, d\sigma_{d-1}.$$

Take $h(\eta) := -\log|\langle e_1, \eta\rangle| \in [0,\infty]$. For $x \in B$ with $x_1 \ne 0$, $h(x/|x|) = -\log|x_1| + \log|x| \le -\log|x_1|$, because $|x| \le 1$. As B lies in the cube $[-1,1]^d$, on which $-\log|x_1| \ge 0$,

$$\frac{1}{d}\int_{S^{d-1}} h\, d\sigma_{d-1} \le \int_{[-1,1]^d} \big(-\log|x_1|\big)\, dx = 2^{d-1}\int_{-1}^1 \big(-\log|t|\big)\, dt = 2^d.$$

So $\log|\langle e_1, \eta\rangle|$ is σ_{d-1}-integrable, and its mean is at most 0 because $|\langle e_1, \eta\rangle| \le 1$. For a unit vector e take $Q \in \mathrm{O}(d)$ with $Qe_1 = e$. Then $\langle e, \eta\rangle = \langle e_1, Q^{\mathsf{T}}\eta\rangle$, and σ_{d-1} is invariant under Q^{T}, so the integral is the same for e as for e_1. At $d = 1$ the sphere is $\{\pm 1\}$, $|\langle e, \eta\rangle| = 1$, and $\lambda_1 = 0$.

Clause (3), the tail integral. For $s > 0$ put

$$T(s) := \int_{1/s}^\infty k(r)\, \frac{dr}{r}.$$

It is finite: it is at most C_k for $s \le 1$ and at most $c\log s + C_k$ for $s \ge 1$. For $s \ge 1$,

$$T(s) = \int_{1/s}^1 k(r)\, \frac{dr}{r} + C_k = c\log s - \int_{1/s}^1 \big(c - k(r)\big)\frac{dr}{r} + C_k = c\log s - \log K(1/s) + C_k. \quad \text{(S.10)}$$

Clause (3), the layer-cake identity. For $0 < y < c$ put $\tau(y) := \sup\{r > 0 : k(r) > y\}$. The set is not empty, because $\sup k = c > y$. It is bounded, because $k(r) \to 0$ as $r \to \infty$ and $y > 0$. So $0 < \tau(y) < \infty$. As k is nonincreasing, the set contains every $r' < r$ with each of its points r, so it is $(0, \tau(y))$ or $(0, \tau(y)]$; and τ is nonincreasing in y, hence Borel. Since $0 \le k \le c$,

$$k(r) = \int_0^c \mathbf{1}\{y < k(r)\}\, dy \qquad (r > 0),$$

and the indicator is a Borel function of (r, y), k being monotone. For a Borel $w : (0,\infty) \to [0,\infty)$, Tonelli's theorem gives

$$\int_0^\infty w(r)\, k(r)\, dr = \int_0^c \Big(\int_0^{\tau(y)} w(r)\, dr\Big) dy.$$

Let $s > 0$. With $w(r) = (1 - \cos rs)/r$ the inner integral is $\mathrm{Cin}(s\,\tau(y))$, by the substitution $u = rs$. With $w(r) = \mathbf{1}\{r > 1/s\}/r$ it is $\log^+(s\,\tau(y))$. So

$$\varphi(s) = \int_0^c \mathrm{Cin}\big(s\,\tau(y)\big)\, dy, \qquad T(s) = \int_0^c \log^+\big(s\,\tau(y)\big)\, dy.$$

Both are finite, and their integrands differ by $G(s\,\tau(y))$, which is bounded. Hence

$$\varphi(s) = T(s) + \int_0^c G\big(s\,\tau(y)\big)\,dy \qquad (s>0). \tag{S.11}$$

This is an identity, with no remainder. It is where the monotonicity of k enters the exponent. Monotonicity is also used for $k \le c$ and $k(r) \to 0$, in clause (1), and in the first term below.

Clause (3), the sphere average. Let $\rho > 0$ and write $\eta_1 := \langle e, \eta\rangle$ for $\eta \in S^{d-1}$. As φ is even, $F_d(\rho e) = \int \varphi(\rho|\eta_1|)\,\bar\sigma(d\eta)$. The set $\{\eta_1 = 0\}$ is $\bar\sigma$-null: it is a great subsphere when $d \ge 2$ and empty when $d = 1$. By (S.11) at $s = \rho|\eta_1|$,

$$F_d(\rho e) - T(\rho) = \int_{S^{d-1}} \big[T(\rho|\eta_1|) - T(\rho)\big]\,\bar\sigma(d\eta) + \int_{S^{d-1}} \int_0^c G\big(\rho|\eta_1|\,\tau(y)\big)\,dy\,\bar\sigma(d\eta), \tag{S.12}$$

where both integrals converge absolutely by the bounds that follow.

The first term. Let $0 < t \le 1$. Then $T(\rho) - T(\rho t) = \int_{1/\rho}^{1/(\rho t)} k(r)\,dr/r$ lies between $k(1/(\rho t))\log(1/t)$ and $c\log(1/t)$. So $-c\log(1/t) \le T(\rho t) - T(\rho) \le 0$ for every $\rho > 0$, and $T(\rho t) - T(\rho) \to c\log t$ as $\rho \to \infty$, because $k(0+) = c$. At $t = |\eta_1|$ the bound $c\,|\log|\eta_1||$ is $\bar\sigma$-integrable by clause (2). By dominated convergence the first term of (S.12) tends to $c\lambda_d$ as $\rho \to \infty$, and it lies in $[c\lambda_d, 0]$ for every $\rho > 0$.

The second term. The integrand lies in $[-3, \frac{13}{4}]$. For every $y \in (0,c)$ and every η with $\eta_1 \ne 0$, $\rho|\eta_1|\tau(y) \to \infty$ as $\rho \to \infty$, because $\tau(y) > 0$, so the integrand tends to g_∞. The function $(\eta, y) \mapsto G(\rho|\eta_1|\tau(y))$ is Borel on $S^{d-1} \times (0,c)$ and bounded: τ is Borel, being monotone, and G is continuous on $[0,\infty)$. So the inner integral $\int_0^c G(\rho|\eta_1|\tau(y))\,dy$ is a Borel function of η, and it lies in $[-3c, \frac{13}{4}c]$ for every $\rho > 0$ and every η. For η with $\eta_1 \ne 0$ it tends to $c\,g_\infty$ as $\rho \to \infty$, by bounded convergence on $(0,c)$. By bounded convergence for $\bar\sigma$, the second term tends to $c\,g_\infty$, and it lies in $[-3c, \frac{13}{4}c]$ for every $\rho > 0$.

Adding the two, $F_d(\rho e) - T(\rho) \to c(\lambda_d + g_\infty) = c\varkappa_d$ as $\rho \to \infty$, and $c(\lambda_d - 3) \le F_d(\rho e) - T(\rho) \le \frac{13}{4}c$ for every $\rho > 0$. As $T(\rho) = \int_{1/\rho}^\infty k(r)\,dr/r$, this is the bound of clause (3) for every $\rho > 0$. For $\rho \ge 1$, (S.10) replaces $T(\rho)$ by $c\log\rho - \log K(1/\rho) + C_k$. This is the limit and the two-sided bound of clause (3) for $\rho \ge 1$. The quotient form follows by taking exponentials: the quotient is exp of minus the difference between $F_d(\rho e) - (c\log\rho - \log K(1/\rho) + C_k)$ and its limit.

Clause (4). F_d is rotation invariant, since σ_{d-1} is, so $F_d(\xi) = F_d(|\xi|e)$. Let $\rho = |\xi| \ge 1$ and write $F_d(\rho e) = c\log\rho - \log K(1/\rho) + C_k + \Delta(\rho)$. Then $\rho^c e^{-F_d(\rho e)} = e^{-C_k - \Delta(\rho)} K(1/\rho)$, and the two-sided bound of clause (3) on $\Delta(\rho)$ gives the display. By the display, $\rho^c e^{-F_d(\rho e)}$ is bounded on $[1,\infty)$ exactly when K is bounded on $(0,1]$, which by clause (1) is the condition on the integral. In that case $K(1/\rho) \to K(0+) < \infty$ and $\Delta(\rho) \to c\varkappa_d$, which gives the limit. □

Proof of Proposition 3.18. A density $p(x) = q(|x|)$ with q nonincreasing exists by Proposition 3.15(2), as in the proof of Lemma S.9. From here on q is any nonincreasing function on $(0,\infty)$ such that $x \mapsto q(|x|)$ is a density of μ. Let e be a unit vector and write $\chi(\rho) := \widehat\mu(\rho e) = e^{-F_d(\rho e)}$ for $\rho > 0$. Since $0 < c \le d < \infty$, Proposition 3.16(3) applies and gives $\chi(\rho)/(A\,\rho^{-c}K(1/\rho)) \to 1$ as $\rho \to \infty$. At $\rho = 1/s$,

$$\frac{\widehat\mu(e/s)}{A\,s^c\,K(s)} \to 1 \qquad (s \downarrow 0). \tag{S.13}$$

Clause (1). Let $c < d$. By Lemma S.9(1), $q(s)/(R(d,c)\,s^{-d}\,\widehat\mu(e/s)) \to 1$ and $\mu(\{|x| \le s\})/(R(d,c)\,|S^{d-1}|\,c^{-1}\,\widehat\mu(e/s)) \to 1$. Multiplying each by (S.13) gives the two limits, with $Q = R(d,c)\,A$.

Clause (2). By clause (1), $q(s)\,s^{d-c}/(Q\,K(s)) \to 1$, and $Q > 0$. By Proposition 3.16(1), as $s \downarrow 0$ the factor $K(s)$ increases to the finite number $K(0+) \ge 1$ if $\int_0^1 (c - k_d(u))\,du/u < \infty$, and $K(s) \to \infty$ if the integral diverges. In the second case $q(s)\,s^{d-c} \to \infty$, and no bound

$q(s) \le C s^{c-d}$ holds near 0. In the first case $q(s)\, s^{d-c} \to Q\, K(0+)$, which is finite and positive, so there is $s_0 > 0$ with $q(s)\, s^{d-c} \le 2Q\, K(0+)$ for $0 < s \le s_0$. For $s \ge s_0$, as q is nonincreasing, $1 \ge \mu(\{|x| < s\}) \ge q(s)\, V_d\, s^d$, so $q(s)\, s^{d-c} \le s^{-c}/V_d \le s_0^{-c}/V_d$. So $C := \max\{2Q\, K(0+),\, s_0^{-c}/V_d\}$ serves for every $s > 0$.

Clause (3), the truncated integral of the transform. Let $c = d$. In polar coordinates the $I(s)$ of Lemma S.9(2) is

$$I(s) = \int_{\{|\xi| \le 1/s\}} \widehat{\mu}(\xi)\, d\xi = |S^{d-1}|\, P(1/s), \qquad P(R) := \int_0^R \chi(\rho)\, \rho^{d-1}\, d\rho.$$

Put $f(\rho) := \chi(\rho)\, \rho^{d-1}$ and $h(\rho) := A\, K(1/\rho)/\rho$ for $\rho \ge 1$. Both are positive and continuous, and $f(\rho)/h(\rho) = \chi(\rho)\, \rho^d/(A\, K(1/\rho)) \to 1$ as $\rho \to \infty$, by Proposition 3.16(3) with $c = d$. The substitution $y = 1/\rho$ gives, for $R \ge 1$,

$$H(R) := \int_1^R h(\rho)\, d\rho = A \int_{1/R}^1 K(y)\, \frac{dy}{y} = A\, \mathcal{L}(1/R),$$

and $H(R) \ge A \log R \to \infty$, because $K \ge 1$. Let $0 < \varepsilon < 1$ and choose $R_0 \ge 1$ with $|f - h| \le \varepsilon h$ on $[R_0, \infty)$. For $R \ge R_0$,

$$\big|P(R) - H(R)\big| \le P(1) + \int_1^{R_0} |f - h|\, d\rho + \varepsilon\, H(R),$$

and $P(1) \le 1/d$ because $\chi \le 1$. Dividing by $H(R)$, which tends to infinity, $|P(R)/H(R) - 1| \le 2\varepsilon$ for all large R. So $P(R)/(A\, \mathcal{L}(1/R)) \to 1$ as $R \to \infty$, that is

$$\frac{I(s)}{|S^{d-1}|\, A\, \mathcal{L}(s)} \to 1 \qquad (s \downarrow 0).$$

Clause (3), the two limits. By Lemma S.9(2), $q(s)/((2\pi)^{-d} I(s)) \to 1$ and $\mu(\{|x| \le s\})/((2\pi)^{-d} V_d\, s^d\, I(s)) \to 1$. Multiplying each by the last display gives the two limits, with $B = (2\pi)^{-d} |S^{d-1}| A$.

Clause (3), the logarithm. Let $0 < s \le 1$. On $(s, 1)$, $1 \le K(y) \le K(s)$, so $\log(1/s) \le \mathcal{L}(s) \le K(s) \log(1/s)$. Let $0 < y_0 < 1$. For $0 < s < y_0$, $\mathcal{L}(s) \ge \int_s^{y_0} K(y)\, dy/y \ge K(y_0) \log(y_0/s)$, and $\log(y_0/s)/\log(1/s) \to 1$ as $s \downarrow 0$. So for every number $\kappa < K(y_0)$, $\mathcal{L}(s)/\log(1/s) \ge \kappa$ for all small s. Now use Proposition 3.16(1) at $c = d$. If $\int_0^1 (d - k_d(u))\, du/u < \infty$, then $K(y_0)$ increases to the finite number $K(0+)$ as $y_0 \downarrow 0$, so the lower bound holds for every $\kappa < K(0+)$; and $\mathcal{L}(s) \le K(s) \log(1/s) \le K(0+) \log(1/s)$. Hence $\mathcal{L}(s)/\log(1/s) \to K(0+)$, and with the first limit of the clause $q(s)/\log(1/s) \to B\, K(0+)$, which is finite and positive. If the integral diverges, then $K(y_0) \to \infty$ as $y_0 \downarrow 0$, so the lower bound holds for every real κ: $\mathcal{L}(s)/\log(1/s) \to \infty$, and with the first limit $q(s)/\log(1/s) \to \infty$, because $B > 0$. □

Proof of Proposition 3.20. Clause (1). Write $F_d(\xi) := F(|\xi|)$ for the exponent of the datum. The clause is proved for two numbers $0 < \ell < L$ in the place of $\chi(s)$ and $\chi(t)$. Put $\kappa(r) := k_d(r/L) - k_d(r/\ell)$ for $r > 0$, and let ν be the measure on $\mathbb{R}^d \setminus \{0\}$ with the density $\kappa(|x|)\, \big(|S^{d-1}|\, |x|^d\big)^{-1}$. Three assertions are proved: (i) $\kappa \ge 0$, the measure ν is rotation invariant, and $\int (|x|^2 \wedge 1)\, \nu(dx) < \infty$; (ii) for every $\xi \in \mathbb{R}^d$,

$$F_d(L\xi) - F_d(\ell\xi) = \tfrac{1}{2} a\, (L^2 - \ell^2)\, |\xi|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle \xi, x\rangle\big)\, \nu(dx);$$

(iii) $\nu(\mathbb{R}^d \setminus \{0\}) = c \log(L/\ell)$, in $[0, \infty]$.

The dilation of a datum. For $\beta > 0$ put $k^{(\beta)}(r) := k_d(r/\beta)$. It is nonnegative and nonincreasing, and

$$\int_0^1 k^{(\beta)}(r)\, r\, dr = \beta^2 \int_0^{1/\beta} k_d(u)\, u\, du < \infty, \qquad \int_1^\infty k^{(\beta)}(r)\, \frac{dr}{r} = \int_{1/\beta}^\infty k_d(u)\, \frac{du}{u} < \infty,$$

by the two conditions on k_d and because k_d is bounded on the interval between 1 and $1/\beta$. So $(a\beta^2, k^{(\beta)})$ is an isotropic origin datum in dimension d. Let $\nu^{(\beta)}$ be its Lévy measure, of density $k_d(|x|/\beta)\,(|S^{d-1}||x|^d)^{-1}$, and ν_d that of (a, k_d). The substitution $y = \beta x$, $dy = \beta^d dx$, gives $\int g(\beta x)\, \nu_d(dx) = \int g\, d\nu^{(\beta)}$ for every nonnegative Borel g. With $g(y) = 1 - \cos\langle \xi, y\rangle$,

$$F_d(\beta\xi) = \tfrac{1}{2} a\beta^2 |\xi|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle \xi, y\rangle\big)\, \nu^{(\beta)}(dy), \tag{S.14}$$

so the exponent of $(a\beta^2, k^{(\beta)})$ is $\xi \mapsto F_d(\beta\xi)$, and the integral is finite, as $\int (1-\cos\langle \xi, y\rangle)\, d\nu^{(\beta)} \le \int (|\xi|^2|y|^2/2 \wedge 2)\, d\nu^{(\beta)} < \infty$ by (3.2) and the two conditions on $k^{(\beta)}$.

Assertion (i). For $r > 0$, $r/L < r/\ell$, and k_d is nonincreasing, so $\kappa(r) = k^{(L)}(r) - k^{(\ell)}(r) \ge 0$. The density of ν is the difference of the densities of $\nu^{(L)}$ and $\nu^{(\ell)}$, so

$$\nu + \nu^{(\ell)} = \nu^{(L)} \tag{S.15}$$

as measures on $\mathbb{R}^d \setminus \{0\}$. In particular $\nu \le \nu^{(L)}$, and $\int (|x|^2 \wedge 1)\, d\nu \le \int (|x|^2 \wedge 1)\, d\nu^{(L)}$, which is finite: by (3.2) it is $\int_0^1 k^{(L)}(r)\, r\, dr + \int_1^\infty k^{(L)}(r)\, dr/r$. The density of ν is a function of $|x|$ and Lebesgue measure is invariant under $\mathrm{O}(d)$, so ν is.

Assertion (ii). Let $g(x) := 1 - \cos\langle \xi, x\rangle \ge 0$. By (S.14) the integrals of g against $\nu^{(L)}$ and $\nu^{(\ell)}$ are finite, so (S.15) gives $\int g\, d\nu = \int g\, d\nu^{(L)} - \int g\, d\nu^{(\ell)}$. By (S.14) at $\beta = L$ and at $\beta = \ell$ this is $F_d(L\xi) - F_d(\ell\xi) - \frac{1}{2} a(L^2 - \ell^2)|\xi|^2$.

Assertion (iii). Let $0 < \varepsilon < A$ with $\varepsilon/\ell \le A/L$, and $B := \{x : \varepsilon < |x| \le A\}$. By (3.2) for the datum $(a\beta^2, k^{(\beta)})$ and $g = \mathbf{1}_B$, and the substitution $u = r/\beta$, $\nu^{(\beta)}(B) = \int_\varepsilon^A k_d(r/\beta)\, dr/r = \int_{\varepsilon/\beta}^{A/\beta} k_d(u)\, du/u$, which is finite. By (S.15), and since the intervals $[\varepsilon/L, A/L]$ and $[\varepsilon/\ell, A/\ell]$ overlap in $[\varepsilon/\ell, A/L]$,

$$\nu(B) = \int_{\varepsilon/L}^{A/L} k_d(u)\, \frac{du}{u} - \int_{\varepsilon/\ell}^{A/\ell} k_d(u)\, \frac{du}{u} = \int_{\varepsilon/L}^{\varepsilon/\ell} k_d(u)\, \frac{du}{u} - \int_{A/L}^{A/\ell} k_d(u)\, \frac{du}{u}.$$

Each of the last two integrals is over an interval whose endpoints have the ratio L/ℓ, so $\int du/u = \log(L/\ell)$ over it. As k_d is nonincreasing, the first lies between $k_d(\varepsilon/\ell)\log(L/\ell)$ and $k_d(\varepsilon/L)\log(L/\ell)$, and both bounds tend to $c\log(L/\ell)$ as $\varepsilon \downarrow 0$, also when $c = \infty$. The second is at most $k_d(A/L)\log(L/\ell)$, which tends to 0 as $A \to \infty$: a nonincreasing nonnegative function with $\int_1^\infty k_d(r)\, dr/r < \infty$ tends to 0 at infinity. With $\varepsilon = 1/n$ and $A = n$ for $n \ge (L/\ell)^{1/2}$, so that $\varepsilon/\ell \le A/L$, the sets B increase to $\mathbb{R}^d \setminus \{0\}$, so $\nu(\mathbb{R}^d \setminus \{0\}) = \lim_n \nu(B) = c\log(L/\ell)$. Since $\log(L/\ell) > 0$, this is finite exactly when c is.

The clause. A gauge is increasing with $\chi(0) = 0$, so $0 < \chi(s) < \chi(t)$. By (3.1) and $F(|\zeta|) = F_d(\zeta)$, $-\log \widehat{\mu}_{s,t}(\xi) = F_d(\chi(t)\xi) - F_d(\chi(s)\xi)$, and assertions (i), (ii) and (iii) with $\ell = \chi(s)$, $L = \chi(t)$ are the clause.

Clause (2). Write $\nu := \nu_{s,t}$, $\ell := \chi(s)$ and $L := \chi(t)$. By clause (1), ν is a finite measure of mass $m = c\log(L/\ell)$, rotation invariant, hence symmetric, with a density, and by (3.8) with $a = 0$, $\widehat{\mu}_{s,t}(\xi) = \exp[-\int (1 - \cos\langle \xi, x\rangle)\, \nu(dx)]$. If c were 0, the nonincreasing nonnegative function k_d would vanish on $(0, \infty)$, the exponent of the datum $(0, k_d)$ would be 0, and F would vanish on $[0, \infty)$. So $c > 0$, $m > 0$ and $0 < e^{-m} < 1$, and $e^{-m} = (\ell/L)^c = \lambda^c$.

Item (a). Put $\pi := e^{-m} \sum_{n \ge 0} \nu^{*n}/n!$. The measure ν^{*n} has the mass m^n, so π is a probability measure. The transform of ν^{*n} is $\widehat{\nu}^{\,n}$ and $|\widehat{\nu}| \le m$, so $\widehat{\pi} = e^{-m} \sum_n \widehat{\nu}^{\,n}/n! =$

$\exp(\widehat{\nu}-m)$. As ν is symmetric, $\widehat{\nu}(\xi)=\int\cos\langle\xi,x\rangle\,\nu(dx)$, and $\widehat{\nu}(\xi)-m=-\int(1-\cos\langle\xi,x\rangle)\,\nu(dx)$. So $\widehat{\pi}=\widehat{\mu}_{s,t}$, and $\mu_{s,t}=\pi$, a probability measure being determined by its transform.

Item (b). Let M be a finite measure and N a Lebesgue-null Borel set. Then $(\nu*M)(N)=\int\nu(N-y)\,M(dy)=0$, because $N-y$ is Lebesgue-null and ν has a density. So $\nu^{*n}=\nu*\nu^{*(n-1)}$ is absolutely continuous for every $n\ge1$, and so is $\eta_{s,t}$, a countable sum of such measures. Its mass is $e^{-m}\sum_{n\ge1}m^n/n!=1-e^{-m}$. The term $n=0$ of item (a) is $e^{-m}\delta_0$. So $\mu_{s,t}=\lambda^c\delta_0+\eta_{s,t}$, and $\mu_{s,t}(\{0\})=\lambda^c+\eta_{s,t}(\{0\})=\lambda^c$, the origin being a Lebesgue-null set. A measure with a density gives the origin no mass, and $\lambda^c>0$, so $\mu_{s,t}$ has no density.

Item (c). The convolution $\mu*f$ is additive in the measure μ, and $\delta_0*f=f$. With item (b), $\mu_{s,t}*f=\lambda^c f+\eta_{s,t}*f$, and $\eta_{s,t}*f=p_{s,t}*f$.

Item (d). Suppose that $\mu_{s,t}$ is self-decomposable, fix $q\in(0,1)$, and let ρ be a residual: a probability measure with $\widehat{\mu}_{s,t}(\xi)=\widehat{\mu}_{s,t}(q\xi)\,\widehat{\rho}(\xi)$ for every ξ, that is $\mu_{s,t}=(qI)_*\mu_{s,t}*\rho$. By item (b), $(qI)_*\mu_{s,t}=\lambda^c\delta_0+(qI)_*\eta_{s,t}$, and $(qI)_*\eta_{s,t}$ is absolutely continuous. As in item (b), $((qI)_*\eta_{s,t}*\rho)(\{0\})=0$. So the mass of the origin under the right side is $\lambda^c\rho(\{0\})$, and under the left side it is λ^c. As $\lambda^c>0$, $\rho(\{0\})=1$, that is $\rho=\delta_0$. Then $\widehat{\mu}_{s,t}(\xi)=\widehat{\mu}_{s,t}(q\xi)$ for every ξ, and by iteration $\widehat{\mu}_{s,t}(\xi)=\widehat{\mu}_{s,t}(q^n\xi)\to\widehat{\mu}_{s,t}(0)=1$, the transform being continuous. So $\mu_{s,t}=\delta_0$, which contradicts $\mu_{s,t}(\{0\})=\lambda^c<1$. The argument of item (d) shows more than the item: on $\mathbb{R}^d$, $d\ge1$, a probability measure that is the sum of a point mass of weight in $(0,1)$ at the origin and an absolutely continuous measure is not self-decomposable.

Clause (3). Write $\nu:=\nu_{s,t}$, $\ell:=\chi(s)$, $L:=\chi(t)$ and $v:=a(L^2-\ell^2)\ge0$. By clause (1), ν gives the origin no mass, is symmetric, and has $\int(|x|^2\wedge1)\,d\nu<\infty$: it is a symmetric Lévy measure.

A symmetric triplet. Let ν_0 be a symmetric Lévy measure on $\mathbb{R}^d$ and $v_0\ge0$. By C3(iii) there is an infinitely divisible law with the triplet $(v_0I,\nu_0,0)$, and its transform is

$$\exp\Big[-\tfrac12 v_0|\xi|^2-\int_{\mathbb{R}^d}\big(1-\cos\langle\xi,x\rangle\big)\,\nu_0(dx)\Big]. \tag{S.16}$$

Indeed the integrand of the representation, $e^{i\langle\xi,x\rangle}-1-i\langle\xi,x\rangle\mathbf{1}_D(x)$, has the real part $\cos\langle\xi,x\rangle-1$ and an imaginary part that is an odd function of x, integrable against ν_0 because it is at most $|\xi|^2|x|^2/2$ in modulus on D (as $|\sin u-u|\le u^2/2$ for every real u) and at most 1 off D; its integral against the symmetric measure ν_0 is 0.

Item (a). By (S.16) with $(v_0,\nu_0)=(v,\nu)$ and (3.8), the infinitely divisible law with the triplet $(vI,\nu,0)$ has the transform of $\mu_{s,t}$, so it is $\mu_{s,t}$. The polar profile and the mass of ν are those of clause (1).

Item (b). Let $a=0$ and $c=\infty$, so that ν has infinite mass (clause (1)). For $\varepsilon>0$ let $\nu_\varepsilon^>$ and $\nu_\varepsilon^\le$ be the restrictions of ν to $\{|x|>\varepsilon\}$ and to $\{0<|x|\le\varepsilon\}$. The first is a finite measure, because $\int(|x|^2\wedge1)\,d\nu<\infty$; it is symmetric and has a density; and its mass m_ε increases to $\nu(\mathbb{R}^d\setminus\{0\})=\infty$ as $\varepsilon\downarrow0$. The second is a symmetric Lévy measure. Let μ^ε be the law of (S.16) for $(0,\nu_\varepsilon^\le)$, and put $\pi_\varepsilon:=e^{-m_\varepsilon}\sum_{n\ge0}(\nu_\varepsilon^>)^{*n}/n!$. As in the proof of items (a) and (b) of clause (2), which uses of the measure only that it is finite, symmetric and has a density, π_ε is a probability measure with the transform $\exp[-\int(1-\cos\langle\xi,x\rangle)\,\nu_\varepsilon^>(dx)]$, and $\pi_\varepsilon=e^{-m_\varepsilon}\delta_0+\eta_\varepsilon$ with η_ε absolutely continuous. The transforms of π_ε and μ^ε multiply to (3.8) with $a=0$, so $\mu_{s,t}=\pi_\varepsilon*\mu^\varepsilon$. Let N be a Lebesgue-null Borel set. Then $(\eta_\varepsilon*\mu^\varepsilon)(N)=\int\eta_\varepsilon(N-y)\,\mu^\varepsilon(dy)=0$, so

$$\mu_{s,t}(N)=e^{-m_\varepsilon}\mu^\varepsilon(N)\le e^{-m_\varepsilon}.$$

Letting $\varepsilon\downarrow0$ gives $\mu_{s,t}(N)=0$. So $\mu_{s,t}$ is absolutely continuous, and the origin is a Lebesgue-null set.

Item (c). Let $a>0$. A gauge is increasing, so $v=a(L^2-\ell^2)>0$. By (3.8), $0<\widehat{\mu}_{s,t}(\xi)\le e^{-v|\xi|^2/2}$, so $\int|\widehat{\mu}_{s,t}(\xi)|\,|\xi|^n\,d\xi<\infty$ for every integer $n\ge0$. By C1 at n, $\mu_{s,t}$ has a density g_n of class C^n whose partial derivatives of orders $0,\dots,n$ tend to 0 at infinity. Two continuous

densities of one measure are equal, so the g_n are one function p. It is of class C^n for every n, and each of its partial derivatives is continuous and tends to 0 at infinity, hence is bounded.

The last sentence. If $a = 0$ and $c < \infty$, item (b) of clause (2) gives the point mass. If $a = 0$ and $c = \infty$, or $a > 0$, items (b) and (c) give a density, which excludes a point mass. □

S3.4 The dimension filtration

The subsection has the proof of the lemma on the shadow, then the separating family: three lemmas on its critical values, the proofs of these and of the lemma on a delay datum, the top of the filtration, and the proof of Theorem 3.25. It ends with a second proof of Proposition 3.22(3).

Proof of Lemma A.5. The integral over the last coordinate. Let $h : (0,\infty) \to [0,\infty)$ be Borel and put $\bar h(r) := C_d \int_{\mathbb R} h(r\sqrt{1+u^2})\,(1+u^2)^{-(d+1)/2}\,du \in [0,\infty]$ for $r > 0$. Let $g \ge 0$ be Borel on $\mathbb R^d$ with $g(0) = 0$, and write $z = (x, y)$ with $x \in \mathbb R^d$ and $y \in \mathbb R$. By Tonelli's theorem,

$$\int_{\mathbb R^{d+1}} g(x)\,\frac{h(|z|)}{|S^d|\,|z|^{d+1}}\,dz = \int_{\mathbb R^d} g(x)\Big[\int_{\mathbb R} \frac{h\big(\sqrt{|x|^2+y^2}\big)}{|S^d|\,(|x|^2+y^2)^{(d+1)/2}}\,dy\Big]dx.$$

For $x \ne 0$ put $r = |x|$ and substitute $y = ru$. The inner integral becomes

$$(|S^d|\,r^d)^{-1}\int_{\mathbb R} h(r\sqrt{1+u^2})\,(1+u^2)^{-(d+1)/2}\,du = \bar h(r)\,(|S^{d-1}|\,r^d)^{-1}.$$

As $g(0) = 0$,

$$\int_{\mathbb R^{d+1}} g(\Pi z)\,\frac{h(|z|)}{|S^d|\,|z|^{d+1}}\,dz = \int_{\mathbb R^d} g(x)\,\frac{\bar h(|x|)}{|S^{d-1}|\,|x|^d}\,dx, \tag{S.17}$$

both sides in $[0,\infty]$.

Clause (1). Take $h = \mathbf 1_{(1,2)}$ and $g = 1$ off the origin in (S.17). In polar coordinates of $\mathbb R^{d+1}$ the left side is $\int_1^2 dt/t = \log 2$; the set $\{\Pi z = 0\}$ is a line, which is null because $d + 1 \ge 2$. In polar coordinates of $\mathbb R^d$, and by Tonelli's theorem, the right side is

$$\begin{aligned}\int_0^\infty \bar h(r)\,\frac{dr}{r} &= C_d\int_{\mathbb R}(1+u^2)^{-(d+1)/2}\Big[\int_0^\infty \mathbf 1_{(1,2)}\big(r\sqrt{1+u^2}\big)\frac{dr}{r}\Big]du\\ &= C_d\int_{\mathbb R}(1+u^2)^{-(d+1)/2}\,du\cdot\log 2,\end{aligned}$$

the inner integral being over an interval whose endpoints have the ratio 2. So $C_d\int_{\mathbb R}(1+u^2)^{-(d+1)/2}du = 1$. As k' is nonincreasing and nonnegative, $0 \le k'(r\sqrt{1+u^2}) \le k'(r)$, so $0 \le \bar k(r) \le k'(r) < \infty$.

Clause (2). For each u, $r \mapsto k'(r\sqrt{1+u^2})$ is nonincreasing, so $\bar k$ is. Let $r_0 > 0$. A nonincreasing function has at most countably many discontinuities, and for each of them, t, at most two real u have $r_0\sqrt{1+u^2} = t$. So for all u outside a countable set, $k'(r\sqrt{1+u^2}) \to k'(r_0\sqrt{1+u^2})$ as $r \to r_0$. For $r \ge r_0/2$ the integrand of $\bar k(r)$ is at most $k'(r_0/2)\,(1+u^2)^{-(d+1)/2}$, which is integrable. By dominated convergence $\bar k(r) \to \bar k(r_0)$. As $r \downarrow 0$, $k'(r\sqrt{1+u^2})$ increases to $k'(0+)$ for every u, so by monotone convergence and clause (1), $\bar k(r) \to k'(0+)$, in $[0,\infty]$.

Clause (3). By Definition 3.10, ν' has the density $k'(|z|)\,(|S^d|\,|z|^{d+1})^{-1}$ on $\mathbb R^{d+1}\setminus\{0\}$. Let $\bar\nu$ be the measure on $\mathbb R^d\setminus\{0\}$ with density $\bar k(|x|)\,(|S^{d-1}|\,|x|^d)^{-1}$. Then (S.17) with $h = k'$ reads $\int g(\Pi z)\,\nu'(dz) = \int g\,d\bar\nu$ for every nonnegative Borel g vanishing at the origin. The set $\{z \ne 0 : \Pi z = 0\}$ lies in a line, which is Lebesgue-null in $\mathbb R^{d+1}$ because $d + 1 \ge 2$, and ν' has a density. So ν' gives it the mass 0, and the image of ν' under Π gives the origin no mass. Take $g(x) = |x|^2 \wedge 1$. As $|\Pi z| \le |z|$,

$$\begin{aligned}\int_0^1 \bar k(r)\,r\,dr + \int_1^\infty \bar k(r)\,\frac{dr}{r} &= \int_{\mathbb R^d}\big(|x|^2\wedge 1\big)\,\bar\nu(dx) \le \int_{\mathbb R^{d+1}}\big(|z|^2\wedge 1\big)\,\nu'(dz)\\ &= \int_0^1 k'(r)\,r\,dr + \int_1^\infty k'(r)\,\frac{dr}{r} < \infty,\end{aligned}$$

the first equality by polar coordinates in $\mathbb{R}^d$ and the last by (3.2) in dimension $d+1$. With clauses (1) and (2), $\bar{k}$ is finite, nonnegative and nonincreasing and satisfies the two conditions of Definition 3.10. So $(a,\bar{k})$ is an isotropic origin datum in dimension d, and its Lévy measure is $\bar{\nu}$. For $\xi \in \mathbb{R}^d$ take $g(x) = 1 - \cos\langle \xi, x\rangle$. Since $\langle(\xi,0),z\rangle = \langle \xi, \Pi z\rangle$ and $|(\xi,0)| = |\xi|$,

$$F'\big((\xi,0)\big) = \tfrac{1}{2}a|\xi|^2 + \int_{\mathbb{R}^{d+1}} \big(1 - \cos\langle \xi, \Pi z\rangle\big)\, \nu'(dz) = \tfrac{1}{2}a|\xi|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle \xi, x\rangle\big)\, \bar{\nu}(dx),$$

which is the exponent (3.3) of $(a,\bar{k})$ at ξ. If $F'(\zeta) = F(|\zeta|)$, the left side is $F(|\xi|)$. □

Lemmas S.10 and S.11 give the strict decrease of the critical values w_d and bounds of their order, and Lemma S.12 gives their limit.

Lemma S.10 (the critical values are strictly decreasing)**.** *For a real parameter $d \geq 1$ put $m = (d+1)/2$, $\lambda_d = 2^{-d/2}$ and $h_d(z) = z^m(e^{-z} - \lambda_d e^{-z/2})$, and let $M_d = \sup_{z>0} h_d(z)$ as in (3.10). Then $M_d \in (0,\infty)$ and the supremum is attained at a point of $(0, d\log 2)$. Consequently $w_d = \Gamma(m)/(2M_d)$ is a well-defined positive number, and when d is an integer, $k_d^{(w)}$ is nonincreasing on $(0,\infty)$ exactly when $w \leq w_d$. Moreover*

$$M_{d+1} > \sqrt{m}\, M_d \qquad \text{and} \qquad w_{d+1} < w_d$$

for every $d \geq 1$. In particular $w_1 > w_2 > w_3 > \cdots > 0$, so $(w_{d+1}, w_d]$ is a nonempty interval for every integer $d \geq 1$.

Lemma S.11 (the critical values tend to 0 at the rate $d^{-1/2}$)**.** *For a real parameter $d \geq 1$ let m, h_d, M_d and $w_d = \Gamma(m)/(2M_d)$ be as in Lemma S.10, and put*

$$\varepsilon_d := 2^{-d/2} e^{m/2} = e^{1/4}\Big(\frac{e}{4}\Big)^{d/4}, \qquad K := \sqrt{\pi/2} + \sqrt{\pi} + 4.$$

Then for every $d \geq 1$

$$w_d \;\geq\; \frac{e^{-1/2}}{2\sqrt{m}},$$

and for every $d \geq 1$ with $\varepsilon_d < 1$

$$w_d \;\leq\; \frac{K}{2(1-\varepsilon_d)\sqrt{m}}.$$

Since $e < 4$, ε_d decreases strictly to 0 as $d \to \infty$. Hence $w_d \to 0$, and w_d is of exact order $d^{-1/2}$: $\sqrt{m}\, w_d \geq e^{-1/2}/2$ for every $d \geq 1$, and $\sqrt{m}\, w_d \leq K$ for every d with $\varepsilon_d \leq \frac{1}{2}$, which is every large d. In particular, for every fixed $w > 0$ there is a D with $w > w_d$ for all $d \geq D$.

Neither constant of the lemma is the limit of $\sqrt{m}\, w_d$. The next lemma gives it.

Lemma S.12 (the limit of the critical values)**.** *For a real parameter $d \geq 1$ let $m = (d+1)/2$, M_d, $w_d = \Gamma(m)/(2M_d)$ and $\varepsilon_d = 2^{-d/2}e^{m/2}$ be as in Lemma S.11, and put*

$$S_m := \Gamma(m)\, m^{1/2-m}\, e^m.$$

1. (Stirling's formula.) *$S_m \to \sqrt{2\pi}$ as $m \to \infty$ through the real numbers.*

2. (The critical value against the Gamma function.) *For every $d \geq 1$,*

$$\sqrt{m}\, w_d \;\geq\; \tfrac{1}{2}\, S_m,$$

and for every $d \geq 1$ with $\varepsilon_d < 1$,

$$\sqrt{m}\, w_d \;\leq\; \frac{S_m}{2\,(1-\varepsilon_d)}.$$

3. (The limit.) *As $d \to \infty$ through the real numbers,*

$$\sqrt{m}\, w_d \to \sqrt{\pi/2} \qquad \textit{and} \qquad \sqrt{d}\, w_d \to \sqrt{\pi}.$$

The limit is approached slowly, and not monotonically. Computed values of $\sqrt{d}\, w_d$ are 12.61 at $d = 1$ and 2.056 at $d = 20$, against $\sqrt{\pi} \approx 1.7725$. They fall below $\sqrt{\pi}$ between $d = 55$ and $d = 56$, are least at $d = 81$, where the value is 1.7661, and increase from there on as far as they were computed, to $d = 300$. The statements are independent of these values.

Proof of Lemma S.10. The closed form and the criterion. [3, App. A], restated (the paragraph *the polar profiles in closed form*, carried out there for every parameter and every integer $d \geq 1$). Let $d \geq 1$ be an integer and $w > 0$, and write $P(s,x) = \Gamma(s)^{-1} \int_0^x v^{s-1} e^{-v} dv$ for the regularized lower incomplete gamma function. Split (3.4) at the two summands of $k_{\mathrm{I}}^{(w)}$. For the power, $k_{\mathrm{I}}(r^2/2v) = (r^2/2v)^{-1/2}$ turns the integral into a Gamma integral and gives $2^{3/2} r^{-1} \Gamma(m)/\Gamma(d/2)$. For the bump, $\mathbf{1}_{[1,2]}(r^2/2v) = \mathbf{1}_{[r^2/4,\, r^2/2]}(v)$. Hence [3, (A.1)]:

$$k_d^{(w)}(r) = \frac{2^{3/2}\,\Gamma(m)}{\Gamma(d/2)}\,\frac{1}{r} + 2w\Big[P\Big(\frac{d}{2}, \frac{r^2}{2}\Big) - P\Big(\frac{d}{2}, \frac{r^2}{4}\Big)\Big], \qquad r > 0, \tag{S.18}$$

a smooth function on $(0,\infty)$. Since $\partial_x P(s,x) = x^{s-1}e^{-x}/\Gamma(s)$, the derivative of $P(d/2, r^2/2b)$ in r is $2\,(r\,\Gamma(d/2))^{-1}(r^2/2b)^{d/2} e^{-r^2/2b}$ for $b > 0$. With $b = 1$ and $b = 2$, and $z = r^2/2$,

$$r^2\,\Gamma(d/2)\,\frac{d}{dr} k_d^{(w)}(r) = -2^{3/2}\Gamma(m) + 4wr\Big(z^{d/2}e^{-z} - \big(\tfrac{z}{2}\big)^{d/2} e^{-z/2}\Big).$$

As $r = \sqrt{2z}$, the last term is $4\sqrt{2}\, w\, z^m \big(e^{-z} - 2^{-d/2} e^{-z/2}\big)$. So the derivative is nonpositive at r if and only if [3, (A.2)] holds:

$$w\, z^m \big(e^{-z} - 2^{-d/2} e^{-z/2}\big) \leq \tfrac{1}{2}\Gamma(m), \qquad z = \tfrac{1}{2} r^2. \tag{S.19}$$

The supremum, and the equivalence. h_d is smooth on $(0,\infty)$ with $h_d(0+) = 0$, and $e^{-z} - \lambda_d e^{-z/2} > 0$ exactly when $e^{-z/2} > 2^{-d/2}$, that is exactly on $(0, d\log 2)$, an interval that is nonempty because $d \geq 1$. So $h_d > 0$ somewhere, $h_d \leq 0$ on $[d\log 2, \infty)$, and $h_d(z) \leq z^m e^{-z} \to 0$ as $z \to \infty$. A positive continuous function vanishing at both ends of $(0,\infty)$ attains its supremum, so $M_d = h_d(z^*) \in (0,\infty)$ for some $z^* \in (0, d\log 2)$. For an integer d, (S.19) reads $w\, h_d(z) \leq \frac{1}{2}\Gamma(m)$ at $z = r^2/2$. So the criterion holds at every $r > 0$ exactly when $w M_d \leq \frac{1}{2}\Gamma(m)$, that is exactly when $w \leq w_d$. A differentiable function on $(0,\infty)$ is nonincreasing exactly when its derivative is nonpositive throughout, which settles the equivalence for $k_d^{(w)}$.

The first-order condition. Let $z^* \in (0, d\log 2)$ be a maximizer and put $q := \lambda_d e^{z^*/2}$, so that $\lambda_d e^{-z^*/2} = q\, e^{-z^*}$. Since $z^* < d\log 2$, $q \in (0,1)$. Then $M_d = (z^*)^m e^{-z^*}(1-q) > 0$. The maximizer is interior, so $h_d'(z^*) = 0$, where

$$h_d'(z) = m z^{m-1}\big(e^{-z} - \lambda_d e^{-z/2}\big) + z^m\big(-e^{-z} + \tfrac{1}{2}\lambda_d e^{-z/2}\big).$$

Dividing $h_d'(z^*) = 0$ by $(z^*)^{m-1} e^{-z^*} > 0$ turns it into $m(1-q) = z^*(1 - q/2)$. As $1 - q/2 > 0$,

$$z^* = m\,\frac{1-q}{1-q/2} \qquad \text{and} \qquad \sqrt{z^*} = \sqrt{m}\,\sqrt{\frac{1-q}{1-q/2}}.$$

Testing the next dimension at that point. Since $\lambda_{d+1} = 2^{-1/2}\lambda_d$ and $m_{d+1} = m + \frac{1}{2}$,

$$h_{d+1}(z^*) = (z^*)^{m+1/2}\big(e^{-z^*} - 2^{-1/2} q\, e^{-z^*}\big) = \sqrt{z^*}\,\big(1 - q/\sqrt{2}\big)\,(z^*)^m e^{-z^*} = \sqrt{z^*}\,\frac{1 - q/\sqrt{2}}{1-q}\, M_d,$$

so that, substituting for $\sqrt{z^*}$,

$$M_{d+1} \;\geq\; h_{d+1}(z^*) \;=\; \sqrt{m}\, M_d \, \frac{1 - q/\sqrt{2}}{\sqrt{(1-q)(1-q/2)}}.$$

The last fraction exceeds 1, because its numerator and denominator are positive and

$$\bigl(1 - q/\sqrt{2}\bigr)^2 - (1-q)\bigl(1-q/2\bigr) = \Bigl(1 - \sqrt{2}\,q + \frac{q^2}{2}\Bigr) - \Bigl(1 - \frac{3}{2}q + \frac{q^2}{2}\Bigr) = \Bigl(\frac{3}{2} - \sqrt{2}\Bigr)q > 0$$

for $q > 0$, because the constant $\frac{3}{2} - \sqrt{2} = (1 - 2^{-1/2})^2$ is positive. (It is the reciprocal of $6 + 4\sqrt{2}$.) Hence $M_{d+1} > \sqrt{m}\, M_d$.

The Gamma factor. Γ is log-convex on $(0, \infty)$: for $x, y > 0$ and $\theta \in (0,1)$, Hölder's inequality applied to $t^{\theta x + (1-\theta)y - 1} e^{-t} = (t^{x-1}e^{-t})^{\theta} (t^{y-1}e^{-t})^{1-\theta}$ with exponents $1/\theta$ and $1/(1-\theta)$ gives $\Gamma(\theta x + (1-\theta)y) \leq \Gamma(x)^{\theta}\Gamma(y)^{1-\theta}$. At $x = m$, $y = m+1$, $\theta = \frac{1}{2}$ this is $\Gamma(m + \frac{1}{2}) \leq \Gamma(m)^{1/2}\Gamma(m+1)^{1/2} = \sqrt{m}\,\Gamma(m)$, by $\Gamma(m+1) = m\Gamma(m)$. Therefore

$$\frac{1}{2w_{d+1}} = \frac{M_{d+1}}{\Gamma(m + \frac{1}{2})} > \frac{\sqrt{m}\, M_d}{\Gamma(m + \frac{1}{2})} \geq \frac{\sqrt{m}\, M_d}{\sqrt{m}\,\Gamma(m)} = \frac{1}{2w_d},$$

which is $w_{d+1} < w_d$. Every w_d is positive, so the chain $w_1 > w_2 > \cdots > 0$ and the nonemptiness of $(w_{d+1}, w_d\,]$ follow. □

Proof of Lemma S.11. Stirling's bounds. Let $m \geq 1$ and $\varphi(u) := u - \log(1+u)$ for $u > -1$, so that $\varphi(0) = 0$ and $\varphi'(u) = u/(1+u)$. The substitution $t = m(1+u)$ in $\Gamma(m+1) = \int_0^\infty t^m e^{-t}\, dt$ gives

$$\Gamma(m+1) = m^{m+1}e^{-m} \int_{-1}^{\infty} e^{-m\varphi(u)}\, du, \qquad \text{so} \qquad \Gamma(m) = m^m e^{-m}\, I_m, \quad I_m := \int_{-1}^{\infty} e^{-m\varphi(u)}\, du,$$

by $\Gamma(m+1) = m\Gamma(m)$. Four comparisons for φ follow by comparing derivatives and integrating from 0. Put $\psi(u) := u^2/(2(1+u))$ for $u \geq 0$.

- On $(-1, 0]$, $\varphi(u) \geq u^2/2$: there $\varphi'(s) = s/(1+s) \leq s$, because $s \leq 0 < 1 + s \leq 1$. Integrate over $[u, 0]$.
- On $[0, \infty)$, $\varphi(u) \geq \psi(u)$: $\psi(0) = 0$ and $\psi'(s) = (s^2 + 2s)/(2(1+s)^2) \leq (2s^2 + 2s)/(2(1+s)^2) = \varphi'(s)$.
- $\psi(u) \geq u^2/4$ on $[0, 1]$, since $1 + u \leq 2$ there, and $\psi(u) \geq u/4$ on $[1, \infty)$, since $u/(1+u) \geq \frac{1}{2}$ there.
- On $[0, \infty)$, $\varphi(u) \leq u^2/2$, because $\varphi'(s) \leq s$ there.

With $\int_0^\infty e^{-as^2} ds = \frac{1}{2}\sqrt{\pi/a}$ for $a > 0$, the lower bounds for φ, on $(-1, 0]$, on $[0, 1]$ and on $[1, \infty)$, give

$$I_m \leq \int_{-\infty}^{0} e^{-mu^2/2}\, du + \int_0^{\infty} e^{-mu^2/4}\, du + \int_1^{\infty} e^{-mu/4}\, du = \sqrt{\frac{\pi}{2m}} + \sqrt{\frac{\pi}{m}} + \frac{4}{m} e^{-m/4} \leq \frac{K}{\sqrt{m}},$$

using $4/m \leq 4/\sqrt{m}$ for $m \geq 1$. The integrand is positive and $1/\sqrt{m} \leq 1$, so the upper bound $\varphi(u) \leq u^2/2$ gives

$$I_m \geq \int_0^{1/\sqrt{m}} e^{-mu^2/2}\, du \geq \frac{1}{\sqrt{m}}\, e^{-1/2}.$$

So $e^{-1/2}\, m^{m-1/2} e^{-m} \leq \Gamma(m) \leq K\, m^{m-1/2} e^{-m}$ for every $m \geq 1$, and $m = (d+1)/2 \geq 1$ because $d \geq 1$.

M_d from above. Since $2^{-d/2}e^{-z/2} > 0$, $h_d(z) < z^m e^{-z}$, and $z \mapsto z^m e^{-z}$ has derivative $z^{m-1}e^{-z}(m-z)$, so it is largest at $z = m$. Hence $M_d \le m^m e^{-m}$, and by Lemma S.10 $M_d > 0$, so

$$w_d = \frac{\Gamma(m)}{2M_d} \ge \frac{e^{-1/2}\, m^{m-1/2} e^{-m}}{2\, m^m e^{-m}} = \frac{e^{-1/2}}{2\sqrt{m}}.$$

M_d from below. Evaluating at $z = m$,

$$M_d \ge h_d(m) = m^m\big(e^{-m} - 2^{-d/2}e^{-m/2}\big) = m^m e^{-m}\,(1 - \varepsilon_d),$$

and $\varepsilon_d = 2^{-d/2}e^{(d+1)/4} = e^{1/4}(e/4)^{d/4}$. When $\varepsilon_d < 1$ this is positive, and

$$w_d = \frac{\Gamma(m)}{2M_d} \le \frac{K\, m^{m-1/2} e^{-m}}{2\, m^m e^{-m}(1-\varepsilon_d)} = \frac{K}{2(1-\varepsilon_d)\sqrt{m}}.$$

The limit. Since $e < 4$, $e/4 \in (0,1)$ and ε_d decreases strictly to 0. In particular $\varepsilon_d \le \frac{1}{2}$ for all large d, and there the upper bound reads $\sqrt{m}\, w_d \le K$. Since $m = (d+1)/2 \to \infty$, $w_d \to 0$ follows. For fixed $w > 0$ the upper bound falls below w from some D on, and so does w_d. □

Proof of Lemma S.12. From the proof of Lemma S.11: with $\varphi(u) := u - \log(1+u)$ for $u > -1$,

$$\Gamma(m) = m^m e^{-m}\, I_m, \qquad I_m := \int_{-1}^{\infty} e^{-m\varphi(u)}\, du,$$

so that $S_m = \sqrt{m}\, I_m$. Also $\varphi(u) \ge u^2/2$ on $(-1, 0]$, $\varphi(u) \ge u^2/4$ on $[0, 1]$ and $\varphi(u) \ge u/4$ on $[1, \infty)$. And $0 < M_d \le m^m e^{-m}$ for every $d \ge 1$, $M_d \ge m^m e^{-m}(1 - \varepsilon_d)$, and $\varepsilon_d \to 0$ as $d \to \infty$.

Clause (1). The substitution $u = v/\sqrt{m}$ gives

$$S_m = \sqrt{m}\, I_m = \int_{\mathbb{R}} f_m(v)\, dv,$$

where $f_m(v) := e^{-m\varphi(v/\sqrt{m})}$ for $v > -\sqrt{m}$ and $f_m(v) := 0$ for $v \le -\sqrt{m}$. *The pointwise limit.* Since $\varphi'(s) = s/(1+s)$ and $\varphi(0) = 0$, $\varphi(u) - \frac{1}{2}u^2 = -\int_0^u s^2/(1+s)\, ds$. For $|u| < 1$ the denominator is at least $1 - |u|$ on the interval of integration, so

$$\big|\varphi(u) - \tfrac{1}{2}u^2\big| \le \frac{|u|^3}{3\,(1-|u|)}.$$

Fix a real v. For $m > v^2$, the point $u = v/\sqrt{m}$ has $|u| < 1$, so $v > -\sqrt{m}$ and

$$\big|m\,\varphi(v/\sqrt{m}) - \tfrac{1}{2}v^2\big| \le \frac{|v|^3}{3\sqrt{m}\,\big(1 - |v|/\sqrt{m}\big)},$$

which tends to 0 as $m \to \infty$. Hence $f_m(v) \to e^{-v^2/2}$ for every real v. *The domination.* Let $m \ge 1$ and $v > -\sqrt{m}$, and apply the three comparisons at $u = v/\sqrt{m}$. For $-\sqrt{m} < v \le 0$, $m\varphi(u) \ge v^2/2$. For $0 \le v \le \sqrt{m}$, $m\varphi(u) \ge v^2/4$. For $v \ge \sqrt{m}$, $m\varphi(u) \ge \sqrt{m}\, v/4 \ge v/4$. So

$$0 \le f_m(v) \le g(v) := e^{-v^2/4} + e^{-|v|/4} \qquad (v \in \mathbb{R},\ m \ge 1),$$

and g is integrable on $\mathbb{R}$. *The limit.* By dominated convergence, along every sequence $m_n \to \infty$, $S_{m_n} \to \int_{\mathbb{R}} e^{-v^2/2}\, dv = \sqrt{2\pi}$. So $S_m \to \sqrt{2\pi}$ as $m \to \infty$.

Clause (2). $w_d = \Gamma(m)/(2M_d) = m^m e^{-m} I_m/(2M_d)$. As $0 < M_d \le m^m e^{-m}$, $w_d \ge I_m/2$, that is $\sqrt{m}\, w_d \ge S_m/2$. If $\varepsilon_d < 1$, then $M_d \ge m^m e^{-m}(1-\varepsilon_d) > 0$ gives $w_d \le I_m/(2(1-\varepsilon_d))$.

Clause (3). As $d \to \infty$, $m = (d+1)/2 \to \infty$ and $\varepsilon_d \to 0$, so $\varepsilon_d < 1$ for all large d, and both sides of clause (2) tend to $\sqrt{2\pi}/2 = \sqrt{\pi/2}$ by clause (1). Hence $\sqrt{m}\, w_d \to \sqrt{\pi/2}$. Finally $\sqrt{d}\, w_d = \sqrt{d/m}\,\sqrt{m}\, w_d$ and $d/m = 2d/(d+1) \to 2$, so $\sqrt{d}\, w_d \to \sqrt{2}\,\sqrt{\pi/2} = \sqrt{\pi}$. □

Proof of Lemma A.6. For $u > 0$ let $g_u(x) = (2\pi u)^{-d/2} e^{-|x|^2/2u}$ be the heat kernel of Lemma S.8. It is even and its transform is $e^{-u|\xi|^2/2}$, so

$$1 - e^{-u|\xi|^2/2} = \int_{\mathbb{R}^d} \bigl(1 - \cos\langle \xi, x\rangle\bigr)\, g_u(x)\, dx.$$

Integrate against $k_{\mathrm{I}}(u)\, du/u$ and exchange the integrals by Tonelli's theorem, which applies because the integrand is nonnegative:

$$F_{\mathrm{I}}\bigl(\tfrac{1}{2}|\xi|^2\bigr) - \tfrac{1}{2} b_0 |\xi|^2 = \int_{\mathbb{R}^d} \bigl(1 - \cos\langle \xi, x\rangle\bigr)\, f(x)\, dx, \qquad f(x) := \int_0^\infty g_u(x)\, k_{\mathrm{I}}(u)\, \frac{du}{u},$$

with f measurable and valued in $[0, \infty]$. Fix $x \neq 0$ and put $r = |x|$. Under the substitution $u = r^2/2v$, du/u becomes dv/v with the orientation reversed, and $g_u(x) = \pi^{-d/2} r^{-d} v^{d/2} e^{-v}$. So, by (3.4) and $|S^{d-1}| = 2\pi^{d/2}/\Gamma(d/2)$,

$$f(x) = \pi^{-d/2} r^{-d} \int_0^\infty v^{d/2-1} e^{-v}\, k_{\mathrm{I}}\Bigl(\frac{r^2}{2v}\Bigr) dv = \frac{k_d(r)}{|S^{d-1}|\, r^d}.$$

This is the identity of exponents. The value $k_d(r)$ is finite. For fixed $x \neq 0$ the factor $g_u(x)/u$ is bounded for $u \in (0, 1]$, because $e^{-r^2/2u}$ decays faster than every power of $1/u$ grows, and $g_u(x) \leq (2\pi)^{-d/2}$ for $u \geq 1$. The two conditions on k_{I} then bound the integral defining $f(x)$ over $(0, 1]$ and over $[1, \infty)$. Measurability of k_d is Tonelli's theorem again. For the integrability conditions, $\int |x|^2 g_u(x)\, dx = du$ and $\int g_u = 1$ give $\int (|x|^2 \wedge 1)\, g_u(x)\, dx \leq \min\{du, 1\}$. By Tonelli's theorem

$$\int_{\mathbb{R}^d} (|x|^2 \wedge 1)\, f(x)\, dx \leq d \int_0^1 k_{\mathrm{I}}(u)\, du + \int_1^\infty k_{\mathrm{I}}(u)\, \frac{du}{u} < \infty,$$

and in polar coordinates the left side is $\int_0^1 k_d(r)\, r\, dr + \int_1^\infty k_d(r)\, dr/r$. If k_d is also nonincreasing, (b_0, k_d) satisfies Definition 3.10. Its Lévy measure has the density f, and its exponent (3.3) is $F_{\mathrm{I}}(\frac{1}{2}|\cdot|^2)$ by the identity above. □

The top of the filtration, $\mathcal{A}_\infty = \mathcal{S}$, is restated from the cone paper as Proposition 2.6(3). The formal development proves it, as the next lemma, from Schoenberg's radial theorem, C6 (Theorem S.6).

Lemma S.13 (the top of the filtration is the subordinated slice)**.** *In the notation of Definition 3.24, $\mathcal{A}_\infty = \mathcal{S}$.*

The route is that of the proof of [3, Prop. 4.9(4)] in the cone paper, which applies the same theorem to the same two functions. Three steps, marked (a), (b) and (c), stand in the place of three facts which that proof cites. Each is stated and proved after the two inclusions.

Proof of Lemma S.13. For $\mathcal{S} \subseteq \mathcal{A}_\infty$, let $F = F_{\mathrm{I}}(\cdot^2/2)$ with F_{I} causally admissible. F is admissible by Proposition 2.5, and since the delay profile of F_{I} is nonincreasing, so is its polar profile (3.4) in every dimension d; Lemma A.6 and Proposition 3.11(1), in its converse direction, then put F in $\mathcal{A}_d$.

For $\mathcal{A}_\infty \subseteq \mathcal{S}$, let $F \in \mathcal{A}_\infty$. Being admissible, F is continuous and nonnegative on $[0, \infty)$, with $F(0) = 0$. Put $f(\sigma) = F(\sqrt{2\sigma})$ and, for $0 < c \leq 1$, $g_c(\sigma) = f(\sigma) - f(c^2\sigma)$. Fix a dimension $d \geq 1$. Proposition 3.11(1), in its forward direction, gives an isotropic origin datum (a_d, k_d) in dimension d with

$$F(|\eta|) = \tfrac{1}{2} a_d |\eta|^2 + \int_{\mathbb{R}^d} \bigl(1 - \cos\langle \eta, x\rangle\bigr)\, \nu_d(dx),$$

where ν_d is its Lévy measure, of density $k_d(|x|)/(|S^{d-1}|\, |x|^d)$. The image $(c\,\cdot)_*\nu_d$ of ν_d under $x \mapsto cx$ has the density $k_d(|x|/c)/(|S^{d-1}|\, |x|^d)$, so $(c\,\cdot)_*\nu_d \leq \nu_d$, because k_d is nonincreasing.

And $F(c|\eta|) = \frac{1}{2}a_d c^2|\eta|^2 + \int(1-\cos\langle\eta,x\rangle)\,(c\,\cdot)_*\nu_d(dx)$. By step (a), $\eta \mapsto F(|\eta|)$ and $\eta \mapsto F(|\eta|) - F(c|\eta|)$ are negative definite on $\mathbb{R}^d$. So are $\xi \mapsto f(|\xi|^2)$ and $\xi \mapsto g_c(|\xi|^2)$, which are these two functions at the point $\sqrt{2}\,\xi$. They are continuous, and in dimension 0 they vanish. Also $f \geq 0$, and $g_c \geq 0$: for a function g with $g(0) = 0$, the inequality of C6 with one point ξ_1, $|\xi_1|^2 = \sigma$, and $c_1 = 1$ reads $2g(\sigma) \geq 0$. So C6 applies to f and to each g_c. It gives $a, b \geq 0$ and a measure m on $(0,\infty)$ with $\int(1\wedge t)\,m(dt) < \infty$ and $f(\sigma) = a + b\sigma + \int(1-e^{-\sigma t})\,m(dt)$ for $\sigma > 0$, and likewise a_c, b_c and m_c for g_c. Here $a = 0$, since $a \leq f(\sigma)$ and $f(\sigma) \to F(0) = 0$ as $\sigma \downarrow 0$. Let $D_\lambda m$ be the image of m under $t \mapsto \lambda t$, so that $f(c^2\sigma) = bc^2\sigma + \int(1-e^{-\sigma t})\,D_{c^2}m(dt)$. Then $f = f(c^2\,\cdot) + g_c$ reads

$$b\sigma + \int \big(1-e^{-\sigma t}\big)\,m(dt) = a_c + (bc^2 + b_c)\,\sigma + \int \big(1-e^{-\sigma t}\big)\,(D_{c^2}m + m_c)(dt) \qquad (\sigma > 0),$$

with both sides finite. By step (b), $m = D_{c^2}m + m_c$, so $D_{c^2}m \leq m$. As c ranges over $(0,1]$, so does c^2: $D_\lambda m \leq m$ for every $\lambda \in (0,1]$. Since $\int(1\wedge t^2)\,m(dt) \leq \int(1\wedge t)\,m(dt) < \infty$, step (c) gives $m(dt) = k(t)\,t^{-1}dt$ with $k \geq 0$ nonincreasing. The integrals $\int_0^1 k(t)\,dt$ and $\int_1^\infty k(t)\,t^{-1}dt$ are at most $\int(1\wedge t)\,m(dt)$. So the function $F_{\rm I}$ with drift b and delay profile k is causally admissible, and $F_{\rm I} = f$ on $(0,\infty)$. Hence $F = F_{\rm I}(\cdot^2/2)$, and $F \in \mathcal{S}$.

Step (a): a Lévy–Khintchine form, and the difference of two with ordered data, is negative definite. Let $\alpha' \leq \alpha$ be real numbers, and let $L' \leq L$ be measures on $\mathbb{R}^d$ such that $\int_{\mathbb{R}^d}(1-\cos\langle\eta,x\rangle)\,L(dx) < \infty$ for every $\eta \in \mathbb{R}^d$. Put

$$\psi(\eta) = \alpha|\eta|^2 + \int_{\mathbb{R}^d}\big(1-\cos\langle\eta,x\rangle\big)\,L(dx), \qquad \psi'(\eta) = \alpha'|\eta|^2 + \int_{\mathbb{R}^d}\big(1-\cos\langle\eta,x\rangle\big)\,L'(dx),$$

and $\phi = \psi - \psi'$. Then for every n, all $\eta_1,\dots,\eta_n \in \mathbb{R}^d$ and all $c_1,\dots,c_n \in \mathbb{C}$,

$$\sum_{j,k=1}^n \big(\phi(\eta_j) + \phi(\eta_k) - \phi(\eta_j - \eta_k)\big)\,c_j\,\overline{c_k} \geq 0.$$

The route uses this with $(\alpha, L) = (\frac{1}{2}a_d, \nu_d)$, and with $(\alpha', L') = (0,0)$ for $F(|\eta|)$ and $(\alpha', L') = (\frac{1}{2}a_d c^2, (c\,\cdot)_*\nu_d)$ for $F(|\eta|) - F(c|\eta|)$. For the proof, first let the coefficients be real, $a_1,\dots,a_n \in \mathbb{R}$. Write $\theta_j = \langle\eta_j, x\rangle$. Since $\cos(\theta_j - \theta_k) = \cos\theta_j\cos\theta_k + \sin\theta_j\sin\theta_k$,

$$(1-\cos\theta_j) + (1-\cos\theta_k) - \big(1-\cos(\theta_j-\theta_k)\big) = (1-\cos\theta_j)(1-\cos\theta_k) + \sin\theta_j\sin\theta_k,$$

and $|\eta_j|^2 + |\eta_k|^2 - |\eta_j - \eta_k|^2 = 2\langle\eta_j,\eta_k\rangle$. Each function $x \mapsto 1-\cos\langle\eta,x\rangle$ is integrable for L by hypothesis, so the integrals may be added, and

$$\sum_{j,k=1}^n a_j a_k\big(\psi(\eta_j) + \psi(\eta_k) - \psi(\eta_j-\eta_k)\big) = 2\alpha\Big|\sum_j a_j\eta_j\Big|^2 + \int_{\mathbb{R}^d} P_a(x)\,L(dx),$$

where

$$P_a(x) = \Big(\sum_j a_j\big(1-\cos\langle\eta_j,x\rangle\big)\Big)^2 + \Big(\sum_j a_j\sin\langle\eta_j,x\rangle\Big)^2 \;\geq\; 0.$$

P_a is integrable for L, being a finite combination of the integrable functions above. Since $L' \leq L$, every function integrable for L is integrable for L', so the same identity holds for ψ', with α' and L'; and $\int P_a\,dL' \leq \int P_a\,dL$, P_a being nonnegative. Subtracting,

$$\sum_{j,k} a_j a_k\big(\phi(\eta_j) + \phi(\eta_k) - \phi(\eta_j-\eta_k)\big) = 2(\alpha-\alpha')\Big|\sum_j a_j\eta_j\Big|^2 + \int P_a\,dL - \int P_a\,dL' \;\geq\; 0.$$

Now let the c_j be complex. The matrix $K_{jk} = \phi(\eta_j) + \phi(\eta_k) - \phi(\eta_j - \eta_k)$ is real and symmetric, because ϕ is even. Writing $c_j = a_j + ib_j$,

$$\sum_{j,k} K_{jk}\, c_j \overline{c_k} = \sum_{j,k} K_{jk}(a_j a_k + b_j b_k) + i \sum_{j,k} K_{jk}(b_j a_k - a_j b_k).$$

The imaginary part vanishes, since exchanging j and k changes its sign and leaves K unchanged, and the real part is the sum of the two real cases just treated, so it is nonnegative.

Step (b): the representation is unique. Let $a, b, a', b' \geq 0$ and let m, m' be measures on $(0,\infty)$ such that, for every $\sigma > 0$,

$$a + b\sigma + \int_{(0,\infty)} \bigl(1 - e^{-\sigma t}\bigr)\, m(dt) = a' + b'\sigma + \int_{(0,\infty)} \bigl(1 - e^{-\sigma t}\bigr)\, m'(dt) < \infty.$$

Then $a = a'$, $b = b'$ and $m = m'$. For the representations of C6 this is the last sentence of Theorem 3.2 of [7] (p. 21), which C6 leaves out. For the proof, write $G(\sigma)$ for the common value. For $t > 0$ and $\sigma > 0$,

$$\bigl(1 - e^{-\sigma t}\bigr) + \bigl(1 - e^{-(\sigma+2)t}\bigr) + e^{-\sigma t}\bigl(1 - e^{-t}\bigr)^2 = 2\bigl(1 - e^{-(\sigma+1)t}\bigr),$$

as expanding the square shows, and $b\sigma + b(\sigma + 2) = 2b(\sigma + 1)$. Integrating against m, every term being nonnegative,

$$G(\sigma) + G(\sigma + 2) + \int_{(0,\infty)} e^{-\sigma t}\bigl(1 - e^{-t}\bigr)^2\, m(dt) = 2G(\sigma + 1),$$

and the same with a', b' and m'. All the values of G are finite, so the measures $n(dt) = (1 - e^{-t})^2\, m(dt)$ and $n'(dt) = (1 - e^{-t})^2\, m'(dt)$ on $(0,\infty)$ have the same Laplace transform $\int e^{-\sigma t}\, n(dt) = 2G(\sigma + 1) - G(\sigma) - G(\sigma + 2)$ at every $\sigma > 0$, and it is finite. Such a transform determines the measure: $n_1(dt) := e^{-t}\, n(dt)$ is finite, its image under $t \mapsto e^{-t}$ is a finite measure on $(0,1)$ whose k-th moment is $\int e^{-(k+1)t}\, n(dt)$, and likewise for n'; so the two images have the same moments, finite measures on $[0,1]$ are determined by their moments (by the Weierstrass approximation theorem), and $n_1 = n_1'$, so $n = n'$. The density $(1 - e^{-t})^2$ is positive and finite on $(0,\infty)$, so dividing by it gives $m = m'$. Then the two integrals in the hypothesis are equal and finite, so $a + b\sigma = a' + b'\sigma$ for every $\sigma > 0$; at $\sigma = 1$ and $\sigma = 2$ this gives $b = b'$ and $a = a'$.

Step (c): a measure that decreases under dilation has a nonincreasing profile. Let m be a measure on $(0,\infty)$ with $\int_{(0,\infty)} (1 \wedge t^2)\, m(dt) < \infty$, and suppose that its image $D_\lambda m$ under $t \mapsto \lambda t$ is at most m for every $\lambda \in (0,1]$. Then $m(dt) = k(t)\, t^{-1} dt$ for a nonincreasing $k : (0,\infty) \to [0,\infty)$. This is the step from the dilation inequality to the profile in the proof of [2, Lem. 7.1], where it is not a numbered statement. For the proof, let $\tilde{m}$ be the image of m under log, a measure on $\mathbb{R}$. Its rays are finite: $\tilde{m}((\theta,\infty)) = m((e^\theta,\infty)) < \infty$, by $\int (1 \wedge t^2)\, m(dt) < \infty$. Dilation by $\lambda = e^{-h}$, $h > 0$, becomes translation by $-h$, so the hypothesis reads $\tilde{m}(A + h) \leq \tilde{m}(A)$ for every Borel A and every $h > 0$: $\tilde{m}$ decreases under every right shift. Put

$$\varphi(\theta) := \sup_{n \geq 0} 2^n\, \tilde{m}\bigl((\theta, \theta + 2^{-n}]\bigr) \in [0,\infty].$$

Each term is nonincreasing in θ, by the shift property, and the sequence is nondecreasing in n, because splitting $(\theta, \theta + 2\delta]$ at its midpoint and shifting the right half gives $\tilde{m}((\theta, \theta + 2\delta]) \leq 2\, \tilde{m}((\theta, \theta + \delta])$; so φ is nonincreasing. For $a < b$, monotone convergence and Tonelli's theorem give $\int_a^b \varphi = \lim_n 2^n \int |(a,b] \cap [x - 2^{-n}, x)|\, \tilde{m}(dx)$, which lies between $\tilde{m}((a + 2^{-n}, b])$ and $\tilde{m}((a, b + 2^{-n}])$; both tend to $\tilde{m}((a,b])$, the second because the rays are finite. So $\tilde{m}(d\theta) = \varphi(\theta)\, d\theta$. And φ is finite at every point: $\varphi(\theta) = \infty$ would force $\varphi = \infty$ on $(\theta - 1, \theta]$, φ being nonincreasing, and so $\tilde{m}((\theta - 1, \theta]) = \infty$, which the finite rays exclude. Returning to $t = e^\theta$, $m(dt) = k(t)\, t^{-1} dt$ with $k(t) := \varphi(\log t)$, finite, nonnegative and nonincreasing. $\square$

Proof of Theorem 3.25. Clause (1), $w \le w_D$ gives membership. Lemma A.6 applies to $D^{(w)}$ in dimension D, since it reads no monotonicity. It gives that the polar profile $k_D^{(w)}$ is finite and satisfies the two conditions of Definition 3.10, and that

$$F^{(w)}(|\xi|) = \int_{\mathbb{R}^D} \bigl(1 - \cos\langle \xi, x\rangle\bigr)\, \frac{k_D^{(w)}(|x|)}{|S^{D-1}|\,|x|^D}\, dx, \qquad \xi \in \mathbb{R}^D.$$

If $w \le w_D$, $k_D^{(w)}$ is nonincreasing (Lemma S.10). So $(0, k_D^{(w)})$ is an isotropic origin datum in dimension D with exponent $\xi \mapsto F^{(w)}(|\xi|)$, and $F^{(w)} \in \mathcal{A}_D$ by Proposition 3.11(2).

Clause (1), $w > w_D$ excludes it. Let $w > w_D$ and suppose $F^{(w)} \in \mathcal{A}_D$. By Proposition 3.11(2), $\xi \mapsto F^{(w)}(|\xi|)$ is the exponent of an isotropic origin datum (a, k_D) in dimension D, with Lévy measure ν_D and k_D nonincreasing. Let ν' be the measure on $\mathbb{R}^D \setminus \{0\}$ with density $k_D^{(w)}(|x|)\,(|S^{D-1}|\,|x|^D)^{-1}$. By polar coordinates in $\mathbb{R}^D$, which need no monotonicity of $k_D^{(w)}$, and the two integrability conditions,

$$\int \bigl(|x|^2 \wedge 1\bigr)\, \nu'(dx) = \int_0^1 k_D^{(w)}(r)\, r\, dr + \int_1^\infty k_D^{(w)}(r)\, \frac{dr}{r} < \infty,$$

and by the display $F^{(w)}(|\xi|) = \int (1 - \cos\langle \xi, x\rangle)\, \nu'(dx)$. By C3(iii) there are infinitely divisible laws with the triplets $(0, \nu', 0)$ and $(aI, \nu_D, 0)$. Both Lévy measures are symmetric, so the odd parts of the integrands integrate to 0, and both laws have the characteristic function $e^{-F^{(w)}(|\xi|)}$. So they are one law, and by the uniqueness of the triplet, C3(ii), $a = 0$ and $\nu' = \nu_D$. The two radial densities then agree almost everywhere on $\mathbb{R}^D$, so $k_D^{(w)} = k_D$ almost everywhere on $(0, \infty)$, by polar coordinates. On the other hand M_D is attained at some $z^* > 0$ (Lemma S.10), and $w > w_D$ means $w\, M_D > \frac{1}{2}\Gamma(m)$. So the criterion of Lemma S.10 fails at z^*: the smooth function $k_D^{(w)}$ has a positive derivative at $r_0 = \sqrt{2z^*}$, and is continuous and strictly increasing on an interval around r_0. Take $r < s$ in that interval. Points $u_n \downarrow r$ and $v_n \uparrow s$ at which $k_D^{(w)} = k_D$ give $k_D^{(w)}(u_n) = k_D(u_n) \ge k_D(v_n) = k_D^{(w)}(v_n)$, and by continuity $k_D^{(w)}(r) \ge k_D^{(w)}(s)$, a contradiction. So $F^{(w)} \notin \mathcal{A}_D$.

Clause (2). This is Lemma S.10: $w_{d+1} < w_d$ for every $d \ge 1$, and every w_d is positive.

Clause (3). By Lemma S.12(3), $\sqrt{d}\, w_d \to \sqrt{\pi}$ as $d \to \infty$ through the real numbers, hence through the integers.

Clause (4). The critical values decrease strictly (Lemma S.10). Let $w \in (w_{d+1}, w_d]$. For $D \le d$, $w_D \ge w_d \ge w$. For $D \ge d+1$, $w_D \le w_{d+1} < w$. If $w > w_1$, then $w > w_D$ for every D. Finally $w_D \to 0$ (Lemma S.11), so for $w \in (0, w_1]$ there is a largest integer $d \ge 1$ with $w_d \ge w$, and it is the one d with $w \in (w_{d+1}, w_d]$. With clause (1), this gives the first three sentences of the clause. For the last, let $d \ge 1$ be an integer. The interval $(w_{d+1}, w_d]$ is nonempty by clause (2), and for w in it $F^{(w)} \in \mathcal{A}_d \setminus \mathcal{A}_{d+1}$. The inclusion $\mathcal{A}_{d+1} \subseteq \mathcal{A}_d$ is part of Proposition 3.22(1). □

A second proof of Proposition 3.22(3). The clause says that for $\tau > 0$ and $b > 0$ the Cin ray $F(\omega) = b\ \mathrm{Cin}(\tau|\omega|)$, the admissible exponent with Gaussian coefficient 0 and folded profile $b\,\mathbf{1}_{(0,\tau)}$ (Proposition 2.7), belongs to no $\mathcal{A}_d$ with $d \ge 2$. The proof above takes it from clause (2) and the nesting of the classes. The proof below projects from $\mathbb{R}^d$ to the line in one step, compares the shadow on the line of a datum in dimension d with the step, and does not use the nesting. It is the route of the formal statement `no_cin_ray_in_the_plane`.

Second proof of clause (3). Let $d \ge 2$ and suppose $F \in \mathcal{A}_d$: then $\xi \mapsto F(|\xi|)$ is the exponent of a self-decomposable law on $\mathbb{R}^d$.

The datum. By Proposition 3.11(1), in its forward direction, $\xi \mapsto F(|\xi|)$ is the exponent F_d of an isotropic origin datum (a, k_d) in dimension d. Write $\nu := \nu_d$ for its Lévy measure. By

(3.3),

$$F(|\xi|) = \tfrac{1}{2}a|\xi|^2 + \int_{\mathbb{R}^d} \bigl(1 - \cos\langle \xi, x\rangle\bigr)\, \nu(dx), \qquad \xi \in \mathbb{R}^d, \tag{S.20}$$

and ν is rotation invariant, with the radial density $f_d(r) = k_d(r)\bigl(|S^{d-1}|r^d\bigr)^{-1}$, where k_d is nonincreasing and finite at every $r > 0$ (Definition 3.10).

The shadow on a line. Let $\Pi(x) = x_1$ and let $\bar\nu$ be the image of ν under Π, restricted to $\mathbb{R} \setminus \{0\}$. Since $|x_1| \le |x|$, $\int (1 \wedge y^2)\, \bar\nu(dy) \le \int (1 \wedge |x|^2)\, \nu(dx) < \infty$, so $\bar\nu$ is a Lévy measure on $\mathbb{R}$. It is symmetric, because ν is invariant under $x \mapsto -x$. Take $\xi = (\omega, 0, \dots, 0)$ in (S.20). The integrand vanishes where $x_1 = 0$, so

$$\tfrac{1}{2}a\omega^2 + \int_{\mathbb{R}} \bigl(1 - \cos(\omega y)\bigr)\, \bar\nu(dy) = b\ \mathrm{Cin}(\tau|\omega|), \qquad \omega \in \mathbb{R}.$$

The right side is the exponent of the datum $(0, b\,\mathbf{1}_{(0,\tau)})$ in dimension 1, whose Lévy measure is $\nu_1(dy) := b\,\mathbf{1}_{\{0<|y|<\tau\}}\,(2|y|)^{-1}dy$, read through (3.2) at $d = 1$: there $S^0 = \{\pm 1\}$ and $\int_{\mathbb{R}}(1 - \cos\omega y)\,\nu_1(dy) = b\int_0^\tau (1 - \cos\omega r)\,dr/r = b\ \mathrm{Cin}(\tau|\omega|)$, which is Proposition 2.7 scaled by b. As $\bar\nu$ is symmetric, the exponent of the triplet $(a, \bar\nu, 0)$ reduces to the left side, so by C3(iii) the law with that triplet has characteristic function $e^{-b\,\mathrm{Cin}(\tau|\omega|)}$, the same as the law of the datum, whose triplet is $(0, \nu_1, 0)$. By the uniqueness in C3(ii), $a = 0$ and $\bar\nu = \nu_1$.

The shadow determines the support. As ν_1 does not charge $\{|y| > \tau\}$, $\nu(\{x : |x_1| > \tau\}) = \bar\nu(\{|y| > \tau\}) = 0$. A reflection carrying e_1 to a unit vector u preserves ν, so $\nu(\{x : |\langle u, x\rangle| > \tau\}) = 0$ for every unit u. Take a dense sequence (u_n) in S^{d-1}. If $|x| > \tau$, some u_n has $|u_n - x/|x|| < (|x| - \tau)/|x|$, and then $\langle u_n, x\rangle \ge |x| - |u_n - x/|x||\,|x| > \tau$. So $\{|x| > \tau\}$ is covered by countably many null slabs, and $\nu(\{|x| > \tau\}) = 0$. By Lemma A.1 the image of ν under $x \mapsto |x|$ is $k_d(r)\,dr/r$ on $(0, \infty)$, so $k_d = 0$ almost everywhere on (τ, ∞), and since k_d is nonincreasing and nonnegative, $k_d(r) = 0$ for every $r > \tau$.

The edge. Let $0 < \varepsilon \le \frac{1}{2}$ and consider the slab $S_\varepsilon = \{x : \tau(1-\varepsilon) < |x_1| < \tau\}$. By (3.2),

$$\nu(S_\varepsilon) = \frac{1}{|S^{d-1}|}\int_{S^{d-1}} \sigma_{d-1}(d\eta) \int_0^\infty \mathbf{1}_{S_\varepsilon}(r\eta)\, k_d(r)\, \frac{dr}{r}.$$

A point $r\eta$ of S_ε with $k_d(r) \ne 0$ has $r \le \tau$, by the previous step, so $r \ge r|\eta_1| > \tau(1-\varepsilon)$, and $|\eta_1| > \tau(1-\varepsilon)/r \ge 1 - \varepsilon$: η lies in the cap $K_\varepsilon = \{\eta : |\eta_1| > 1 - \varepsilon\}$ and $r \in (\tau(1-\varepsilon), \tau]$, an interval of length $\tau\varepsilon$ on which $k_d(r)/r \le M := 2k_d(\tau/2)/\tau$. Hence

$$\nu(S_\varepsilon) \le \frac{\sigma_{d-1}(K_\varepsilon)}{|S^{d-1}|}\, M\, \tau\varepsilon.$$

On the other side, $\nu(S_\varepsilon) \ge \bar\nu(\{\tau(1-\varepsilon) < |y| < \tau\}) = \nu_1(\{\tau(1-\varepsilon) < |y| < \tau\}) = b\int_{\tau(1-\varepsilon)}^\tau dr/r \ge (b/\tau)\,\tau\varepsilon$. Dividing by $\tau\varepsilon$, $b/\tau \le M\,\sigma_{d-1}(K_\varepsilon)/|S^{d-1}|$ for every $\varepsilon \in (0, \frac{1}{2}]$. As $\varepsilon \downarrow 0$ the caps decrease to the two poles $\{|\eta_1| = 1\} = \{\pm e_1\}$, which lie on the great subsphere $\{\eta_2 = 0\}$ because $d \ge 2$, and which are therefore σ_{d-1}-null (a great subsphere is null, the cone over it lying in a hyperplane). So $\sigma_{d-1}(K_\varepsilon) \to 0$, and $b/\tau \le 0$, a contradiction. Hence $F \notin \mathcal{A}_d$. □

S4 The formal development

Appendix B of the paper says where the formal development is, in which versions, and under which names it holds the cited facts C1, C2, C3, C4, C5, C6 and C7. This section has the declarations statement by statement, Lean's report on them, and the formal examples.

The statements covered. The list below gives the declarations of the formal development that state and prove each statement of the paper and of this supplement that has a formal counterpart: those of §3, of Appendix A and of §S3, in that order. They are the first list of §1.1. In the blueprint the declarations are the names in the `\lean` tag of the statement. They are in the namespace `AffineHemigroup` unless another is written. For a definition the status means that the formal statements are written in its terms. The item of Lemma 3.3 holds the tags of two blueprint statements, and with them three declarations, `operators_to_kernels_gw`, `operators_to_kernels_affine` and `kernels_to_operators_gw`, for clauses that this paper does not print. The formal development also has two statements of the blueprint that this paper prints in another form. The strictness of the filtration between consecutive critical values, with the bound $6 + 4\sqrt{2}$ of the cone paper, is `filtration_strict`, `filtration_witness` and `filtration_witness_exists`; here it is Theorem 3.25(4) and Proposition 3.22(2). And clause (3) of that proposition, as a statement of its own with the proof by one projection, is `no_cin_ray_in_the_plane` with `cin_ray_scaled`.

- Def. 3.1, formalised in Lean: `IsIsotropicOperatorFamily`, `IsSingleOperator`.
- Def. 3.2, formalised in Lean: `IsIsotropicCascadeFamily`, `IsIncreasingBijection`, `IsGauge`, `ScaleSpace.IsSelfDecomposable`, `ScaleSpace.isSelfDecomposable_iff_charFun`, `push`, `charFun_push`, `isGauge_iff_isIncreasingBijection`.
- Lemma 3.3, machine-checked: `operators_to_kernels_repr`, `operators_to_kernels_unique`, `operators_to_kernels_converse`, `operators_to_kernels_transport`, `operators_to_kernels_transport_iff`, `operators_to_kernels_continuity`, `operators_to_kernels_gw`, `operators_to_kernels_affine`, `operators_to_kernels_isotropic`, `gaussDensity`, `gaussL1`, `kernels_to_operators_continuity`, `kernels_to_operators_gw`, `kernels_operators_correspondence`, `kernels_to_operators_isotropic`.
- Lemma 3.4, machine-checked: `radial_marginal`, `radial_marginal_eq_dirac`, `radial_marginal_ops`, `marginal`.
- Lemma 3.5, machine-checked: `isotropic_marginal_family`, `isotropic_marginal_cascadeData`, `ScaleSpace.CascadeFamily`, `ScaleSpace.CascadeData`.
- Thm. 3.6, machine-checked: `isotropic_classification`, `isotropic_classification_origin`, `isotropic_classification_unique`.
- Cor. 3.7, machine-checked: `isotropic_classification_operators`, `isotropic_classification_operators_unique`.
- Lemma 3.8, machine-checked: `isotropic_realisation`.
- Cor. 3.9, machine-checked: `scale_zero_kernels`, `scale_zero_operators`, `scale_zero_families`, `scale_zero_relabelling`.
- Def. 3.10, formalised in Lean: `OriginDatum`, `OriginDatum.levyMeasure`, `OriginDatum.exponentL`, `OriginDatum.exponent`, `OriginDatum.profileAtOrigin`, `OriginDatum.IsSubordinated`, `IsLawOf`, `sphereArea`, `polarProfileOf`, `sphereArea_eq_toSphere`, `OriginDatum.tendsto_profileAtOrigin`, `OriginDatum.polar_decomposition`.
- Prop. 3.11, machine-checked: `polar_reading`, `polar_reading_unique`, `polar_reading_admissible`, `classA_iff_originDatum`, `polar_classification`.
- Prop. 3.12, machine-checked: `isotropic_bridge_family`, `isotropic_bridge_matern`, `isotropic_bridge_stable`, `isotropic_bridge_student_transform`, `isotropic_bridge_student_family`, `IsCausalKernels`, `bridgeFamily`, `gaussVar`, `measurable_gaussVar`.
- Thm. 3.13, machine-checked: `origin_decay_exponent`, `origin_density_smooth`, `origin_density_bounded_smooth`, `subordinated_profileAtOrigin`, `subordinated_law`, `delayLaw_zero`, `subordinated_density_limit`, `subordinated_unbounded`, `origin_unbounded`, `origin_unbounded_rate`, `origin_dichotomy`, `measurable_gaussVar`.
- Cor. 3.14, machine-checked: `exponent_gaussian_limit`, `exponent_profile_limit`, `transform_profile_limit`, `origin_profile_dimension_free`.
- Prop. 3.15, machine-checked: `radial_monotone_family`, `radial_monotone_law`.
- Prop. 3.16, machine-checked: `slowFactor_basic`, `slowFactor_ratio`, `slowFactor_tendsto_of_integrable`, `slowFactor_tendsto_atTop`, `cin_sub_log_limit`, `sphere_log_mean`, `sphereLogMean_one`, `exponent_profile_asymptotic`, `exponent_profile_bounds`, `transform_power_comparison`, `transform_power_bounded_iff`, `transform_power_limit`, `OriginDatum.slowFactor`, `OriginDatum.slowFactorZero`, `sphereLogMean`, `exponentConst`.
- Prop. 3.18, machine-checked: `origin_rate_profile`, `origin_rate_profile_critical`, `origin_density_rate_profile`, `origin_ball_rate_profile`, `origin_density_power_of_integrable`, `origin_density_power_of_not_integrable`, `origin_density_rate_profile_critical`, `origin_ball_rate_profile_critical`, `slowLog_bounds`, `slowLog_div_log`, `origin_density_log_of_integrable`, `origin_density_log_of_not_integrable`,

`OriginDatum.profileConst`, `OriginDatum.slowLog`.

- Prop. 3.20, machine-checked: `OriginDatum.stageProfile`, `OriginDatum.stageMeasure`, `OriginDatum.stageProfile_nonneg`, `OriginDatum.stageMeasure_zero`, `OriginDatum.stageMeasure_absolutelyContinuous`, `OriginDatum.stageMeasure_rotation`, `OriginDatum.stageMeasure_integrable`, `OriginDatum.exponent_smul_sub_smul`, `OriginDatum.stageMeasure_univ`, `OriginDatum.stageMeasure_univ_ne_top_iff`, `isotropic_stage_transform`, `isotropic_stage_mass`, `isotropic_stage_compoundPoisson`, `isotropic_stage_exp_neg_mass`, `profileAtOrigin_pos`, `isotropic_stage_atom`, `isotropic_stage_operator`, `isotropic_stage_not_selfDecomposable`, `isotropic_stage_triplet`, `isotropic_stage_absolutelyContinuous`, `isotropic_stage_no_point_mass`, `isotropic_stage_smooth_density`, `isotropic_stage_dichotomy`.
- Prop. 3.22, machine-checked: `profile_one_dimension_down`, `profile_one_dimension_down_of_classA`, `not_classA_succ_of_profile_discontinuous`, `step_datum_separates`, `cin_ray_not_classA`, `stepOriginDatum`.
- Def. 3.24, formalised in Lean: `classS`, `classA`, `classAInf`, `DelayDatum`, `delayExponentL`, `witnessProfile`, `witnessExponent`, `witnessF`, `polarProfileOf`, `thresholdFun`, `thresholdSup`, `threshold`, `witness_isDelayDatum`.
- Thm. 3.25, machine-checked: `separating_family_critical_values`, `witness_classA_iff`, `criticalValues_strictAnti`, `criticalValues_tendsto`, `witness_classA_iff_le`, `witness_not_classA_of_gt`, `criticalValues_existsUnique_window`, `separating_family_strict`.
- Lemma A.1, machine-checked: `OriginDatum.polar_decomposition`.
- Lemma A.2, machine-checked: `isotropic_exponent_log_bound`, `isotropic_exponent_log_limit`, `OriginDatum.tailConst`.
- Prop. A.3, machine-checked: `origin_lower_bound`, `origin_lower_bound_density`, `origin_lower_bound_ball`, `lowerBoundConst`, `lowerBoundTau`, `lowerBoundKappa`.
- Prop. A.4, machine-checked: `origin_upper_bound`, `origin_upper_bound_of_heat`, `origin_density_upper`, `origin_density_lower`, `origin_density_log_order`, `origin_ball_log_order`, `origin_density_order`.
- Lemma A.5, machine-checked: `shadow_profile_mean`, `shadow_profile_regular`, `shadow_datum`, `lintegral_shadow`, `OriginDatum.shadow`, `shadowProfile`.
- Lemma A.6, machine-checked: `delay_polar_profile`, `delay_polar_profile_datum`.
- Lemma S.8, machine-checked: `positive_transform_heat`, `positive_transform_bound`, `density_bound_ball`, `heatKernel`.
- Lemma S.9, machine-checked: `origin_ball_rate`, `origin_density_rate`, `origin_ball_rate_critical`, `origin_density_rate_critical`, `origin_transform_ball_log`, `origin_density_log_lower`, `origin_rate`, `origin_rate_critical`.
- Lemma S.10, machine-checked: `thresholdSup_spec`, `threshold_pos`, `thresholdSup_succ`, `threshold_strictAnti`, `witness_polar_closed_form`, `witness_threshold_criterion`, `witness_antitone_iff`.
- Lemma S.11, machine-checked: `threshold_limit_bounds`, `threshold_limit`, `thresholdEps`, `stirlingK`.
- Lemma S.12, machine-checked: `tendsto_stirlingRatio`, `threshold_asymptotic_bounds`, `threshold_asymptotic`, `stirlingRatio`.
- Lemma S.13, machine-checked: `classAInf_of_classS`, `classS_of_classAInf`, `filtration_top`.

The report for every declaration. The list above has 222 names, of which 219 are distinct, three of them standing under two statements. Lean's report for these 219 declarations is the file `records/affine/axioms-iso-all-declarations-output.txt` of the repository. Every one of them depends on the three logical axioms, 160 depend on nothing else, 59 depend also on one or more of the nine, and none depends on any other axiom. The file also has the report for the declarations of the two statements of the blueprint that this paper prints in another form. For each statement the cited facts that its declarations name are those of Table 4. The script `records/affine/axioms-iso-all-declarations.py` writes the file of requests from which Lean prints the report, one request for each declaration, and compares the report with the table.

The guard of the repository. The development repository, from which this one is taken, also holds the formal development of the companion paper on affine covariance, and its guard serves both papers. The guard is the file `Formalization/CIAxiomGuard.lean`, run by `lake env lean CIAxiomGuard.lean`. It admits the twelve axioms listed in `blueprint/trust-boundary.txt`:

the nine, and three cited facts of the companion paper, on which no declaration of this paper depends. In the development repository the guard holds for the main declarations a copy of Lean's report and fails when a report differs from its copy, so that none of them gains or loses an axiom unnoticed, and for these it is silent as long as the copy holds. In the public repository the guard is the part of that list that names declarations of this repository, without the copies: it prints Lean's report for each, and the file `axioms.txt` at the root of the repository is what it printed when the repository was built.

Two standard steps proved in the development. Besides Karamata's theorem (C7), the literature takes two steps of the proof of Lemma S.9 from the theory of regular variation: a bound on the ratios of a regularly varying function, and the passage from a distribution function to a monotone density. The formal development proves both, for the functions at hand, so the rate cites C7 alone from that theory.

Formal examples. The review of the formal statements (§S1) asked that the hypotheses of the main statements can be met. For the classification, the threshold, the rate, the strict filtration and the Cin rays, the development holds a formal example that rests on no cited fact. They are in the file `Formalization/AffineHemigroup/Witnesses.lean`, under names that begin with `witness_`: the declarations `witness_isotropic_classification`, `witness_origin_boundedness`, `witness_origin_rate`, `witness_origin_rate_critical`, `witness_filtration` and `witness_no_cin_ray`.

The build checks the six, and the guard of the repository holds Lean's report for `witness_origin_rate` and `witness_origin_rate_critical`. The example $w = 4$ of §3.4 is in the development as far as §S1 says. The file `Probes.lean` of the same directory has two declarations for it. The first, `threshold_three_lt`, is the inequality $w_3 < 4$ and rests on no cited fact. The second, `probe_witness_four`, says that $F^{(4)}$ belongs to no $\mathcal{A}_D$ with $D \geq 3$, and rests on C2, C3 and C4. The build checks the two. The other inequality of the example, $4 \leq w_2$, is proved in print in the cone paper: the proof of [3, Lem. 4.12(2)] shows the criterion (S.19) at $w = 4$ and $d = 2$ for every z.

Lean's report. The block below is Lean's report for eight of the declarations, as it stands in the file of the report named above: those of the classification, on kernels and on operators, of the threshold (its clause (5)), of the rate through the transform, Lemma S.9 (`origin_rate` for its clause (1) and `origin_rate_critical` for its clause (2), the only two of the block that name C7), of the dimension filtration (`filtration_strict`, the strictness of every step, and `filtration_top`, which is Lemma S.13, the only one that names C6), and of the Cin rays (`no_cin_ray_in_the_plane`, the second proof of Proposition 3.22(3)). On each of the eight Lean prints its three logical axioms and names of the cited facts, and nothing else:

```
'AffineHemigroup.isotropic_classification' depends on axioms: [propext,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_unique,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.isotropic_classification_operators' depends on axioms: [propext,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_unique,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.origin_dichotomy' depends on axioms: [propext,
 AffineHemigroup.density_of_integrable_charFun,
 Classical.choice,
 Quot.sound]
```

```
'AffineHemigroup.origin_rate' depends on axioms: [propext,
 AffineHemigroup.karamata_tauberian_zero,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.sato_polar_criterion,
 AffineHemigroup.wolfe_symmetric_unimodal,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.origin_rate_critical' depends on axioms: [propext,
 AffineHemigroup.karamata_tauberian_zero,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.sato_polar_criterion,
 AffineHemigroup.wolfe_symmetric_unimodal,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.filtration_strict' depends on axioms: [propext,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_exists,
 AffineHemigroup.levy_khintchine_unique,
 AffineHemigroup.sato_polar_criterion,
 AffineHemigroup.selfDecomposable_infinitelyDivisible,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.filtration_top' depends on axioms: [propext,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_exists,
 AffineHemigroup.levy_khintchine_unique,
 AffineHemigroup.sato_polar_criterion,
 AffineHemigroup.schoenberg_radial_bernstein,
 AffineHemigroup.selfDecomposable_infinitelyDivisible,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.no_cin_ray_in_the_plane' depends on axioms: [propext,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_exists,
 AffineHemigroup.levy_khintchine_unique,
 AffineHemigroup.sato_polar_criterion,
 AffineHemigroup.selfDecomposable_infinitelyDivisible,
 Classical.choice,
 Quot.sound]
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